

1. HCF & LCM (2-3 qns)

HCF (Highest Common Factors)

$$6 = (2 \times 3)$$

$$= (2 \times 3)$$

$$= (2 \times 3)$$

Factors of 6 (these no.s that can completely divide 6)

HCF (out of all factors, what is the greatest factor)

$$\text{eg} = 6 = 1, 2, 3, (6)$$

HCF

also
write
as

→ GCD (Greatest Common Divisor)

• Eg - 42 54 36 → HCF??

$$2 \mid 42, 54, 36$$

$$3 \mid 21, 27, 18$$

$$7, 9, 6$$

$$3, 3, 2$$

HCF (all 3 no.s should have to be divided by the same no.)

$$\text{HCF} = 2 \times 3 = 6$$

LCM (Least Common multiple)

$$\text{eg} - 2 \rightarrow 4, 6, 8, 10, (12), 14, 16, 18, \dots$$

$$3 \rightarrow 6, 9, (12), 15, (18), 21, \dots$$

$$6 \text{ is least}$$

$$\therefore \text{LCM}(2, 3) = 6$$

$$12 \div 2 = 6$$

$$12 \div 3 = 4$$

LCM

(2, 3)

multiples that

are common b/w

2 & 3

• Eg - 250 100 125 → LCM?

$$25 \mid 250, 100, 125$$

$$5 \mid 10, 4, 5$$

$$2 \mid 2, 4, 1$$

$$\text{LCM} = 500$$

$$125 \times 2 = 250$$

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(for 3 no 40)

$$2 \times 2 \times 2 \times 2 \times 2 = 500$$

$$2 \times 2 \times 2 \times 2 \times 2 = 500$$

TIPS
 HCF ① 18, 24, 30 → HCF?
 divide by 2, 3, 6 → HCF = 6
 3, 4, 5 → not possible to divide
 LCM ② 18, 24, 30 → LCM?
 (mul of 18) (mul of 24) (mul of 30)
 LCM = 360

③ 24, 36, 48
 1st way: 12L 36, 48 → LCM = 144
 2nd way: 48 ÷ 12 = 4, 36 ÷ 12 = 3
 12 × 4 × 3 = 144
 Apply this to 6
 for nos that are
 easy to mul either
 higher like take time for
 multiplying

④ $(\frac{1}{30} - \frac{1}{45})$ → 0.4 in this case how to easily find the LCM & solve the problem?

$$= \frac{45 - 30}{30 \times 45}$$

$$= \frac{15}{1350}$$

$$= \frac{1}{90}$$

(known to me)

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⑤ $(\frac{9}{30} - \frac{2}{45})$
 $= \frac{180 - 80}{1350} = \frac{100}{1350} = \frac{2}{27}$
 (known to me)

① What is LCM of $\frac{36}{225}, \frac{48}{150}, \frac{72}{65}$?

$$\frac{36}{225}, \frac{48}{150}, \frac{72}{65}$$

LCM of fraction = LCM numerators
 HCF Denominators

$$\frac{36}{225}, \frac{48}{150}, \frac{72}{65}$$

$$= \frac{144}{5}$$

Ans

③ What is LCM of $\frac{36}{75}, \frac{48}{150}, \frac{72}{135}$?

$$\frac{36}{75}, \frac{48}{150}, \frac{72}{135}$$

$$LCM = \frac{144}{15}$$

Ans

HCF of fraction = HCF N
 LCM D

$$= \frac{12}{1350} = \frac{2}{225}$$

Ans

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③ What greatest no. divides 17, 42 & 93 leaves remainders 1, 3, 15 respectively?

Let, the greatest no. is x

$$\frac{17}{x} \rightarrow \text{rem} = 1$$

$$17 - 1 = 16 \text{ (prime)}$$

$$\frac{42}{x} \rightarrow \text{rem} = 3$$

$$42 - 3 = 39 \text{ (prime)}$$

$$\frac{93}{x} \rightarrow \text{rem} = 15$$

$$93 - 15 = 78 \text{ (prime)}$$

$$\text{LCM} \rightarrow 13, 39, 78$$

$\therefore$ the greatest no. $= x = 13$ (Ans)
 $\therefore \text{HCF} = 13$

④ What least no. when divided by 36, 24 & 16 leaves 11 as remainders in each case?

$$\text{LCM} \rightarrow 144$$

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$$\frac{15}{24} = \text{quotient} = 0$$

$$\frac{135}{24} = \text{quotient} = 5$$

$$\frac{153}{16} = \text{quotient} = 9$$

Whenever there is this type of Qns i.e. What is the least no. that is divided by some given no. & leaves the same remainders =
① Find the LCM of given no.s
② Add the rem to the LCM
③ The result will be the ans of the Qn.

⑤ What least no. when divided by 20, 48, 36 leaves the remainders 13, 14, 29 respectively?

$$\text{LCM} = 720$$

Whenever there is the Qn like this, i.e. What the least no. that is divided by some given no.s & leaves diff. remainders =
① Find the LCM of given no.s
② The diff. b/w given no.s & remainders same (if so then)
③ Subtract that rem from LCM
④ The result will be our ans

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③ [Whereas the num is same we add it, & when
- even the num are different we subtract it] (the difference)

④ What least possible 4-digit number, when divided
by 12, 15, 18 & 20 leaves 21 as remainder?
Rem = same (so need to add to result of the Lcm)

$$\begin{aligned} \text{Lcm}(12, 15, 18, 20) &= 12 \times 15 \times 18 \times 20 \\ &= 5 \times 8 \times 9 \times 10 \\ &= 3 \times 4 \times 9 \times 5 \\ &= 144 \times 5 \\ &= 720 \end{aligned}$$

$\rightarrow 720 + 21 = 741$ (least 4 digit no.)
now 720 is divisible by 12, 15, 18, 20 (as it is their Lcm), i.e. the multiples of 720
will be also divisible by 12, 15, 18, 20 (as 720
itself is divisible by them)

$\therefore$ no. multiples of 720 = $720 \times 2 = 1440$

$\therefore$ now $\rightarrow 1440 + 21 = 1461$ (Ans)

⑤ The ratio of 2 no.s is 5:6 & their Lcm is 480,
then their HCF is?

It, Here are 2 no.s A B

$$AB = \text{HCF} \times AB \times \text{Lcm} \div AB$$

$$\begin{array}{r} 5x \\ \times 6x \\ \hline 30x \\ 30x \\ \hline 180x \end{array}$$

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A	B
15	20
HCF = 5	HCF = 5
Lcm = 60	Lcm = 60

Now, $L.H.S = B \times (3) = 15 \times 20 = 300$
 $R.H.S = 5 \times 60 = \text{HCF} \times \text{Lcm} = 300$

$\therefore L.H.S = R.H.S$ (Proved)
 $\therefore AB = \text{HCF} \times AB \times \text{Lcm} \div AB$

Ratio of 2 no.s = 5:6 (as H's ratio we can't
let, & be the common factor use them directly)

$\therefore$ Two no.s are = $5K$ & $6K$ (as it's ratio)

Lcm = 480 (given)

HCF = ?

$5K, 6K \therefore \text{HCF} = K$

$\therefore$ Now, (Whenever there will
be the ratio, the common factor
of the ratio is HCF (always))

$5K + 6K = K + 480$

$\Rightarrow 11K = 480$

$\Rightarrow K = 16$

$\therefore \text{HCF} = K = 16$ (Ans)

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Q. HCF & LCM of 2 no.s is 8 & 96. Sum of these no.s is 56. Then what is sum of their reciprocals?

A B

$$AB = \text{HCF} \times \text{LCM} \quad \text{--- (i)}$$

Let the no.s are A & B

$$\text{Given, } A+B = 56 \quad \text{--- (ii)}$$

$$\text{HCF} = 8$$

$$\text{LCM} = 96$$

$$AB = 8 \times 96 \quad \text{[from (i)]} \quad \text{--- (iii)}$$

Now, the req. is $\frac{1}{A} + \frac{1}{B}$

$$\frac{1}{A} + \frac{1}{B} = \frac{A+B}{AB} = \frac{56}{8 \times 96} \quad \text{[from (ii) & (iii)]}$$

$$= \frac{7}{96} \quad \text{--- Ans}$$

Q. What largest no. will divide 47, 35 & 27 leaving same remainder in each case? What will be the common remainder?

largest = GCD/No.

47 - 35 27 [I don't know the common value]

$$\text{So, } 47 - 35 = 12$$

$$35 - 27 = 8$$

$$27 - 47 = -20$$

Subtract them from e/o

align the size

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$$\text{HCF}(12, 8, 20) = 4 \mid 12, 8, 20$$

$\therefore \text{HCF} = 4 \rightarrow$ that is the largest no. (ans. Greater too - 100)

47	rem = 3	35	27
4		4	4
		3	3

$\therefore$ the common rem = 3 --- Ans

Q. There are 3 triangular triangles with sides 114 cm, 76 cm & 152 cm. What maximum size scale can measure them exactly?

$$114 \quad 76 \quad 152$$



$$\text{HCF}(114, 76, 152)$$

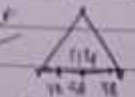
$$= 2 \mid 114, 76, 152$$

$$1 \mid 57, 38, 76$$

$$3 \quad 2 \quad 4$$

$$\therefore \text{HCF} = 38 \text{ cm}$$

$\therefore$ The max. size scale is = 38 cm --- Ans



$$38 \times 3 = 114$$

$$38 \times 2 = 76$$

$$38 \times 4 = 152$$

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Whenever there is this kind of sum, i.e. the sides of some equilateral triangles are given & need to find out the max side site that can measure them perfectly then we need to do - ① HCF of the given sides / no. ② the result will be our ans

[this can be asked by many ways, i.e. instead of equilateral triangle, there can be anything else, just remember "max side site to measure least side" is the clue] → it would always be HCF

11) If $(n-4)$ is the HCF of $x^2 - 8x + 15$ & $n^2 - kn + 1$, then what is k ?
 HCF = $(n-4)$ i.e. $(n-4)$ is the factor of both eqs. It means it can divide both eqs perfectly with 0 rem.
 HCF = can divide (factor)
 LCM = can be divided (multiple)

So, $n-4$ is HCF, then if we put $n=4$ in both eqs $n^2 - 8n + 15$ & $n^2 - kn + 1$ they would satisfy i.e. the result of both eqs will be 0.
 i.e. $x = 4$ in $x^2 - 8x + 15 = 0$
 $\Rightarrow 16 - 32 + 15 = 0$
 $\Rightarrow 16 - 32 + 15 = n^2 - kn + 1$ when $x=4$
 $\Rightarrow 16 - 32 + 15 = 16 - 4k + 1$
 $\Rightarrow -17 + 15 = -4k + 1$
 $\Rightarrow -2 = -4k + 1$
 $\Rightarrow -3 = -4k$
 $\Rightarrow k = \frac{3}{4}$

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Topic: LCM

traffic signals, ... together ...

② 5 clocks ring automatically at intervals of 12 mins, 8 mins, 3 mins, 4 mins & 10 mins, respectively. In 8 hours from the moment they start, how many times will they ring together?

Always remember, whenever we have to find out how many times they ring together after the start, it will always be the next instance & it will always be LCM.

Ex: can ring together at 12, 24, 36, 48, 60, 72, 84, 96, 108, 120, 132, 144, 156, 168, 180, 192, 204, 216, 228, 240, 252, 264, 276, 288, 300, 312, 324, 336, 348, 360, 372, 384, 396, 408, 420, 432, 444, 456, 468, 480, 492, 504, 516, 528, 540, 552, 564, 576, 588, 600, 612, 624, 636, 648, 660, 672, 684, 696, 708, 720, 732, 744, 756, 768, 780, 792, 804, 816, 828, 840, 852, 864, 876, 888, 900, 912, 924, 936, 948, 960, 972, 984, 1000

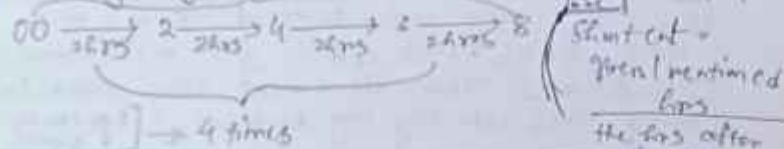
It would always be LCM (It is future, i.e. next instances, the next instances can be given by LCM)

LCM(12, 8, 3, 4, 10) = 2 | 12, 8, 3, 4, 10
 2 | 6, 4, 3, 2, 5
 3 | 2, 1, 1, 5
 5 | 1, 1, 1, 1
 LCM = 120 min

120 mins = 2 hrs
 120 mins = 2 hrs

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In 8 hrs, from the moment they start, how many times will they ring together.



∴ 4 times (Ans)

(Same concept as Ex 9)
 ③ 3 cyclists cycle along the circumference of a jungle. They complete 1 round in 27 minutes, 45 mins & 63 mins respectively. Since they start together, when will they meet again at the starting position?

need LCM instance → LCM to find

$$\begin{array}{r} 9 \overline{) 27, 45, 63} \\ \underline{3 } \\ 3 \end{array}$$

LCM = 9 × 3 min

∴ They will meet again at the starting position after 9 × 3 mins (Ans)

④ Mary wants to paste wallpaper on the wall of his room. The wall is 4 meters & 50 cm in length & 3 meters & 50 cm in height.

Example

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But the wall should be covered completely only by the same pieces of wallpaper having same size that is the no. of maximum sized wallpaper squares needed to cover the walls completely? (The sum is given for max sized square. So need to find HCF/GCD)

4m 50cm = 450cm



3m 50cm = 350cm

$$\text{HCF}(450, 350) = \begin{array}{r} 50 \overline{) 450, 350} \\ \underline{9 } \\ 50 \end{array}$$

Area = 450 × 350

Area of 1 square wallpaper = 50 × 50

∴ The no. of max sized wallpaper squares needed to cover the walls completely =

$$\frac{450 \times 350}{50 \times 50} = 63 \text{ (Ans)}$$

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⑤ The sum of 2 no.s is 156 & their HCF is 13. The no.s of such no pairs is? (can be solved by logic)

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In 3 hrs
 1st B $A+B = 156$
 $HCF = 13$ (13) 13 can divide this 2 nos
 i.e. this no's are multiple of 13

multiple of 13
 13 26 39 52 65 78 91 104 117 130

4th
 3 Cys
 now the sum of any no. can be 156, but as this 2 nos is already given, so we need to find only the multiples of 13 to find such no. whose sum is 156

$A+B = 156$ (HCF = 13) → 1 pair (now 13 can be combined with any no. as we already have the actual no. which is 13. we want to add with other no. so always $2x < 156$)

$x(36, 130)$ (HCF = 26) → 2nd pair
 $x(39, 117)$ (HCF = 39) → 3rd pair
 $x(52, 104)$ (HCF = 52) → 4th pair
 $x(65, 91)$ (HCF = 13) → 5th pair
 $x(78, 78)$ (HCF = 78)

14th
 2 pairs = (13, 143) & (65, 91)

Q. What is the least no. which when divided by numbers 3, 5, 6, 8, 10 & 12 leaves in each case a remainder 2, but which when divided by 13 leaves no remainder?

Trick (when in exam time is constraint, & we are not able to solve it by normal method) now, we can see that when the no. is divided by 13, it leaves no rem, so, the no. is perfectly divisible by 13.

Here we are given 9 or 14 or 20 or 31 or 52 or 65 or none of the above

now, another info is given in the qn = the no. when it's divided by 3, 5, 6, 8, 10 & 12, leaves rem = 2

∴ $312 - 2 = 310$ → It has to be perfectly div by 3, 5, 6, 8, 10, 12
 But it is not div by 3 so $312 \times$

$362 - 2 = 360$ →

It is perfectly div by 3, 5, 6, 8, 10, 12 ∴ $360 \times$

$1586 - 2 = 1584$ →

But it is not div by 10
 ∴ $1586 \times$

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 ∴ Ans is 362

1)
2)
3)

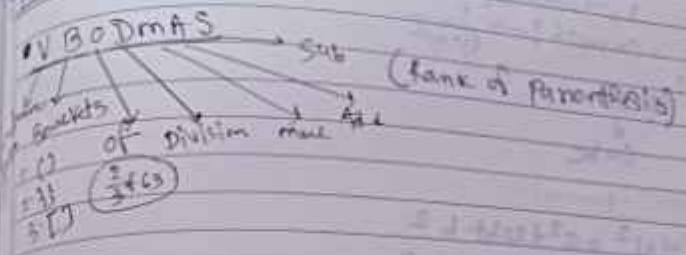
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[Faint handwritten notes on the left page, mostly illegible]

Simplification (5-6 sums)

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$$\begin{aligned} \text{Ex } 17 + [8 - \{15(30 - 8 - 12)\}] \\ = 17 + [8 - \{15(90)\}] \\ = 17 + [8 - \{1350\}] \\ = 17 - 1342 \\ = -1325 = \text{Ans} \end{aligned}$$

Rationalization

$$\begin{aligned} \text{Ex } \frac{1}{\sqrt{28} - \sqrt{24}} &= \frac{(\sqrt{28} + \sqrt{24})}{(\sqrt{28} - \sqrt{24})(\sqrt{28} + \sqrt{24})} \\ &= \frac{(\sqrt{28} + \sqrt{24})}{28 - 24} \\ &= \frac{(\sqrt{28} + \sqrt{24})}{4} = \text{Ans} \end{aligned}$$

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10

100

100

100

100

• Squares & Cubes

$$\begin{aligned} a &\rightarrow 26 \rightarrow 26^2 = 676 \\ b &\rightarrow 27 \rightarrow 27^2 = 729 \\ c &\rightarrow 28 \rightarrow 28^2 = 784 \end{aligned}$$

• Basic formulae

$$1) (a+b)^2 = a^2 + 2ab + b^2$$

$$2) (a-b)^2 = a^2 - 2ab + b^2$$

$$3) a^2 - b^2 = (a+b)(a-b)$$

$$4) (a+b)^3 = a^3 + b^3 + 3ab(a+b)$$

$$5) (a-b)^3 = a^3 - b^3 - 3ab(a-b)$$

$$6) a^3 + b^3 = (a+b)(a^2 - ab + b^2)$$

$$7) a^3 - b^3 = (a-b)(a^2 + ab + b^2)$$

$$8) (a+b+c)^2 = a^2 + b^2 + c^2 + 2(ab+bc+ca)$$

$$9) a^3 + b^3 + c^3 - 3abc = (a+b+c)(a^2 + b^2 + c^2 - ab - bc - ca)$$

$$10) (a + \frac{1}{a})^2 = (a^2 + \frac{1}{a^2}) + 2 = (a - \frac{1}{a})^2 + 2 + \frac{4}{a^2}$$

$$11) (a - \frac{1}{a})^2 = (a^2 + \frac{1}{a^2}) - 2 = (a + \frac{1}{a})^2 - 4$$

$$12) (a + \frac{1}{a})^3 = (a^3 + \frac{1}{a^3}) + 3(a + \frac{1}{a})$$

$$13) (a - \frac{1}{a})^3 = (a^3 - \frac{1}{a^3}) - 3(a - \frac{1}{a})$$

needed to remind
st - 1 - 30
leave - 1 - 10

• Simplifying series

$$\begin{aligned} 2+4+8+120 \\ = 2(1+2+3+10) \quad [1 \text{ trick} = \text{error}] \end{aligned}$$

$$\begin{aligned} 2(n+1) &= \\ 2(n+1) &= \end{aligned}$$

2nd trick
① (Not all to get the ans)

Problem
Type

$$\begin{aligned} 3608 \\ 1404 \end{aligned}$$

div from the Smallest no = 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20

Start

$$\text{if } \frac{360}{135} \text{ div starts from } 5 \quad (\text{Immediately div by } 5)$$

$$\text{if } \frac{360}{120} \text{ div } = 10 \quad (\text{Immediately div by } 10)$$

Problem
Type

WORD Problem

1 Detailed explanation
in video problem
now calculate

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① Find value of $m - \frac{1}{m}$ if $m + \frac{1}{m} = 4$

1st way (Algebraic eqn way)

$$m + \frac{1}{m} = 4$$

$$\frac{(m + \frac{1}{m})^2}{m} = 16$$

$$\frac{m^2 + 2 + \frac{1}{m^2}}{m} = 16$$

$$m^2 + 2 + \frac{1}{m^2} = 16m$$

$$\frac{1}{m^2} + 2 - 16m + m^2 = 0$$

$$\frac{1}{m^2} + 15m^2 - (16 - \frac{2}{m})m = 0$$

$$\frac{1}{m^2} + 15m^2 - 16m + \frac{2}{m} = 0$$

$$\frac{1}{m^2} + 15m^2 - 16m + \frac{2}{m} = 0$$

$$\frac{1}{m^2} + 15m^2 - 16m + \frac{2}{m} = 0$$

$$m = \frac{1}{3} \text{ or } -\frac{1}{5}$$

2nd way $(m + \frac{1}{m})^2 = 16$

$$\Rightarrow (m + \frac{1}{m})^2 = 16$$

$$\Rightarrow (m^2 + \frac{1}{m^2}) + 2 = 16$$

$$\Rightarrow (m^2 + \frac{1}{m^2}) + 2 = 16$$

$$\Rightarrow (m^2 + \frac{1}{m^2}) + 2 = 16$$

$$\Rightarrow (m^2 + \frac{1}{m^2}) + 2 = 16$$

$$\Rightarrow m - \frac{1}{m} = \sqrt{12} = 2\sqrt{3}$$

$$m = \frac{1}{3}$$

$$m - \frac{1}{m} = \frac{1}{3} - 3 = -\frac{8}{3}$$

$$m = -\frac{1}{5}$$

$$m - \frac{1}{m} = -\frac{1}{5} + 5 = \frac{24}{5}$$

$$(m - \frac{1}{m})^2$$

$$= (m^2 + \frac{1}{m^2}) - 2$$

$$= (m^2 + \frac{1}{m^2}) - 2 - 2$$

$$= (4)^2 - 4$$

$$= 12$$

$$\therefore m - \frac{1}{m} = 2\sqrt{3}$$

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② Find value of $\frac{a^3 + 2a + b^2}{a^3 - 2a^2}$ if $a + \frac{b^2}{a} = 2$

$$\frac{a^3 + b^2}{a} = 2$$

$$\frac{a^3 + b^2}{a} = 2$$

$$\frac{a^3 + b^2}{a} = 2$$

$$\frac{a^3 + b^2}{a} = 2$$

$$\frac{a^3 + b^2}{a} = 2$$

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$$\frac{a^3 + b^2}{a} = 2$$

$$\frac{a^3 + b^2}{a} = 2$$

$$\frac{a^3 + b^2}{a} = 2$$

③ Find value of $a^3 + b^3 + 3ab$, when value of $a + b = 1$

$$a^3 + b^3 + 3ab = ?$$

$$a^3 + b^3 + 3ab = ?$$

$$a^3 + b^3 + 3ab = ?$$

$$a^3 + b^3 + 3ab = ?$$

$$a^3 + b^3 + 3ab = ?$$

$$a^3 + b^3 + 3ab = ?$$

$$a^3 + b^3 + 3ab = ?$$

$$a^3 + b^3 + 3ab = ?$$

$$a^3 + b^3 + 3ab = ?$$

④ What is value of $[(3 \times 3 \times 3 \times 3 \times 3 \times 3) \div (3 \times 3 \times 3 \times 3) \times 3]$

Concept: Indices & surds

$$A^m \times A^n = A^{m+n}$$

$$A^m \div A^n = A^{m-n}$$

$$A^m \div A^n = A^{m-n}$$

$$A^m \div A^n = A^{m-n}$$

$$A^m \div A^n = A^{m-n}$$

$$A^m \div A^n = A^{m-n}$$

$$A^m \div A^n = A^{m-n}$$

$$A^m \div A^n = A^{m-n}$$

$$A^m \div A^n = A^{m-n}$$

$$(3^6) \div (3^4)^2 \times 3$$

$$= 3^6 \div 3^8 \times 3$$

$$= 3^6 \div 3^8 \times 3$$

$$= 3^6 \div 3^8 \times 3$$

$$= 3^6 \div 3^8 \times 3$$

$$= 3^6 \div 3^8 \times 3$$

$$= 3^6 \div 3^8 \times 3$$

$$= 3^6 \div 3^8 \times 3$$

$$= 3^6 \div 3^8 \times 3$$

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⑤ What will be the value of $\sqrt[3]{0.0016} + \sqrt[3]{2000000}$
 $\sqrt[3]{0.000512} + \sqrt[3]{0.064}$?

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$$= \sqrt[3]{0.0016} + \sqrt[3]{2000000}$$

$$\sqrt[3]{0.000512} + \sqrt[3]{0.064}$$

$$= 0.08 + (2 + 100) = 10$$

⑥ What will be value of c, if a(mtr)

$$= 85, b(a(mtr)) = 26, c(a+b(mtr)) = 108$$

$$a(a+b(mtr)) + b(a+b(mtr)) + c(a+b(mtr))$$

$$\rightarrow a+b(mtr) = \sqrt{289} = 17$$

$$\rightarrow c(a+b(mtr)) = 108$$

$$\rightarrow c + 17 = 108$$

$$\rightarrow c = 108 - 17$$

⑦ What is value of $(13.8 + 1.5 \div 5.7 + 11.2 \text{ of } \frac{1}{16} + \frac{1}{20})$?

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$$11.2 \text{ of } \frac{1}{16} = 11.2 \times \frac{1}{16} = 0.7$$

$$(13.8 + 1.5 \div 5.7 + 0.7 - \frac{1}{20})$$

$$= (13.8 + \frac{1.5}{5.7} + 0.7 - 0.05)$$

$$= 13.8 + \frac{1}{8} + 0.7 - 0.05$$

$$= 4.6 + 0.7 - 0.05$$

$$= 5.3 - 0.05$$

$$= 5.25$$

⑧ What will be value of (mtr) if he finds that $\sqrt{28-6\sqrt{3}} = \sqrt{3m+n}$

$$\sqrt{28-6\sqrt{3}} = \sqrt{3m+n}$$

$$\rightarrow 28-6\sqrt{3} = (\sqrt{3m+n})^2$$

$$\rightarrow \frac{28}{3} - 2\sqrt{3} = 3m^2 + 2\sqrt{3}mn + n^2$$

$$\rightarrow (3\sqrt{3})^2 - 2 \times 3\sqrt{3} + 1 = (\sqrt{3m+n})^2$$

$$\rightarrow (1)^2 - 2 + 1 + 3\sqrt{3} + (3\sqrt{3})^2 = (\sqrt{3m+n})^2$$

$$\rightarrow (3\sqrt{3} - 1)^2 = (\sqrt{3m+n})^2$$

$$\rightarrow (3\sqrt{3} - 1)^2 = (\sqrt{3m+n})^2$$

$$\rightarrow 3\sqrt{3} - 1 = \sqrt{3m+n}$$

$$m=3, n=-1 \text{ mtr } = 2$$

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On simplification of the following, the result will be

$$(1-\frac{1}{2})(1-\frac{1}{3})(1-\frac{1}{4}) \dots (1-\frac{1}{100})$$

$$= \frac{1}{2} \times \frac{2}{3} \times \frac{3}{4} \times \dots \times \frac{99}{100}$$

$$= \frac{1}{100} = 0.01$$

(10) $4^0 + 4^1 + 4^2 + 4^3 + 4^4 + 4^5$ is divisible by

$$4^0 + 4^1 + 4^2 + 4^3 + 4^4 + 4^5$$

$$= 4^0(1 + 4^1 + 4^2 + 4^3 + 4^4 + 4^5)$$

$$= 4^0(1 + 4 + 16 + 64 + 256)$$

$$= 4^0(341)$$

Ans = 11

(11) If $a = 4, 36, b = 2, 39, c = 1, 37$, then the value of

$$a^3 - b^3 - c^3 - 3abc = (a+b+c)(a^2 + b^2 + c^2 - ab - bc - ca)$$

$$a^3 - b^3 - c^3 - 3abc$$

$$= (a+b+c)(a^2 + b^2 + c^2 - ab - bc - ca)$$

$$= (4+36+2)(4^2 + 36^2 + 2^2 - 4 \cdot 36 - 36 \cdot 2 - 4 \cdot 2)$$

$$= 42(16 + 1296 + 4 - 144 - 72 - 8)$$

$$= 42(1200 - 120) = 42(1080) = 45360$$

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(8) If $a+b = 2a+3b$, then the value of

$a+b \neq a+b$
Hence, just symbolizing that $a+b = 2a+3b$
Hence, $a+b$ can be calculated by $2a+3b$
 $\therefore a+b = 2a+3b$

$$= 2+3+3+3+2+3+4 \times b$$

(13) If $1^2 + 2^2 + 3^2 + \dots + 10^2 = 385$, then $3^2 + 6^2 + 9^2 + \dots + 30^2$ is equal to?

$$3^2 + 6^2 + 9^2 + \dots + 30^2$$

$$= 3^2(1^2 + 2^2 + 3^2 + \dots + 10^2)$$

$$= 3^2(385)$$

$$= 3 \times 1465$$

(14) Simplify:

$$0.72 + 0.72 + 0.72 + 0.39 + 0.39 + 0.39$$

$$= (0.72)^3 + (0.39)^3$$

$$= (0.72)^3 + (0.39)^3$$

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$$= (a^3 - b^3)$$

$$a^2 + ab + b^2$$

$$= (a-b)(a^2 + ab + b^2)$$

$$(a^2 + ab + b^2)$$

$$= a-b$$

$$= 0.42 - 0.30$$

$$= 0.33$$

(15) For what value of x , the statement $\frac{x}{15} \times \frac{1}{35} = 1$ is true?

$$\frac{x}{15} \times \frac{1}{35} = 1$$

$$x(1) = 15 \times 35$$

$$x = 45$$

(16) The value of

$$\frac{1}{1+\sqrt{2}} + \frac{1}{1+\sqrt{3}} + \frac{1}{1+\sqrt{4}} + \dots + \frac{1}{1+\sqrt{16}}$$

$$= \frac{(1-\sqrt{2})}{1-2} + \frac{(\sqrt{2}-\sqrt{3})}{2-3} + \frac{(\sqrt{3}-\sqrt{4})}{3-4} + \dots + \frac{(\sqrt{15}-\sqrt{16})}{15-16}$$

$$= -1 + \sqrt{2} - \sqrt{2} + \sqrt{3} - \sqrt{3} + \sqrt{4} + \dots - \sqrt{15} + \sqrt{16}$$

$$= -1 + \sqrt{16}$$

$$= -1 + 4$$

$$= 3$$

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(17) The value of $25.25 - 23.23 + 24.24$ is?

$25.25 - 23.23 + 24.24$ (do both first
store the nos before decimal) add, sub)

$$= 25 - 23 + 24$$

$$= 26$$

(have nos after decimal point)

$$25 - 23 + 24 = 26 \Rightarrow 26.26$$

$$26.26 - \text{Ans}$$

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2. Fractions & decimals (4-5 ms)

① What will be fraction form of $0.3\overline{523}$?

- $\rightarrow$ Base means the no. it represents, i.e. base

0.35 282323

0.3523 Repeating

$$n \sqrt{3523} = 35$$

3 200

For you	For no one
3489	0000

 $0.35 \frac{23}{51}$

10/5
20/10

below
the line

So we have

Two-05
a p p e n d i x[illegible]

For d n, write the full no (i.e. after point what even the no. will be) & subtract the no.s that aren't below the bar.

$$\frac{K_f}{P} \rightarrow 0.45 \rightarrow 0.45666 \dots$$
$$\therefore \text{frame fin } \frac{a}{b} = 46 - 4$$
$$11) 0.\overline{25} \rightarrow 0.25252525 \dots$$
$$\therefore \frac{25-0}{99} = \frac{25}{99}$$

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fraction times d. no. = $\frac{p}{n} = \frac{4}{1}$

The no. of $\overline{X}$ is 100
← (Pop. sz) that

$$F(x) = \int_a^x f(t) dt = F(x) - F(a)$$

25-fer 25-fer

no. of numbers

Below the	number
Lower	below

ndition

② What is value of $3.35 + 5 + 5 + 5 + 5.55 + 1.67$?

5:550

5. 0.000

5-500

5.555

50.60

$$+ 5400$$

31695

$$\therefore 31.655$$

② What will be value of $2.4 + 0.72 + 4.5$
 $0.18 + 0.06 + 0.02 + 2$

2-4-07 2 4-5

0-18 + 4-06 + 0-5

$\frac{1}{2}$

72 14 10

10. $\frac{1}{100} \times \frac{1}{100} = \frac{1}{10000}$

10/10/10

3400

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② What will be the value of $1 + 0.3 + 0.01 + 0.003$?

$$1 + 0.3 + 0.01 + 0.003$$

$$1 + 3 + 1 + 3 = 9$$

$$0.000000$$

③ What will be the value of

$$0.2146?$$

$$0.000465 \text{ if } \frac{1}{4.65} = 0.2146$$

$$\frac{1}{0.000465} = \frac{1}{4.65 \times 10^{-4}}$$

$$\frac{1}{4.65} \times 10^4$$

$$= 0.2146 \times 10^4$$

$$= 2146$$

④ $4.142 + 100.8 + 7 + 0.053 = 105.153$
? stands for what?

$$4.142 + 100.8 + 7 + 0.053 = 105.153$$

$$x = 105.153 - 4.142 - 100.8 - 0.053$$

$$05 - 4 - 100 - 0 = 1$$

$$153 - 142 - 800 - 153 = 0 - 842 \rightarrow -0.842$$

$$1 + (-0.842) = 1 - 0.842 = 0.158$$

decimal places

$$\frac{1}{4.65} = 0.2146$$

while shifts
10⁻ⁿ to the
no. of
decimal
places
You are
shifting

$$\text{let } = 3.467$$

$$1 \text{ decimal place}$$

$$= 34.67 \times 10^{-1}$$

$$\text{let } = 3.46705$$

$$3 \text{ decimal place}$$

$$(3.46705) \times 10^{-3}$$

for shifting
the 1. For now
i.e. while shifts
10⁻ⁿ to the no. of
decimal places
You are
shifting

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→ This is the way to do add/subtraction (shorthand)

$$x(2) = 0.158$$

$$10.375 + 10.33 + 7.82 + 3.11 = ?$$

$$40.660$$

$$40.650$$

$$90.850$$

$$40.500$$

Shorthand

$$10.10 + 7.33$$

→ 10.10 + 7.33 = 17.43 (there are 3 d.p.s, so calculating it shorthand)

⑪ What is approximate value of $8450 \div 11.98$

$$-23.99 \div 7$$

140. They have approximate val.

So, while calculating we can also consider approximate val.

$$8450 \div 11.98 = 23.99 \div 7$$

$$8460 \div 12 = 24$$

$$= 8460 \div 12 = 24$$

$$= 8460 \div 12 = 24$$

$$= 205 - 24 \div 7$$

$$= 705 - 480$$

$$= 225$$

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If there is such critical calculation & given make word not mentioned, then if there are many options & there are huge diff. in the b/w the options, then we can take approximate of the given no's & can calculate the approximate ans. & then can choose the ans from the opt that is nearest to our calculated ans.

(12) What is value of $245.34 - 1042.752 + 875.435 + 31.035$?

a) 308.004

b) 110.008

c) 295.630

d) 807.000

$245 - 1042 + 875 + 31$

$\approx 245 - 1045 + 875 + 31$

≈ 806

Ans. As there are 2 ops & calculate it give that i.e. just take the last single digit from the digits after $\therefore$

$$1 - 2 + 5 + 5 = 9$$

(13) Which of the following is longest fraction amongst $\frac{5}{8}, \frac{3}{7}, \frac{2}{3}, \frac{4}{5}$? Compare $\rightarrow$ $\frac{5}{8} > \frac{3}{7}$ $\frac{3}{7} > \frac{2}{3}$ $\frac{2}{3} > \frac{4}{5}$ $\frac{4}{5} > \frac{5}{8}$

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1st way (Ratio & Proportion's way) If $D > 10$ then use this method

2nd way, if the Denominator is < 10 then we can divide directly to find the ans

2nd way,

$$1) \frac{5}{8} = 0.625 \quad 2) \frac{3}{7} = 0.428 \quad 3) \frac{2}{3} = 0.666 \quad 4) \frac{4}{5} = 0.8$$

0.8 - Ans

This is largest

1st way,

then are 2 fractions,

$$\frac{a}{b} \text{ and } \frac{c}{d}$$

now just cross multiply

$$\frac{a}{b} > \frac{c}{d}$$

If $ab > bc$ then,

(LHS) $>$ (RHS)

There is $\frac{a}{b}$ in LHS

$$\text{So } \frac{a}{b} > \frac{c}{d}$$

ii)

If $ab < bc$ then,

LHS $<$ RHS

$$\frac{a}{b} < \frac{c}{d}$$

iii)

If $ab = bc$

then LHS = RHS

$$\frac{a}{b} = \frac{c}{d}$$

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$$\frac{5}{8} > \frac{3}{7}$$

$$\frac{3}{7} > \frac{2}{3}$$

$$\frac{2}{3} > \frac{4}{5}$$

$$35 > 24$$

$$10 < 15$$

$$25 < 32$$

$$\frac{5}{8} > \frac{3}{7}$$

$$\frac{3}{7} > \frac{2}{3}$$

$$\frac{2}{3} > \frac{4}{5}$$

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14) The least of the following $0.\overline{2}$, $(0.\overline{2})^2$, $0.\overline{2}$, $1 \div 0.\overline{2}$

$0.\overline{2} \rightarrow$ least

$(0.\overline{2})^2$ [after 1 num
0.2 $\times$ 2 $\rightarrow$ after
not after
Ans

$0.\overline{2} = \frac{2}{10} = \frac{1}{5}$
 $1 \div 0.\overline{2} = \frac{1}{\frac{1}{5}} = 5$

15) What is increasing order of the fractions

$\frac{6}{7}, \frac{8}{9}, \frac{2}{3}, \frac{3}{10}$

$D \times 10$
Method: Just simply div them, arrange them in a ascending order. Another is we can use ratio technique (that is smallest which is larger so arrange them)

Trick: $D > N$

For each $\frac{p}{q}$, $D = q$ same

now whenever $D > N$ & $D = N$ = same, then we can say that in all fractions

Smaller the $N \rightarrow$ smaller the fraction

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8, [greater # $N \rightarrow$ greater fraction]

$\frac{6}{7} < \frac{8}{9} < \frac{2}{3} < \frac{3}{10}$

16) What is increasing order of the fractions

$\frac{14}{17}, \frac{10}{12}, \frac{6}{9}, \frac{18}{22}$

$D > N$

$17 - 14 = 3$ not sure
 $12 - 10 = 2$

Here we can divide or we can do the ratio technique.

Trick: $D > N$

1) Arrange the p from lowest to largest

$6^{th}, 10^{th}, 14^{th}, 18^{th}$

$7^{th}, 12^{th}, 9^{th}, 22^{th}$

2) Write the corresponding D also

3) If N is increasing by some definite fixed val. & p is also increasing by some definite fixed val. then

Smaller $N \rightarrow$ Smaller fraction
or $N \rightarrow$ a time

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$$\therefore \frac{6}{7} < \frac{140}{12} < \frac{14}{12} < \frac{18}{22}$$

In such cases we always named them in order of their size (small to large) (small to large)

⑦ $\frac{4}{7}$ of $\frac{2}{3}$ of $\frac{5}{6}$ of $\frac{6}{8}$ of 1008 is?

$$\frac{4}{7} \times \frac{2}{3} \times \frac{5}{6} \times \frac{6}{8} \times 1008$$

$$= 200$$

⑧ If a fraction's denominator is decreased by 50% & numerator is increased by 300%, the fraction becomes $\frac{2}{3}$. what is the fraction?

$$\text{let } \frac{a}{b} = \frac{n}{d}$$

$$\frac{n}{d} \rightarrow \frac{a + \frac{300a}{100}}{b - \frac{50b}{100}} = \frac{4a}{0.5b}$$

$$\frac{4a}{0.5b} = \frac{2}{3} \Rightarrow \frac{4a}{b} = \frac{2}{3} \times 0.5b$$

$$\frac{4a}{b} = \frac{1}{3}$$

$$\frac{4a}{b} = \frac{1}{3}$$

$$\frac{4a}{b} = \frac{1}{3}$$

⑨ In 3 fractions when the largest fraction is divided by smallest fraction, the result $\frac{5}{4}$ is greater than the middle fraction by $\frac{1}{2}$.

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If sum of the 3 fractions is $2\frac{2}{3}$ what will be the difference b/w largest & smallest fraction?

Let: Largest = A, middle = B, Smallest = C

$$A + B + C = 2\frac{2}{3}$$

$$\text{Given, } \frac{A}{C} = \frac{5}{4} \rightarrow A = \frac{5C}{4} \quad \text{--- (i)}$$

also, $\frac{5}{4} = B + \frac{1}{2}$ [$\frac{5}{4}$ is greater than the mid by $\frac{1}{2}$]

$$\Rightarrow \frac{5C}{4} - \frac{1}{2} = B$$

$$\Rightarrow B = \frac{5C}{4} - \frac{1}{2} \quad \text{--- (ii)}$$

$$\text{Given, } A + B + C = 2\frac{2}{3} = \frac{8}{3}$$

$$\Rightarrow \frac{5C}{4} + \frac{5C}{4} - \frac{1}{2} + C = \frac{8}{3}$$

$$\Rightarrow C = \frac{2}{3} \quad \text{--- (iii)}$$

now, from (i),

$$A = \frac{5}{4} \times \frac{2}{3} = \frac{5}{6} \quad \text{--- (iv)}$$

$$\therefore \text{The diff. b/w largest & smallest fr. } = A - C = \frac{5}{6} - \frac{2}{3}$$

$$= \frac{1}{6} - \frac{2}{3} = -\frac{1}{2}$$

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(2) Express $\frac{2}{3}$ or $\frac{1}{4}$ of Rs. 25.20 as a fraction of $\frac{1}{2}$ of Rs. 36.

Let fraction = m

$$\frac{2}{3} \times \frac{1}{4} \times 25.20 = m \times \frac{1}{2} \times 36$$

$$\therefore m = \frac{1 \cdot 40}{18} = \frac{0.7}{9} = \frac{1}{90}$$

(21) $8 - 8 + \frac{2}{5} - \frac{1}{7}$ is equal to?

$$= 8 - 8 + \left\{ \frac{11}{5} - \frac{8}{7} \right\}$$

$$= 8 - 8 + \frac{32}{35} - \frac{35}{35} = 8 - 8 + \frac{1}{35} = \frac{1}{35}$$

(22) $8\frac{1}{6} + 5\frac{1}{8} + 4\frac{2}{3} = ?$

$$= \frac{49}{8} + \frac{41}{8} + \frac{14}{3}$$

one method to sum

Trick

$$8 + 5 + 4 = 17$$

$$\frac{1}{6} + \frac{1}{8} + \frac{2}{3} = \frac{23}{24}$$

$$\therefore 17\frac{23}{24}$$

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4. Ratio & Proportion (4-5 hrs)
tips: Units & definition
Meaning of Ratio & Proportion

• Whenever there are 2 quantities & whenever we are comparing them in more than their common sense is expressed in the form of ratios

Let, P = 50 & Q = 10 (P is 5 times faster than Q)

$\therefore$ the ratio of speed of P & Q = 50:10 = 5:1

P:Q = 5:1 (another way (mix))
P = 5x, Q = 1x
P:Q = 5:1
5:1 $\rightarrow$ Another way of writing ratio

i.e. If $a:b = \frac{a}{b}$, $b:a = \frac{b}{a}$ like that

Ratio can be considered as simple fraction

• Ratio is the simplified form of related values.

• Whenever we compare 2 ratios or equate 2 ratios it is called as proportion.

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• $a:b :: c$ can always be written as $a:b :: b:c$

• $a:b :: c:d \rightarrow a:b :: b:c$
 $\Rightarrow \frac{a}{b} = \frac{c}{d}$

$a:b :: c:d$

if whenever there is some ratio, say $\frac{5}{6}$ in $5:6$

How to find actual vals. (not actual vals. from the simplified vals. they are simplified vals.)

$5:6 :: \frac{5}{x} = \frac{6}{x} \Rightarrow x(1d)$ [K = common factor]

Then we know K & add it with simplified ratios (another way)

$5K, 6K \leftarrow$ actual values

[Always get same common factor & get the actual values]

① Which of the following 2 ratios is greater $17:18$

1) $10:11$?

$\frac{17}{18} \times \frac{10}{11}$
 $= 17 \times 11 = 187$
 $= 18 \times 10 = 180$
 $\therefore \frac{17}{18} > \frac{10}{11}$

2) $\frac{4}{5} \times \frac{3}{2} = \frac{12}{10}$

$\frac{4}{5} > \frac{3}{2}$

1st, 2nd, 3rd, 4th
 compare
 compare
 ans

oxford

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② The 3rd Proportional to 18 & 54 is?

$a:b :: c:d \rightarrow a:b :: b:c$
 $\frac{a}{b} = \frac{c}{d} \Rightarrow \frac{18}{54} = \frac{b}{c}$
 $b = ac$

If 54? $\rightarrow$
 $(54)^2 = 18 \times c$
 $c = 162$

If 18 is one, then find out b. If, asked a/c/b then this way. If, asked extreme then find a/c give both.

③ What is 4th Proportional in $13, 13, 15, 3$?

$a:b :: c:d \rightarrow a:b :: b:c$
 $\frac{a}{b} = \frac{c}{d} \Rightarrow \frac{13}{13} = \frac{15}{d}$
 $d = \frac{bc}{a}$

$d = \frac{13 \times 15}{13} = 15$
 $d = 15$

④ Find the mean Proportional between 7 & 63

$a:b :: b:c \Rightarrow b = \sqrt{ac}$
 $b = \sqrt{7 \times 63}$
 $b = 21$

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⑤ $\frac{10}{15} = \frac{11}{25} = \frac{21}{11} = \frac{12}{11} = x$, what is x ?

$\rightarrow \frac{10+11+21+12}{15+25+11+11} = K$ [Simply just add NS & DS & then simplify the ans]
 $\rightarrow K = \frac{54}{63} = \frac{6}{7}$
 $\therefore x = \frac{6}{7}$

⑥ Income of Ramesh & Suresh is 5:6. Their spending ratio is 7:9. Ramesh saves Rs. 4000 & Suresh saves Rs. 3000. Income & Spending respectively Ramesh & Suresh are?

I. Ratio R:S = 5:6

Let common factor = K

∴ Actual I of R = $5K$
 S = $6K$

Now Spend R = $5K - 4000$ — (i)

Spend S = $6K - 3000$ — (ii)

Given, S. Ratio R:S = 7:9

∴ $\frac{\text{Spd R}}{\text{Spd S}} = \frac{7}{9}$

$\rightarrow \frac{5K - 4000}{6K - 3000} = \frac{7}{9}$; $\therefore K = 5000$ ∴ Spd R = $25000 - 4000 = 21000$
 Spd S = $30000 - 3000 = 27000$

$21000 : 27000 = 7 : 9$

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⑦ a:b = 3:7 & b:c = 2:5 what is a:b:c?

a:b = 3:7, b:c = 2:5

a:b = 3:7 = 24:63

b:c = 2:5 = 63:315

a:b:c = 24:63:315

⑧ How to divide 3395 in ratio of 42:32:23?

2 ways

1st way - Partnership method

$\rightarrow \frac{42}{97} \times 3395 = 42 \times 35 = 1470$

$\rightarrow \frac{32}{97} \times 3395 = 32 \times 35 = 1120$

$\rightarrow \frac{23}{97} \times 3395 = 23 \times 35 = 805$

2nd method

1st, common factor = K

∴ Actual val. S are $42K, 32K, 23K$

According to 1st way

$42K + 32K + 23K = 3395$

$\therefore K = 35$

∴ now $42 \times 35 = 1470$
 $32 \times 35 = 1120$
 $23 \times 35 = 805$

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③ 285 is summation of 3 numbers. Ratio between 2nd & 3rd no is 6:5. Ratio b/w 1st & 2nd no is 3:7. The 3rd no is?

$$1st : 2nd = 2nd : 3rd$$

$$= 3 : 4 \quad = 6 : 5$$

$$1st : 2nd : 3rd$$

$$= 18 : 42 : 35$$

$$= \frac{35}{3} + 285 = 3rd \text{ no}$$

$$\frac{35}{3} + 285 = 105$$

⑩ Ratio of 2 no is 2:3. On adding 5 to both no. the ratio becomes 2:5, which is the smaller no. out of the two?

$$1d. A, B \quad A:B = 2:3$$

$$\text{Actual no.} = 3x, 5x \text{ (H.C.F. common factor)}$$

$$\frac{3x+5}{5x+5} = \frac{2}{5}$$

$$3x = 15$$

$$\therefore x \text{ is same for both } 3x = 3 \times 5 = 15$$

$$\therefore \text{smaller no.} = 3x = 3 \times 15 = 45$$

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11) Find A:B:C:D when A:B = 2:3, B:C = 4:5, C:D = 5:7?

Trick



to 105, 105, 180

multiplication of all 15

mul of all D.

1st 2nd then start from the 2nd no. in the ratio & jump to 5 & mul all

Start from 2nd, then jump to 4 (just opposite of 3) & then jump to 5 (like the 2nd no) & mul all.

from 1st no. →

→ (mul all 5)

from last no. →

→ (mul all 5)

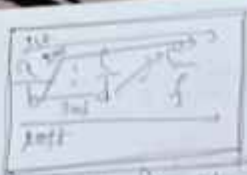
from 2nd no. →

→ (start from 2nd & then jump then 5 & mul all)

from 3rd no. →

→ (start from 2nd & then jump to 4. Then 5 & mul all)

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Generalized formula for 1st part,
 $A:B = a:b$ $R:C = c:d$ $C:D = e:f$ a b c d e f
 2nd part, $A:B:C = ace: bce: bdf$

(2) Price of each article of type R & S & P is
 Rs. 300, Rs. 100 & Rs. 120 respectively. Sumesh
 buys articles of each type in the ratio 3:2:3
 in Rs. 6400. How many articles of type R did he
 purchase?

RS 6400 $\rightarrow$ P S R
 Rs. 300 Rs. 100 Rs. 120

The ratio of the quantities of the article S:P:R
 $= 3:2:3$
 Let common factor $= k$, $S = 3k$, $P = 2k$, $R = 3k$
 $\therefore$ Actual Vals $= 3k, 2k, 3k$

P is 1 article 300/-
 3P $= 3 \times 300 = 900k$ /-
 S is 300/-, R is 300/-

According to Q,
 $900k + 300k + 300k = 6400$
 $1500k = 6400$
 $k = 4$

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no. of articles of type S $= 2k = 2 \times 4 = 8$

(3) May & Raj together have Rs. 100 initially.
 R. 100 from May. May had same amount as what Raj had initially. Find the ratio of amounts with May & Raj initially.

Let May has A Rs & Raj has B Rs initially

$$A + B = 100 \quad \text{--- (i)}$$

$$A = 100 - B$$

$$A - B = 150 \quad \text{--- (ii)}$$

$$(i) + (ii)$$

$$2A = 1200$$

$$A = 600$$

$$R = 150$$

$$A : R = 600 : 150 = 4 : 1$$

(14) If $x:y = 3:4$, then $(7x+3y):(7x-3y)$ is equal to?
 24/5

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Indira

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ISI say we
(Pre-simplified)

$$\frac{21}{7} = \frac{3}{1}$$

Putting the value of $x=3$ in eq

$$\frac{2x+3y}{7x-3y} = \frac{21+12}{21-12} = \frac{33}{9} = \frac{11}{3}$$

2nd method

$$\frac{x}{y} = \frac{3}{4}$$

$$\Rightarrow \frac{7x}{3y} = \frac{7 \times 3}{3 \times 4}$$

$$\Rightarrow \frac{7x}{3y} = \frac{21}{12}$$

$$\frac{7x+3y}{7x-3y} = \frac{21+12}{21-12} \left[\begin{array}{l} \text{Compare to} \\ \text{3} \\ \text{diff terms} \end{array} \right]$$

$$= \frac{33}{9} = \frac{11}{3}$$

(15) If $a:b=5:7$ & $c:d=29:36$, then $ac:bd$ is?

$$\frac{a}{b} = \frac{5}{7} \quad \text{--- (i)} \quad \frac{c}{d} = \frac{29}{36} \quad \text{--- (ii)}$$

$$\text{--- (i) \& (ii)} \quad \frac{a \times c}{b \times d} = \frac{5 \times 29}{7 \times 36} = \frac{145}{252}$$

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$$\Rightarrow \frac{ac}{bd} = \frac{145}{252} = \frac{10}{18} = \frac{5}{9} \quad \text{[from (i)]}$$

$$= \frac{50}{147}$$

$$ac:bd = 50:147$$

(16) The 3 nos are in the ratio $\frac{1}{2} : \frac{2}{3} : \frac{3}{4}$.
The difference b/w greatest & smallest nos.
is 36. Find the nos.

$$\text{Ratio} = \frac{1}{2} : \frac{2}{3} : \frac{3}{4}$$

Let x be the common factor

$$\therefore \text{Let nos } = \frac{x}{2}, \frac{2x}{3}, \frac{3x}{4}$$

$$= 0.5x, 0.66x, 0.75x$$

$$\text{Given, } 0.75x - 0.5x = 36$$

$$\Rightarrow \frac{3x}{4} - \frac{x}{2} = 36$$

$$\Rightarrow 3x - 2x = 36 \times 4$$

$$\Rightarrow x = 36 \times 4$$

oxford

$$\frac{x}{2} = \frac{36 \times 4}{2} = 72 \quad \text{nos } = 72, 96, 108$$

$$\text{1st} = 72, 96, 108$$

$$\text{2nd} = \frac{3 \times 36 \times 4}{4} = 108$$

Teacher's Signature

15 (7) The ratio of market prices of wheat & Paddy is 2:3 & the ratio of quantities consumed in a family is 5:4. And the ratio of expenditure on wheat & Paddy.

Ratio of W: Price of Paddy = 2:3

actual WTS = 2K, 3K Rs (C.F. = K)

Ratio wheat consumed: Paddy consumed = 5:4

Actual WTS = 5A, 4A (C.F. = A)

Expenditure = Price + Quantity consumed (Price)

Exp W = Price W + Q.W

Exp P = Price P + Q.P

$$\begin{aligned} \text{According to the Qn, } \frac{\text{Exp W}}{\text{Exp P}} &= \frac{R.W + Q.W}{P.P + Q.P} \\ &= \frac{2K + 5A}{3K + 4A} \\ &= \frac{5}{6} \end{aligned}$$

$\therefore \text{Exp W} : \text{Exp P} = 5:6$

15 Question why do common factors get cancelled when we find the ratio itself? $\text{Exp} = R + QP$ (ratio)

$$\frac{\text{Exp W}}{\text{Exp P}} = \frac{\frac{R}{5} + \frac{Q}{4}}{\frac{R}{3} + \frac{Q}{2}} = \frac{5}{6}$$

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18 RS 8400 is divided among A, B, C & D in such a way that the shares of A & B, B & C, C & D are in the ratios of 2:3, 4:5 & C:D respectively. The share of A is

A:B = 2:3, B:C = 4:5, C:D = 6:7

A, B, C, D x (Don't have to calculate)

$$\frac{2}{3} = \frac{4}{5} = \frac{6}{7}$$

A : B : C : D

= 48 : 72 : 90 : 105

$$\text{Share of A} = \frac{48}{315} \times 8400 = 1280$$

19 In a library, the ratio of no. of two story books to that of non-story books was 4:3 & total no. of story books was 1248. When some more story books were bought, the ratio became 5:3. Find the no. of story books bought.

Ans: S: N = 4:3, total S. Books = 1248

$$\frac{S}{N} = \frac{4}{3}$$

$\therefore 1248 \times \frac{4}{3}$

$$\therefore N = 936$$

Teacher's Signature _____

Let no. of string bags were bought
 $S = (1248 + m)$
 1. $\frac{S}{N} = \frac{5}{3}$
 $\Rightarrow \frac{1248 + m}{5K} = \frac{5}{3}$
 $\Rightarrow m = 312$ no. of S. Bags were bought

Ans
 $S: N = 4:3$
 Let C.F. = K
 $\therefore$ no. of S & N = $4K + 3K$
 $4K = 1248$
 $\Rightarrow K = 312$
 $1248 + m = 5K$ ($\because 4K + 3K$)
 $\Rightarrow 1248 + m = 5 \times 312$
 $\therefore m = 5 \times 312 - 1248$
 $= 312$

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5. Percentage (4-6 hrs)
 TIPS: Thomas

• Percentage meaning
 It means $\rightarrow$ symbol of Percentage = %
 Part (i.e. Part of something) (e.g. 25% = $\frac{25}{100}$)
 (e.g. 30% is a part of whole is 1% of whole)
 • tons
 80% = 80% of something
 10% of 80% = $\frac{10 \times 80}{100} = 8\%$ (10% of 80% is 8% of whole)
 • Common Percentages
 Brand $\rightarrow$ 100% $\rightarrow$ 1 entire brand
 Shreemast. he/she wants $\frac{1}{2}$ of brand $\rightarrow \frac{1}{2}$ of 100%
 $\frac{1}{2} \rightarrow 50\% \rightarrow \frac{1}{2}$
 $\frac{1}{3} \rightarrow 33\% \rightarrow \frac{1}{3}$
 $\frac{1}{4} \rightarrow 25\% \rightarrow \frac{1}{4}$

Teacher's Signature

$$\frac{10\%}{7 \times 100\%} = 10\%$$

Teacher's Signature _____

$$= 32.5$$

When we want to find f' , we treat y as

$$= \frac{10}{5} = 2 \text{ s.f.}$$
OXFORD**Quelche**

251. ~~is y?~~ 2. 62

$$1. \frac{18.4}{100} = \frac{9.2}{100} = \frac{18.4}{200} = \frac{18.4}{200} + \frac{18.4}{200} = \frac{18.4}{100} + \frac{18.4}{100}$$

$$\therefore 12.5\% \text{ of } 8 = \underline{17.5\% \text{ of } 3}$$

⑪ If price of rice is 30% less than that of wheat, then price of wheat is how much % more than that of rice?

In this kind of sum always consider an arbitrary value as the best value for this kind of sum is 100.

Let's say, price of wheat = Rs. 100

∴ Price of rice = $100 - (30\% \text{ of } 100)$

$\varphi(10) = 30$

Teacher's Signature _____

we need the % so, follow the
prev method,

The price of wheat = 100 RS (A)
The price of rice = 90 RS (B)

we need the price of wheat is how
much % more than that of rice

wheat's % so $\rightarrow$ A m/s
we'll find it considering unit in

price $\rightarrow$ So B m/s

we know, $\frac{A}{B} + 100 = \frac{100}{90} + 100$
 $= 142.85\%$

if 100 less
means 30%
more $\rightarrow$ no
we can't say
that the %
value is
diff

this will be
as the % that
is how much it is
more / less, then
calculate it

$\therefore$ The price of A is 142.85 times of rice's
price (ie, $\frac{1}{90}$ of 100 (A's) of 90 (B's))

$\rightarrow$ (How much is price of A in relation
to rice)

Entire quantity = 100% (as 100 is max)

How much is extra = $(142.85 - 100)\%$
 $= 42.85\%$ more

price of A is 42.85% more than that
of price of rice

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$\frac{58}{20} \times 100 = 42.85\%$ (Range 100)
method: the value of Price of A is 100

(5) The price of apple is first increased by
10% & then decreased by 10%. What is the
change in the price of apple?

Let the price of apple = 100/- (original price)
710% $\rightarrow$ 710 RS ($\because$ 10% of 100)

now = $100 + 10 = 110/-$
410% $\rightarrow$ 41 RS (10% of 110)

new price = $110 - 11 = 99$ (final price)
change in price = $100 - 99$
 $= 1$ RS

Now if we want to represent it in % form
What % is RS 1 of RS 100

$= \frac{1}{100} + 100$
 $= 100 + 1 = 101\%$

change in the price of apple = RS 1
or 1% (this is the final answer)

we have to calculate the % as we don't
know the actual money price & also in
in any given question

oxford

Teacher's Signature

Q If the price of a sugar is raised by 25%.
Then by how much percent should be a
person reduce his consumption of sugar so
that his expenditure remains same?

Ans: Same

Price of
sugar

Let the price of sugar = 100 RS per kg
Let the person used to consume 1 kg of sugar
now, the new price = 125 RS [1 kg of sugar
is used to consume by the person]
Per kg is = 100/-

now 25% price has increased,
initially 100 RS

100 + 25
100

= 125/-

The price increases by 25/-
the new price = 100 + 25 = 125/-

Let us the new consumption (reduced consumption,
so that the exp remain same) = C

According to the m

(125 * C = 100 * 1)

∴ C = 0.8 kg

Initial	Reduction in consumption
100%	25%
100%	25%
20%	

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Initially he consumed 1 kg
now he need to consume 0.8 kg
∴ the person need to reduce his consumption
quantity by = 0.2 kg (1 - 0.8)
now this 0.2 kg is how much % of 1 kg
= $\frac{0.2}{1} \times 100$
= 20%

∴ the consumption of sugar should be reduced
by 20%.

Q If a student has to score 40% marks to pass. He
gets 20 marks & fails by 40 marks. The maximum
marks of the exam are?

Let the max marks = M

40% of M
= $\frac{40M}{100}$

According to the m, if gets 20 marks, it
he would get more 40 marks he would pass.

So, $40 + 20 = \frac{40M}{100}$

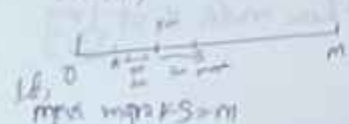
∴ $60 = \frac{40M}{100}$

∴ M = 150

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Q A scores 10% & fails by 30 marks B gets 40% marks & gets 30 marks more than the minimum marks needed to pass the exam. There are the maximum marks for the exam?



A $\rightarrow 10\% \times \frac{m}{100} = \frac{m}{10}$; $\frac{m}{10} + 30 \rightarrow$ Passing marks

B $\rightarrow \frac{40m}{100}$

marks of B = $\frac{m}{10} + 30 = \frac{40m}{100} - 30$

(i.e., if B gets 30 marks more than passing marks then if we reduce 30 marks it would be the passing marks)

$\therefore \frac{m}{10} - \frac{40m}{100} - \frac{m}{10} = 60$

$\therefore \frac{4m - m}{10} = 60$

$\therefore \frac{3m}{10} = 60 \times 20$

$\therefore m = 200$

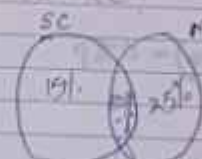
mine
his friend scoring (say)
= + 30 marks more
gets more than pass marks
(error) = 10

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Q In a class, 15% of total no. of students failed in science, 25% of total no. of students failed in maths & 10% of total no. of students failed in both. How much percentage of students passed in both maths & science?



Failed = 15% + 25% - 10%
= 30%

$\therefore$ Pass = 100% - 30% = 70%

Q By 20% decrease in the price of rice, people can buy 1 kg more rice in Rs. 100. What is the original price of 1 kg of rice?

Let, the original price of rice = P Rs/kg
into people used to buy (constly) 1 kg of rice
for Rs. 100.

Exp (constly) = 100
Exp (constly) = 100

Teacher's Signature

∴ new handy exp = $A + P$
 now, price decrease of 20%

$$\text{Current Price} = P - \frac{P \times 20}{100}$$

$$= \frac{80P}{100}$$

$$= 0.8P$$

∴ now, the current exp = $(A + 10) - 0.8P$

According to the ms

$$AP = (A + 10) - 0.8P \quad [\text{exp remains same}]$$

$$\frac{1}{2}A = (A + 10) \frac{8}{10}$$

$$\Rightarrow A - \frac{8A}{10} = 8$$

$$\Rightarrow \frac{2A}{10} = 80$$

$$\Rightarrow A = 400$$

∴ any of this was possible from people to initially by initially in 100

And we know the handy exp = 100

$$\therefore 100 = A + P$$

$$100 = 40 + P$$

$$\Rightarrow P = \frac{100}{40} = 2.5 \text{ RS}$$

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(11) In an election which was contested by 3 candidates, one candidate got 40% of total votes & yet lost by 1000 votes. What is the total no. of votes casted in the election?

Let the 1st vote be A

40% A	60% of A
40A	60A
100	100

According to the ms. (2nd one yet 1000 more votes as 1st one lost by 1000 votes)

$$\therefore \frac{40A}{100} - \frac{60A}{100} = 1000$$

$$\Rightarrow \frac{CA - 4A}{100} = 1000$$

$$\Rightarrow \frac{2A}{100} = 1000$$

$$\Rightarrow A = 50,000$$

(1st candidate got 1000 less votes to win, i.e. if he get 1000 votes then he will win) can 2 (above)

(12) In a country 55% population is female & 50% of the male population is literate. How much of females are literate if total literacy is 50%?

Let population = 100

Teacher's Signature

$$f = 55$$

i.e. 55% of total population is f

$$\frac{55}{100} \times 100 = 55 \text{ females}$$

$$\therefore \text{males} = 100 - 55 = 45 \text{ males}$$

Literate male = $\frac{80}{100}$ of male population

$$= \frac{80}{100} \times 45$$

$$= 36 \text{ males are literate}$$

total literacy = 58% (out of 100 people 55 people are literate)

= 58% of 100 (total population)

= 58 people

(36 male literate out of 58, so rest are f. literates)

$$f_L + m_L = 58$$

$$f_L + 36 = 58$$

$$\Rightarrow f_L = 58 - 36$$

$$= 22$$

now, we have to find the percentage (how much of females are literate) [As no concrete val. of population is given, everything is given in % so ans has to be in % also]

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i.e. how much % is 22 of 55

$$\therefore \frac{22}{55} \times 100 = 40\%$$

(13) If 20% of an electricity bill is deducted, then Rs. 100 is still to be paid. How much was the original bill?

From original 20% is deducted i.e. 80% bill has to be paid

Now given, after reducing, the person still have to pay 100

i.e. 80% of original bill = 100 Rs.

$$\frac{80}{100} \times B = 100 \text{ (let } B = \text{original bill)}$$

$$\therefore B = \frac{100 \times 100}{80} = 125 \text{ Rs.}$$

(14) A's salary is 60% more than B's. How much percent is B's salary less than A's?

let, B's Sal = 100 Rs.

$$A's \text{ Sal} = \frac{100 + 60}{100} \times 100$$

$$= 160 \text{ Rs.}$$

Teacher's Signature

Change
Recent
Sol.
50
150

How much is B's salary less than A's

$$B's\ salary = 100/-$$

$$A's\ salary = 150$$

i.e., B's salary is 50% less

(now we have to convert it into %)

$$\frac{50}{150} \times 100 = 33.33\% \text{ less than that of A.}$$

(5) Two nos are less than a third no. by 30% & 37% respectively. How much is 2nd no less than the 1st no.

$$\text{Let } 3rd\ no = 100$$

$$1st\ no = 100 - \frac{30}{100} \times 100$$

$$= 70$$

$$2nd\ no = 100 - \frac{37}{100} \times 100$$

$$= 63$$

2nd no is $(70 - 63) = 7$ less than 1st no.

$$\frac{7}{70} \times 100 = 10\%$$

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normal

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(10) 10% of inhabitants of a village having died of cholera, a famine set in during which 25% of remaining inhabitants left village. the population is then reduced to 4050. the no. of inhabitants existing was?

Let, Population of the village = m

10% died of cholera, let the remaining = $m - 10\% \times m$
Remaining = $m - \frac{10}{100} \times m$
 $= \frac{90}{100} m$

Let, total inhabitants existing in = m

now, 10% of in. died in cholera,
Remaining = $m - \frac{10}{100} \times m$
 $= \frac{90}{100} m$

Out of this 25% of them (remaining) left the village.

Still remaining = $\frac{90}{100} m - \frac{25}{100} \times \frac{90}{100} m$

$$= \frac{36}{100} m - \frac{225}{10000} m$$

$$= \frac{3600}{10000} m - \frac{225}{10000} m$$

Teacher's Signature

25% of remaining left

now, we don't exclude from 25% say
30% we always need to exclude from 100%.

∴ now remaining is 100%

$$\therefore \text{now left} = 100\% - 25\%$$

= 75% of remaining

$$= \frac{75}{100} \times \frac{5000}{100}$$

∴ according to the In, after that the population
reduced to 4050 i.e., after that in/pres = 45

$$\frac{75}{100} + \frac{20m}{100} = 4050$$

$$\Rightarrow m = 6000 \text{ inhabitants}$$

$$\therefore \text{original In/pres} = 6000$$

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6. Problems on Ages (imp)

• Note

Read the In carefully, then the ans lies in the
In's in such problems (i.e., everything is in
the problem, we do constant find & find the ans)

① Ram is three times as old as Sam. 2 years
ago he was five times as old as Sam.
What is the present age of Ram?

Present
age

2 years ago

Ram - R

R - 2

Sam - S

S - 2

Union

Union

$$(R-2) = 5(S-2)$$

$$R = 3S \quad \text{--- (i)}$$

$$\Rightarrow R-2 = 5S-10$$

$$\Rightarrow R = 3 + 4 \text{ (from (i))}$$

$$= 12 \text{ yrs}$$

$$\Rightarrow 3S-2 = 5S-10 \text{ [from (i)]}$$

$$\Rightarrow 2S = 8$$

$$\Rightarrow S = 4 \text{ yrs --- (ii)}$$

crosscheck:
The ans is right

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② 12 years ago age of P was 3 times the age of Q. After 12 years ratio of ages of S to P will be 2:3. What is the present age of P?

Present age: $\xrightarrow{12 \text{ years ago}}$ $\xrightarrow{12 \text{ years after}}$

P: $P-12$ yrs $\rightarrow$ (P-12) yrs $\rightarrow$ (P+12) yrs
 Q: $Q-12$ yrs $\rightarrow$ (Q-12) yrs $\rightarrow$ (Q+12) yrs

Given: $(P-12) = 3(Q-12)$

$\Rightarrow P-12 = 3Q - 36$
 $\Rightarrow P = 3Q - 24$ — (1)

from (1)

$\Rightarrow P = 3Q - 24$
 $\Rightarrow 3Q - 24 = 24$
 $\Rightarrow 3Q = 48$
 $\Rightarrow Q = 16$

$\Rightarrow Q + 12 = 16 + 12 = 28$
 $\Rightarrow 3Q - 24 + 12 = 28$
 $\Rightarrow 3Q - 12 = 28$
 $\Rightarrow 3Q = 40$
 $\Rightarrow Q = 13.33$

$\Rightarrow Q + 12 = 16 + 12 = 28$
 $\Rightarrow 3Q - 24 + 12 = 28$
 $\Rightarrow 3Q - 12 = 28$
 $\Rightarrow 3Q = 40$
 $\Rightarrow Q = 13.33$

$\Rightarrow 3Q - 24 = 3(13.33) - 24 = 39.99 - 24 = 15.99$

$\Rightarrow 3Q = 40$
 $\Rightarrow Q = 13.33$

③ Rohan is as much younger than Ajay as he is older than Meena. The sum of ages of Ajay & Meena is 105 years. How old is Rohan?

EXERCISE

Sum of ages of Rohan & Meena is 105 years.

$R + M = 105$

$R = 105 - M$

$R = 105 - M$

$R = 105 - M$

$R = 105 - M$

$R = 105 - M$

$R = 105 - M$

$R = 105 - M$

$R = 105 - M$

$R = 105 - M$

$R = 105 - M$

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10 7 24
 10 + 24 = 34
 2 = 17

10 7 24
 10 + 24 = 34
 2 = 17

10 7 24
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10 7 24
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10 7 24
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10 7 24
 10 + 24 = 34
 2 = 17

10 7 24
 10 + 24 = 34
 2 = 17

④ Rohan's age is five times Ajay's & seven-eighths of Meena's age. The sum of the ages of all three is 102 years. How much younger is Ajay to Meena?

Let Ajay's age = A yrs, Meena's age = M

Rohan's age = $5A$ & $\frac{7}{8}M$ also

$R = 5A$ & $R = \frac{7}{8}M$

$5A = \frac{7}{8}M$

$40A = 7M$

$M = \frac{40}{7}A$

$M = \frac{40}{7}A$

$M = \frac{40}{7}A$

Representing age in terms of A (Ajay's age)

Teacher's Signature _____

$$\text{avg of } F \text{ \& } T \& R = 132$$

$$\frac{F + T + R}{3} = 132$$

$$F + T + R = 396$$

$$R = 599 = 35 \text{ yrs}$$

$$m = \frac{\text{sum}}{2} = 90 \text{ yrs}$$

$$A \text{ is } (30 - 35) = 5 \text{ yrs younger to } m.$$

$$A \text{ is age } = 25 \text{ yrs } m \text{ is age } = 90 \text{ yrs}$$

$$A \text{ is } (30 - 3) = 83 \text{ \& } \text{ younger than } m.$$

Q Ram \& shyam's average age is 65 years
the average age of Ram, shyam \& John is 53
years what is the age of John? (check average video)

Average
Avg = $\frac{\text{Sum}}{\text{Total observations}}$

$$\text{Avg Ram \& shyam} = 65 \text{ (given)}$$

let, age of Ram = R

shyam = S

John = J

$$\frac{R + S}{2} = 65$$

$$R + S = 130 \text{ --- (I)}$$

$$\frac{R + S + J}{3} = 53$$

$$R + S + J = 159$$

$$J = 159 - 130 \text{ (from I)}$$

$$J = 29 \text{ yrs}$$

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Q The average of 10 students \& their teacher is 15 years. the average age of first seven students is 15 years. what of the last three is 11 years. what is the teacher's age?

$$\text{Avg age} = 15 \text{ (Avg age of 10 students \& their teacher is 15)}$$

$$\therefore \text{Age of 10 st + Age of their teacher} = \text{Total no. of people } (10 + 1)$$

$$\text{let, the avg age of 10 students} = S_1 \text{ (Sum of the age of 10 students)}$$

$\therefore$ according to the Q,

$$\frac{S_1 + 1}{11} = 15$$

$$S_1 + 1 = 165 \text{ --- (I)}$$

$$\text{let, avg age of 1st 7 students} = S_2$$

$$\text{avg age of last 3 students} = S_3$$

$$\frac{S_2}{7} = 15, \therefore S_2 = 105 \text{ --- (II)}$$

$$\frac{S_3}{3} = 11, \therefore S_3 = 33 \text{ --- (III)}$$

$$S_1 = S_2 + S_3$$

$$\therefore \text{from (I)}, S_1 + 1 = 165$$

$$\therefore 105 + 33 + 1 = 165$$

$$\therefore T = 27 \text{ yrs}$$

Teacher's Signature

⑦ The average age of group of 4 friends is 36 years. The youngest friend amongst them is 6 years old. What was the average age of the group at the time of the birth of the youngest friend?

Given: $\text{Avg}_4 = 36$

Let, Sum of the ages of 4 friends = S_4

$$\frac{S_4}{4} = 36$$

$S_4 = 144$ y. $\therefore$ we need the avg. of the group at the time of birth of the youngest friend.

6 years ago

$$-6 - 6 - 6 - 6$$

$$-24$$

total reduction age

$$\therefore \text{Sum of the ages of 4 friends 6 years ago} = 144 - 24 = 120 \text{ y}$$

Now, there are 4 friends & 6 years ago then the sum of their ages = 120 y. 6 years ago the youngest friend was just born, so his age at 6 years ago = 0. That means '120 y' is the sum of the ages of rest of 3 friends as that youngest friend was just born at that time.

$\therefore$ The avg age of the group 6 years ago =

$$\frac{120}{3} = 40 \text{ y}$$

Example

⑧ Average age of a family of 4 members was 10 years 5 years back.

Birth of a new child kept the avg age of the family same. Now today, how old the child is?

Given: $\text{Avg}_4 = 10 \frac{1}{2}$ (4 years ago)

4 years back $\rightarrow$ Present
ages of 4 years ago

Let sum of total family now = S_4

$$\therefore \frac{S_4}{4} = 10 \frac{1}{2} \therefore S_4 = 42 \frac{1}{2}$$

$\therefore$ After 4 years (Present)

$$+4 + 4 + 4 + 4$$

$$= 16 \text{ (total increment)}$$

$$\therefore \text{Sum of the ages of 4 members today (4 years after)} = 42 + 16 = 58 \frac{1}{2}$$

Let present age of child =

'K'

$$\text{Avg (Present)} =$$

$$\text{Known: } \frac{52 + K}{5}$$

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$$\text{Present age of child} = \text{same} = 13$$

$$\frac{52 + K}{5} = 13$$

$$\therefore K = 13 \text{ y}$$

NAME - Rajasree

CLASS - XII

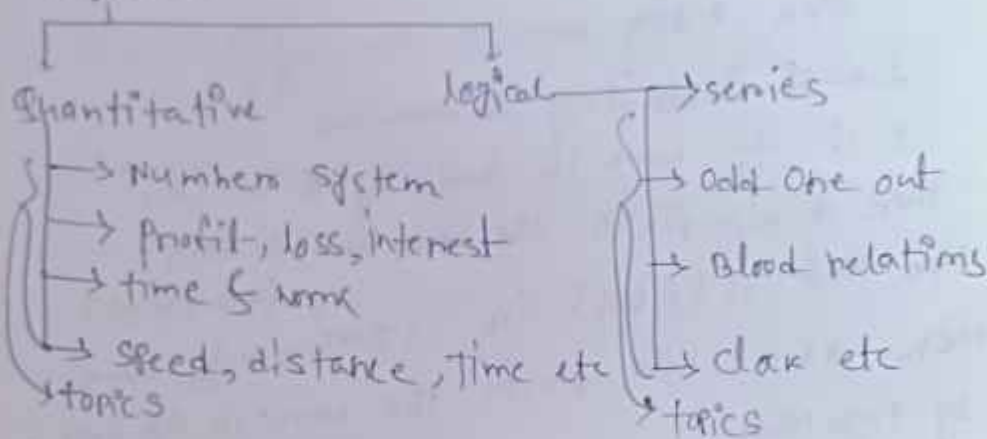
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ISTRATION N

B - Physics

tool - Balanagar

Attitude



Quantitative Attitude

Time & Work

$$A \rightarrow 2 \text{ day} \quad A \rightarrow 3 \text{ day}$$

$$\frac{1}{2} + \frac{1}{2} \quad \frac{1}{3} + \frac{1}{3} + \frac{1}{3}$$

$$B \rightarrow x \text{ hours / days etc} \quad [B \text{ do a work in } x \text{ hrs}]$$

$$\text{ie, } x \text{ hrs} \rightarrow 1 \text{ work, } 1 \text{ hr} \rightarrow \frac{1}{x} \text{ work}$$

$$\therefore \text{at } 1 \text{ hr / day or etc } B \text{ does } \frac{1}{x} \text{ work}$$

Ex: A can do a piece of work in 12 hrs & B alone can do it in 15 hrs. In how much time will they finish the whole work, working together?

$$A \rightarrow 12 \text{ hrs} \quad B \rightarrow 15 \text{ hrs}$$

$$A \text{ in } 1 \text{ hrs} \rightarrow \frac{1}{12} \quad B \text{ in } 1 \text{ hrs} \rightarrow \frac{1}{15}$$

$$(A \& B) \text{ in } 1 \text{ hrs} \rightarrow \frac{1}{12} + \frac{1}{15} \rightarrow \frac{5+4}{60}$$

$$\rightarrow \frac{9}{60} = \frac{3}{20}$$

$$3 \overline{) 12.15}$$

$$\underline{4.5}$$

like previous, if $A \rightarrow 12 \text{ hrs}$

then A doing work in 1 hr = $\frac{1}{12}$

i.e. if A ~~15~~ days work in

i.e. if $\frac{1}{12}$ work is done by A in 1 hr

then A will finish the work in 12 hrs.

like that if (A+B) in 1 hr $\rightarrow \frac{3}{20}$ work

then (A+B) will finish the work in $\frac{20}{3}$ hrs by together.

Q2) A & B $\rightarrow 12 \text{ days}$

B $\rightarrow 30 \text{ days}$

A $\rightarrow ?$

A & B $\rightarrow 12 \text{ days}$

(A+B) in 1 day $\rightarrow \frac{1}{12}$ work

B $\rightarrow 30 \text{ days}$

B in 1 day $\rightarrow \frac{1}{30}$ work

$\therefore A + B \xrightarrow{\text{in 1 day}} \frac{1}{12}$

$\Rightarrow A + \frac{1}{30} = \frac{1}{12}$ [$\because B$ in 1 day $\rightarrow \frac{1}{30}$ work]

$\Rightarrow A = \frac{1}{12} - \frac{1}{30}$

$= \frac{5-2}{60}$

$= \frac{3}{60}$
 $= \frac{1}{20}$

$\begin{array}{r} 3 \overline{) 12, 30} \\ 2 \overline{) 4, 10} \\ 2, 5 \end{array}$

A in 1 day $\rightarrow \frac{1}{20}$ work

A will finish the work in 20 days by alone.

Q3) A & B $\rightarrow 18 \text{ days}$, B & C $\rightarrow 24 \text{ days}$, C & A $\rightarrow 36 \text{ days}$

A, B & C $\rightarrow ?$

A $\rightarrow ?$

(A+B) in 1 day = $\frac{1}{18}$ work

B & C = $\dots = \frac{1}{24}$

C & A = $\dots = \frac{1}{36}$

2(A+B+C) in 1 day = $\frac{1}{18} + \frac{1}{24} + \frac{1}{36} = \frac{4+3+2}{72} = \frac{9}{72} = \frac{1}{8}$

$\Rightarrow A+B+C \xrightarrow{\text{(in 1 day)}} \frac{1}{8}$

A, B & C will finish the work by 8 days

$A+B+C = \frac{1}{8}$

$\Rightarrow A + \frac{1}{24} = \frac{1}{8}$

$\Rightarrow A = \frac{1}{8} - \frac{1}{24}$

$= \frac{1}{48}$

$\therefore A$ will $\dots \dots \dots$ 48 days

NAME - Rajasree Jaha

CLASS - XI

Roll - 23 SEC - A

ALSTRATION No - 4193

1B - Physics

Roll - Rajasree Jaha

34) A & B can do a piece of work in 40 days respectively. They began the work together but A leaves after some days & B finished the remaining work in 23 days. After how many days did A leave?

In 1 day, A $\rightarrow \frac{1}{40}$ work

B $\rightarrow \frac{1}{40}$ "

$$A \& B \rightarrow \frac{1}{40} + \frac{1}{40} = \frac{17}{360}$$

work A & B did together

B in 1 day $\rightarrow \frac{1}{40}$

B has worked for 23 days by alone

B in 23 days $\rightarrow 23 \times \frac{1}{40}$ work

$$\text{work A \& B did together } \left(1 - \frac{23}{40}\right) = \frac{17}{40} \text{ work}$$

A & B in 1 day $\rightarrow \frac{17}{360}$ work

A & B did $\frac{17}{360}$ work together

A & B did $\frac{17}{360}$ work $\rightarrow$ 1 day

$$1 \rightarrow \frac{360}{17}$$

$$\frac{17}{360} \rightarrow \frac{360}{17} \times \frac{17}{40} = 9 \text{ days}$$

35) Let A & B did the work together by x days

$$\frac{17 \times x}{360} = \frac{17}{40}$$

$$\Rightarrow x = 9 \text{ days}$$

A & B did the work together for 9 days. After 9 days A left & B did the work for 23 days.

Ans - 9 days

36) If 4 men or 6 boys can finish a piece of work in 20 days, in how many days can 6 men & 11 boys finish it?

4 men or 6 boys $\rightarrow$ 20 days

50 ~~men~~ 4 men $\rightarrow$ 20 days

4 men in 1 day $\rightarrow \frac{1}{20}$ day work

$$1 \rightarrow \frac{1}{20} \text{ day work} \rightarrow \frac{1}{80}$$

In 1 day

$$4 \text{ men} = \frac{1}{20}$$

$$1 \text{ man} = \frac{1}{80}$$

$$6 \text{ boys} = \frac{1}{20}$$

$$1 \text{ boy} = \frac{1}{120}$$

$$6 \text{ men \& 11 boys } \frac{6}{80} + \frac{11}{120} = \frac{18+22}{240} = \frac{40}{240}$$

$$6 \text{ men \& 11 boys in 1 day} \rightarrow \frac{1}{6} \text{ work}$$

NAME - Rajasree Laha

CLASS - XII

Roll - 23 SEC - A

REGISTRATION NO. - 41923

18 - Physics

School - Baleshwar Girls H.S.

5. They can finish the work together in 4 days.

Q. If Roger can do a piece of work in 8 days & Antony can do a piece of work in 5 days, in how many days will both of them do it together?

In 1 day,

Roger $\rightarrow \frac{1}{8}$ work

Antony $\rightarrow \frac{1}{5}$ "

Roger & Antony in 1 day $\rightarrow \frac{1}{8} + \frac{1}{5}$
 $\rightarrow \frac{5+8}{40}$
 $\rightarrow \frac{13}{40}$ work

Both of them will finish the work together in $\frac{40}{13}$ days.

Q. To complete a piece of work, A takes 50% more time than B. If together they take 18 days to complete the work, how much time shall B take to do it alone?

14. B takes x days to complete a work.

A $\rightarrow \frac{150}{100}x = \frac{3x}{2}$

In 1 day, B $\rightarrow \frac{1}{x}$ work
A $\rightarrow \frac{2}{3x}$

Given, $\frac{1}{x} + \frac{2}{3x} = \frac{1}{18}$ [18 days
1 day $\rightarrow \frac{1}{18}$ work]
 $\rightarrow \frac{3+2}{3x} = \frac{1}{18}$

$\rightarrow \frac{5}{3x} = \frac{1}{18}$

$\rightarrow 3x = 5 \times 18$

$\rightarrow x = 30$ days

B will take 30 days to do it alone.

Q. 8. A takes 3 days to complete a work while B takes 2 days. Both of them finish a work & earn RS. 150. What is A's share of money? (Hint: money should be divided in the ratio of how much work a person does in 1 day)

RS 75 to B RS 30 to A RS 60 to B RS 75 to A

In 1 day, A $\rightarrow \frac{1}{3}$ work
B $\rightarrow \frac{1}{2}$ work

Let A's share of money

Given, money in 1 day

$\therefore$ their work ratio $= \frac{1}{3} : \frac{1}{2} = 2 : 3$

NAME - Rajasree John
CLASS - 10

ROLL - 23 SEC - A
DISTRIBUTION NO - 415

18 - Physics
tool - Balamangal

Many of the lines in this note
 are about the way the paper is
 made.

The paper was produced and the
 quality was analyzed. Some of the
 things that were found in the
 analysis of the paper were:
 1. The paper was made of 2 or
 3 layers of paper. The paper
 was made of lignin.

I was able to show some of the
 things in the paper using 5 lines.
 The following are the things I found:

- | | |
|------------|------------|
| $P_1 = 1$ | $P_2 = 2$ |
| $P_3 = 2$ | $P_4 = 20$ |
| $P_5 = 1$ | $P_6 = 1$ |
| $P_7 = 20$ | $P_8 = 20$ |

$$\frac{P_1 + P_2 + P_3 + P_4 + P_5 + P_6 + P_7 + P_8}{8} = \frac{1 + 2 + 2 + 20 + 1 + 20 + 20 + 20}{8}$$

$$\frac{1 + 2 + 2 + 20 + 1 + 20 + 20 + 20}{8} = \frac{86}{8} = 10.75$$

Now, the paper was made of 2 or 3
 layers of paper. The paper was made
 of lignin.

The paper was made of 2 or 3
 layers of paper. The paper was made
 of lignin.

$$\frac{1 + 2 + 2 + 20 + 1 + 20 + 20 + 20}{8} = \frac{86}{8} = 10.75$$

The paper was made of 2 or 3
 layers of paper. The paper was made
 of lignin.

we can go $D \rightarrow W$ by
 $(\textcircled{P}) W \rightarrow D$ just simple inversion

we know, if 1 man, then 1 time, 1 days
 1 men, then 1 time, 1 days, 1 time

If A can work 5 times faster than B & takes 60 days less than B to complete the work. In how many days does A & B individually can complete the work?

let $B \xrightarrow{\text{work}} n \text{ days}$
 $A \rightarrow n-60$
 $A \rightarrow B \leftarrow A$ 5 times faster
 as B is 5 times slower than A then it will take 5 times more time than A

as $A \rightarrow B$
 so time taken by A will be $\frac{1}{5}$ of time taken by B

$$\therefore n-60 = \frac{n}{5}$$

$$\frac{1}{5}n = \frac{n}{5} = 60$$

$$\frac{1}{5}n = 300$$

$$\therefore n = 75 \text{ days} \quad \therefore n-60 = 15 \text{ days}$$

B takes 75 days to complete a work
 A $(75-60)=15$ days

Work 1 time 1 days
 1 man 1 time 1 days
 more men, less time, less day

more amount of work 1 time 1 days
 more men $\propto \frac{1}{\text{amount of work}} \propto \frac{1}{\text{days taken}}$

If 24 men can finish a work in 10 days, then find the no. of days required to complete the same work by 30 men?

~~24 men $\rightarrow$ 10 days~~
~~by $\rightarrow$ 10 days work~~

24 m 10 days $\frac{1 \text{ day}}{10}$ amount of work
 30 m ?

24 $\leftarrow \frac{1 \text{ day}}{10}$
 30 $\rightarrow 7 \frac{1}{2} \text{ (let } x)$

$$24 \times x = \frac{30}{10}$$

$$\therefore x = \frac{30}{24 \times 10}$$

$$x = \frac{30}{240}$$

$$x = \frac{1}{8}$$

$\therefore 8 \text{ days}$ [1 day $\rightarrow \frac{1}{8}$ work
 can finish work by 8 days]

NAME - Rajasree Latha

CLASS - XII

ROLL - 25 SEC - A

REGISTRATION NO. - 41923152

IB - Physics

SCHOOL - Balurangan Girls' High

3) A can do work in 3 days. B can do the same work in 6 days & C can do the same work in 7 days. If they work together, in how many days will they take to complete the work?

A → 3 days $\frac{1 \text{ day}}{3}$ amt of work
B → 6 → $\frac{1}{6}$
C → 7 → $\frac{1}{7}$

$$A+B+C = \frac{1}{3} + \frac{1}{6} + \frac{1}{7}$$

$$\frac{14+21+12}{42} = \frac{47}{42}$$

$$= \frac{47}{42} \text{ amt. of work}$$

1 day $\frac{42}{47}$

They will complete the work in $\frac{42}{47}$ days together.

4) P & S can do work in 12 days. S & R can do the same work in 16 days, & R & P can do it in 24 days. Find the time in which P, S & R can finish the work together?

P & S → 12 days $\frac{1 \text{ day}}{12}$ amt of work
S & R → 16 → $\frac{1}{16}$
R & P → 24 → $\frac{1}{24}$

P & S → $\frac{1}{12}$
S & R → $\frac{1}{16}$
R & P → $\frac{1}{24}$

$$2(P+S) = \frac{1}{12} + \frac{1}{16} + \frac{1}{24}$$

$$= \frac{4+3+2}{48}$$

$$\Rightarrow P+S+R = \frac{18}{96} = \frac{3}{32}$$

P & S & R → $\frac{1 \text{ day}}{32}$ amt. of work

P, S & R can finish the work together in $\frac{32}{3}$ days.

$\begin{aligned} 1 \text{ day} &= \frac{3}{32} \text{ work} \\ 7 \text{ days} &= \frac{21}{32} \text{ (remaining)} \\ \rightarrow 2 \times \frac{3}{32} &= \frac{6}{32} \\ \rightarrow 2 \times \frac{3}{32} &= \frac{6}{32} \\ \hline &= \frac{12}{32} \end{aligned}$	For total work = 1 1 day → $\frac{3}{32}$ work before the work is over, the work is not over. It needs $\frac{20}{32}$ more work.
--	--

NAME - Rajasree Isha

CLASS - XII

ROLL - 23 SEC - A

REGISTRATION NO - 415231583

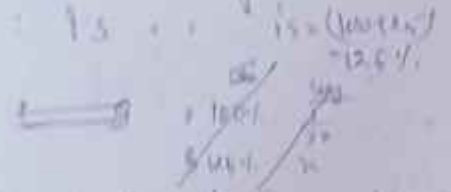
IB - Physics

School - Baluranga Girls' High

Q) P can do a work in 30 days, Q is most efficient than P in completing the same work in how many days will Q complete the work?

P → 30 days $\frac{1}{30}$ amt. of work

Q is 25% more efficient than P
So, if P efficiency is 100%
Q's = 125% (100+25)



P can do $\frac{1}{30}$ amt. of work in 1 day
Q = 125% of $\frac{1}{30}$ [∵ 125% = 1.25]

$$= \frac{125}{100} \times \frac{1}{30} = \frac{5}{24}$$

$\frac{1}{24}$ amt. of work

∴ S work is finished in 24 days in total

Q) If 3 men can do a work in 2 days & 4 boys can do the same work in 6 days, then in how many days will the same work be completed by 8 men & 8 boys?

3m → 2 days
4b → 6 days

4b are doing 3 times more than 3m
i.e. 3m are doing more work than 4b, so they are taking 3 times less than the time taken by 4b

the working speed of the 3m = 3x(4b)

i.e. 3m are working 3 times more than 4b

$$\therefore 3m = 3 \times 4b$$

$$\therefore [1m = 4b] \rightarrow \text{[chain rule]}$$

$$8m + 8b$$

$$= 8 \times 4b + 8b$$

$$= 40b$$

$$4b \rightarrow 6d \quad \frac{1 \text{ day}}{1/6}$$

$$40b \rightarrow ?$$

$$4b \quad \frac{1 \text{ day}}{1/6}$$

$$40b \quad ?(d)$$

$$x \times x = \frac{40 \times 6}{4} = 60$$

$$\therefore x = \sqrt{60} \quad (1 \text{ day})$$

∴ 40b can do the entire work in $\frac{6}{5}$ days

$$(1 \text{ day})$$

NAME - Rajasree Laha

CLASS - XII

Roll - 23 SEC - A

REGISTRATION NO. - 410231580

18 - Physics

MOOL - Baborangon girls' High

3 times the
time as
3 times more
speed

7) Sita & Aita can do a work in 20 days & 25 days, respectively. Both begin together but after a few days, Sita leaves. When Aita finishes the remaining in 10 days. After how many days did Sita leave?

Line diagram
[Total time]

S → 20 days $\frac{1}{20}$ amt. of work

A → 25 days $\frac{1}{25}$

$$\therefore 10 = \frac{10}{\frac{1}{25}} = \frac{2}{5}$$

$\therefore 1 - \frac{2}{5} = \frac{3}{5}$ amt. of work is done by Sita

(Total amt. of work)

$$Sita = \frac{1}{20} + \frac{1}{25} = \frac{9}{100}$$

Wg → $\frac{1}{n}$ amt. of work
 $n \rightarrow \frac{9n}{100} = \dots$

We know, Sita = $\frac{3}{5}$

$$\frac{3n}{100} = \frac{3}{5}$$

$$\therefore n = \frac{20}{3} \text{ days}$$

S & A have done the work together for $\frac{20}{3}$ days

8) After that he worked alone.

i.e. Sita left the work after 20 days

9) When P alone does a work, he takes 25 days more than the time taken by P & Q working together to complete the work. But Q alone takes 9 days more than the time taken by P & Q working together to complete the work.

In what time, P & Q together finish this work?

P & Q → Let's say take 'n' days together to finish a work.

P → (n + 25) days

Q → (n + 9) days

Formula: $n = \sqrt{\text{Extra days of P} \times \text{Extra days of Q}}$

$$\therefore n = \sqrt{25 \times 9} = \sqrt{5 \times 3} = 15 \text{ days}$$

$\therefore$ P & Q together takes 15 days to finish the work.

10) A can complete a work in 12 days & B can complete in 8 days. A works for 8 hours every day while B works for 10 hrs every day. If A & B together start working 8 hrs per day, in how many days will they complete the work?

NAME - Rajasree Laha

CLASS - XII

Roll - 23 Sec - A

REGISTRATION No. - 4192315

18 - Physics

Host - Balurangpur girl's high

→ 2 days to decide a winner
and 1 day allows for errors

A number from $12 \times 8 \text{ hrs}$
Completing the form is $12 \times 8 \text{ hrs}$

$B \rightarrow 8$ days to complete a work
And, if 1 day B works then 10 days

B reaches completes the work in = 80 to
A → 120 hrs $\frac{120}{80} = 1.5$ hrs

A → 66 hrs $\frac{16m}{hr}$
B → 80 hrs $\frac{1}{hr}$ amt. of water
 $\frac{1}{80}$

$$A+B = \frac{1}{\frac{1}{56} + \frac{1}{80}}$$

$$= \frac{136}{56 \times 80} \text{ amt of work in 1 hr}$$

* A & B together will ~~do~~ complete the work in $\frac{70 \times 80}{120}$ (65) ~~75~~

now, given A + B will work 8 hrs/day
at 1 day A + B working for 18 hrs.

8 hrs $\rightarrow$ 1 day

$$t \rightarrow \frac{t}{\delta}$$
$$\frac{9 \times 10^8}{140} \rightarrow \frac{3 \times 10^8}{8 \times 140} = \frac{60}{11} \text{ days}$$

In $\frac{60}{11}$ days A & B can complete the work together.

10. Raj can build a house alone in 10 days but Suraj alone can build it in 12 days. Raj & Suraj work on alternate days. If Raj works on first day, the house will be built in how many days?

R → 16 days $\frac{1 \text{ day}}{16 \text{ amt. of work}}$
S → 12 " $\frac{1}{12}$ "

(R S P S P S) ...

1st day = $\frac{1}{6}$ and of work is done as Raj is working
2nd day = $\frac{1}{12}$ as Raj is sitting

In 2 days $\Rightarrow \frac{1}{12} + \frac{1}{12}$
 $\Rightarrow \frac{2}{12}$ amt of work

In $2 \times 6 = 12$ days $\rightarrow \frac{7 \times 4}{588} = \frac{7}{8}$ amt of work

[Now why it does times 8 times]

Ans. If we read 7 days then,

As if he do 7 times repeat, then,
 $2 \times 7 = 14$ days

$$\frac{7}{48} \times 7 = \frac{49}{48} > 1 \times \left[\because \text{Nucleons are} \right. \\ \left. \text{never } < 1 \right]$$

$\therefore$ So we have to take such a multiple that will give the remainder $= 1$. (i.e. 6 times)

6 times
we need to
repeat this
every 2 days
to get out
of our sick
room in
2-3 days

NAME - Rajasree Isha

CLASS- ALL

Roll - 23 Sec - 11

REGISTRATION No. - 4152

28- Physics

Haut - Bekleidung vordr. S.

the work left = $1 - \frac{7}{8}$ [∵ $\frac{7}{8}$ work has been done already]
 $= \frac{1}{8}$ part of work

12 days have already passed.
 on 13th day Raju will come & he will do $\rightarrow \frac{1}{16}$ part of work

now, $\frac{1}{8} > \frac{1}{16}$
 So, Raju can do $\frac{1}{16}$ part of work & the work left

$= \frac{1}{8} - \frac{1}{16}$
 $= \frac{1}{16}$ part of it is still left

on 14th day Subu come & he will do $\rightarrow \frac{1}{16}$ part of work

now, $\frac{1}{16} > \frac{1}{12}$ [∵ S can work more the work left] from

∴ he can finish that left part of work in,

$\frac{1}{16} \div \frac{1}{12} = \frac{1}{12} \times 2$ [∵ he types in days to complete]

$\therefore 18 = \frac{18 \times 3}{24}$ days [1 day $\rightarrow \frac{1}{12}$ part
 2 days $\rightarrow \frac{1}{6}$ part
 3 days $\rightarrow \frac{1}{4}$ part
 4 days $\rightarrow \frac{1}{3}$ part
 6 days $\rightarrow \frac{1}{2}$ part]

∴ He will complete $\frac{1}{16}$ part of work in $\frac{3}{4}$ days.
 13 days are completed & 14th day only $\frac{3}{4}$ day is used.

∴ total days taken = $13 + \frac{3}{4} = \frac{55}{4}$ days

∴ total no. days / amt. of time taken by P & S to complete the house = $13 \frac{3}{4}$ days

Work & Wages

HPS - 17th Tommy

2) A - 2 days $\rightarrow$ fast (as it takes Tommy less days to complete)

B - 3 days
 Both of them are doing the same work

$\rightarrow$ slow (as it takes more days (time) to complete)

∴ 1 days 1 money

∴ Income $\propto \frac{1}{\text{days}}$

Ratio ^{days} A : B = 2 : 3

Ratio Income A : B = 3 : 2 [∵ Income $\propto \frac{1}{\text{days}}$]

exactly $\frac{1}{2} : \frac{1}{3}$
 opposite as = 3 : 2
 exactly inverse

3) Share of B = $\frac{2}{5} \times (\text{total income})$
 out of total income

Share of A = $\frac{3}{5} \times (\text{total income})$
 out of total income

NAME - Rajasree Laha

CLASS - 5th

Roll - 25 Sec - A

REGISTRATION NO. - 4152

SB - Physics

Host - Bhubaneswar

Q. 74. Many

considering R/S concept solve the problem then
 A - 20
 B - 30
 both of them are doing the same work
 So then how the work is distributed, we have to consider work here not the no. of days. At
 (2) we were considering no. of days.

A $\rightarrow 20 \rightarrow \frac{1}{20}$ out of work (time & work concept)
 B $\rightarrow 30 \rightarrow \frac{1}{30}$ " " " "

Ratio. A/B = $\frac{1}{20} : \frac{1}{30}$ (in 1 day) = 3:2

Ratio of income = 3:2 [Income $\propto$ work]

Ex) Invert,
 total work is done by n days
 In 1 day = $\frac{1}{n}$ out of work (just invert)

Similarly,
 in 1 day $\frac{1}{n}$ out of work is done

total work would be done in n days (invert)

So, eg - A $\rightarrow \frac{1}{8}$ out of work $\rightarrow$ 8 days
 total work would be done in 8 days.

1) Ramesh can do a work in 5 days. Sunesh do the same work in 7 days. The total amount given for this work to them is Rs. 480. If both are working together, then what will be the share of Sunesh.

1st TM
 [but here the work is same, so we need to see the time taken by both & we know,]

7d 1m

R $\rightarrow$ 5d $\rightarrow$ fast & money

S $\rightarrow$ 7d $\rightarrow$ slow & money

Ratio of days = 5:7

Ratio of income = $\frac{5}{7} : \frac{7}{5}$
 = 7:5

Income Ramesh: Sunesh = 7:5

$\therefore$ Share of Sunesh = $\frac{5}{7+5} \times 480$
 = 200/-

Share of Ramesh = $\frac{7}{12} \times 480$ (480 - 200)
 = 280/-

2) Ramesh can do a work in 15 days. Vijay & Ramesh together do the same work in 10 days. They received Rs. 1155 for that work. What is share of Ramesh & Vijay?

NAME - Rajasree Laha

CLASS - XII

Roll - 25 SEC - A

REGISTRATION No. - 4192315855

JB - Physics

Host - Bhatnagar vidyalaya

7d 1m
 1st TM - this concept we will get the same ans

2 → 150 → $\frac{100}{15}$ out of work

2 → 100 → $\frac{100}{10}$

100 → 100 → $\frac{100}{10}$

$$100 = \frac{100}{10}$$

$$\Rightarrow \frac{1}{10} + \frac{1}{15} = \frac{1}{10}$$

$$\frac{10+15}{15 \times 10}$$

$$\Rightarrow \frac{1}{10} = \frac{1}{15} + \frac{1}{15}$$

$$= \frac{3+2}{30}$$

$$= \frac{1}{30} \therefore \text{in 30 days}$$

work done by vijay in 1 day = $\frac{1}{30}$

the no. of days taken by vijay to complete the work = (total work) = 30 days

ratio of days = P:V = 15:30 = 1:2

$$\therefore \frac{P}{V} = \frac{1}{2}$$

$$\text{Ratio of income} = \frac{P}{V} = \frac{1}{2}$$

$$= 2:1$$

$$\therefore \text{share of P} = \frac{2}{2+1} \times 1155 = \frac{2}{3} \times 1155 = 770/-$$

$$\therefore \text{share of V} = \frac{1}{2+1} \times 1155 = \frac{1}{3} \times 1155 = 385/-$$

3) wages of 44 women for 56 days comes to RS. 29568. How many men are needed for 47 days to receive RS. 16920, if the daily wages of a man being 5 times those of a woman?

$$44W \text{ for } 56d = 29568 \text{ RS}$$

$$44W \text{ for } 1d = \frac{29568}{56}$$

$$1W \text{ for } 1d = \frac{29568}{56 \times 44} \text{ RS (for } 1W \rightarrow 1d) = \text{RS. } 12$$

1M = 5412 = 860 [∵ wages of a man is 5 times than those of a woman]

let the no. of men = M

$$1M \text{ for } 1\text{ day} = \text{RS. } 60$$

$$M \therefore 1\text{ day} = \text{RS. } 60M$$

$$M \therefore 47\text{ days} = \text{RS. } 60M \times 47$$

We know that According to the Q.

$$60M \times 47 = 16920$$

$$\Rightarrow M = 6 \text{ men}$$

NAME - Pooja Singh

CLASS - 10

Roll - 03 SEC - A

REGISTRATION NO. - 40923152

SB - Physics

Host - Bhuvan Singh

Q. P, Q & R take a job for RS. 640.
P & Q together finish $\frac{1}{5}$ th of the work
& rest is done by R alone. What is the share
of R?

$$\frac{1}{5} \text{ work done by P \& Q} \\ \frac{1}{5} \rightarrow 1 - \frac{4}{5} \rightarrow \frac{1}{5} \text{ alone}$$

For money [Here we are applying this
as we can see directly
 $\frac{3}{5}$ th of the money is done by them, it is fair
(more work) (more money) to say by this method]

How much work, that much money

$$\text{Share of R} = \frac{3}{5} \times 640 \\ = 384 \text{ RS}$$

$$\text{Ratio of work} = \frac{2}{5} : \frac{3}{5} \\ = 2 : 3$$

Ratio of money = 2 : 3 (Work & money)

$$\text{Share of R} = \frac{3}{5} \times 640 \\ = 384 \text{ RS}$$

S/P can do a job in 30 days, while Q alone
can do it in 45 days. They work together
for 15 days & rest of the job is done by
R in 6 days. They get RS. 15000 for the whole
job. What will be R's share?

or we can't apply normal approach to money in this way

$$P \rightarrow 30 \text{ d} \rightarrow \frac{1}{30} \text{ amt of work}$$

$$Q \rightarrow 45 \text{ d} \rightarrow \frac{1}{45}$$

$$P \& Q \rightarrow 15 \text{ d} \rightarrow$$

$$\frac{1}{30} + \frac{1}{45} = \frac{5}{90} = \frac{1}{18}$$

$$\text{In 1 day P \& Q} \rightarrow \frac{1}{18}$$

$$15 \text{ d} \rightarrow \frac{15}{18} = \frac{5}{6}$$

Rest of work is done by R

Work done by R = $1 - \frac{5}{6} = \frac{1}{6}$ amt of work
[Given R completed that remaining work in
6 days, hence we can see in 1 day R can do
 $\frac{1}{6}$ amt of work, so R can complete the
rem work in 6 days]

$$\therefore \frac{1}{6} \times 25000 \\ = 2500/-$$

NAME - Rajasree John

CLASS - 10

Roll - 23 Sec - A

REGISTRATION No - 419231529

JB - Physics

Host - Babington Girls High

A man & a boy received RS. 1800/- for 3 days for a job they did together. man's efficiency in work was 5 times that of the boy. what is the daily wages of the

$$M+B = 1800/- \text{ for 3 days} = 1800/-$$

$$M+B \text{ 1 day} = \frac{1800}{3} = 600/-$$

$$M = 5B$$

∴ man's 5 times more work than that of B

∴ 5M = 10B
5 times 5 times

∴ Ratio of man to B

$$= 5:1 \quad \left[\begin{smallmatrix} \text{E} \\ \text{S} \\ \text{M} \\ \text{B} \end{smallmatrix} \right]$$

∴ Share of B = $\frac{1}{5+1}$ (total daily wages)

$$= \frac{1}{6} \times 600 \text{ for 1800/- for 3 days 2 at 1 day 600/-}$$

∴ Pankaj was appointed for a 100 days. the condition was that he will be paid RS. 24 for every working day. But he will also be fined RS. 12 for every day he is absent. At the end, he got RS. 420 for how many days he was absent?

The job was for 100 days.

The condition was that he will be paid RS. 24 for every working day.

∴ He should have gotten RS. $24 \times 100 = 2400$

But he got RS. 420 only.

Let us assume that he was absent for x days.

∴ He was present for $(100 - x)$ days. For every present day he got 24/-

$$\therefore (100 - x) 24 - 12x = 420 \quad \begin{matrix} 1d \rightarrow 12 \\ 12d \rightarrow 12x \end{matrix}$$

$$\Rightarrow 2400 - 420 = 36x$$

$$\Rightarrow x = 55 \text{ days}$$

∴ P was absent for 55 days.

87 total wages of 6 men, 4 women & 8 boys is RS. 26. If the wages of 6 men is equal to that of 8 women & the wages of 4 women is equal to that of 6 boys, then find out the total wages of 8 men, 6 women & 4 boys?

$$\begin{aligned} 1 \text{ man} &\rightarrow 3 \text{ boys} \\ 1 \text{ W} &\rightarrow 2 \text{ WPS} \\ 1 \text{ B} &\rightarrow 1 \text{ BPS} \end{aligned}$$

$$\begin{aligned} 6M &= 8W & 4W &= 6B \\ \Rightarrow M &= \frac{4}{3}W & \Rightarrow B &= \frac{2}{3}W \\ 6M + 4W + 8B &= 26 \\ \Rightarrow 6 \times \frac{4}{3}W + 4W + 8 \times \frac{2}{3}W &= 26 \end{aligned}$$

NAME - Rajasree Laha
CLASS - 5th
Roll - 23 Sec - A
REGISTRATION No. - 41923158
UB - Physics
Hool - Bhubaneswar 751013

Problem on numbers

It can be solved in least time

$$1. \quad 12W + 16W = 26 \quad \text{[from (1)]}$$

$$\Rightarrow 28W = 26 \times 3$$

$$\Rightarrow W = 1.5 \text{ Rs}$$

$$M = \frac{3 \times 1.5}{6} = 2.5$$

$$6 = \frac{4 \times 1.5}{6} = 1.25$$

$$\text{Sum} = 460?$$

$$\Rightarrow 8 + 2 + 4 + 15 + 4 + 1$$

$$= 29 \text{ Rs}$$

2. If P, Q and R get Rs. 10800 for doing a work in 18 days. P & R get Rs 3600 for doing the same work in 10 days, while Q & R get 6080 for doing the same work in 20 days. Find the amount received by R per day.

$$P + Q + R \rightarrow 10800 \rightarrow \text{Total}$$

$$18 \text{ days} \rightarrow 10800/-$$

$$1 \rightarrow \frac{10800}{18} \text{ (in 1 day)}$$

$$= 600 \text{ (1 day)}$$

$$P + R \rightarrow 3600 \rightarrow 10 \text{ days} \rightarrow \frac{3600}{10}$$

$$= 360 \text{ (1 day)}$$

$$Q + R \rightarrow 6080 \rightarrow 20 \text{ days}$$

$$\rightarrow \frac{6080}{20}$$

$$= 304 \text{ (1 day)}$$

$$P + Q + R = 600 \text{ (1 day)}$$

$$\Rightarrow 360 + 304 = 600$$

$$\Rightarrow R = 224 \text{ (1 day)}$$

$$P + R = 360 \text{ (1 day)}$$

$$\Rightarrow P + 224 = 360$$

$$\Rightarrow P = 136 \text{ (1 day)}$$

$$P + Q = 600 \text{ (1 day)}$$

$$\Rightarrow 136 + 224 = 600$$

$$\Rightarrow R = 80 \text{ (1 day)}$$

[from (1)]

$$\therefore R = 80 \text{ Rs per day (in 1 day)}$$

10. The amount of money with which A's wage can be paid for 18 days, when A is working alone is enough for paying B's wage for 12 days, when B is working alone. If A & B start working together, then the same amount would be enough for wages of both for how many days?

$$\text{Let, Amount} = X \text{ Rs}$$

$$\text{(Total amount)}$$

$$A \rightarrow 18 \text{ days, i.e. in 18 days X Rs is enough to pay wage to A}$$

$$1 \text{ day} \rightarrow \frac{X}{18}$$

$$B \rightarrow 12 \text{ days, i.e. in 12 days X Rs is enough to pay wage to B}$$

$$1 \text{ day} \rightarrow \frac{X}{12}$$

$$A + B = \frac{X}{18} + \frac{X}{12} = \frac{5X}{36} \text{ Rs are paid working together}$$

working together

for X Rs

$$\frac{X}{18} + \frac{X}{12} = \frac{5X}{36}$$

$$\Rightarrow X = \frac{5X}{36} \times 36$$

$$\Rightarrow Y = \frac{36}{5} \text{ days}$$

For $\frac{36}{5}$ days the

same amount would be

enough for wages of both A & B

NAME - Rajasree John

CLASS - 10

ROLL - 23 SEC - A

REGISTRATION No. - 41523152

UB - Physics

Model - Bhuvanagiri 15 High

16 Jibing employs 5 workers to work for 6 hrs in day. In total they pay them Rs. 130 in a week. How much should Jibing pay 13 workers working 4 hrs per day for a week?

Let x = cost of 1 hr of 1 worker for 1 day
 $\therefore$ total hrs = $6 \times 5 = 30$ hrs
 $\therefore 30x = 130$

$5W \rightarrow 6 \text{ hrs} \rightarrow 130 \text{ RS}$
 $5W \rightarrow 1 \text{ hr} \rightarrow \frac{130}{6}$
 $1W \rightarrow 6 \text{ hrs} \rightarrow \frac{130}{5}$
 $\therefore 1W \rightarrow 1 \text{ hr} \rightarrow \frac{130}{30}$

$13W \rightarrow 1 \text{ hr} \rightarrow \frac{130}{30}$
 $13W \rightarrow 4 \text{ hrs} \rightarrow \frac{130}{30} \times 4 = \frac{520}{15} \text{ RS}$
 $13W \rightarrow 4 \text{ hrs} \rightarrow 135 \text{ RS}$ (1 worker / week)
 $4 \text{ hrs/day} \times 7 = 28 \text{ hrs}$ (1 worker / day)
 $13W \rightarrow 28 \text{ hrs} \rightarrow \frac{135}{7} \times 28 = 540 \text{ RS}$

If Ram & Syam have been given a job of painting a house for Rs. 800. With the help of Pitu they complete the job in just 3 days. Had Ram alone be doing the task, he would need 6 days. If Syam alone would be doing the task, he would need 8 days. How much money will Pitu get?

$R \rightarrow 6 \text{ d} \rightarrow \frac{1}{6}$ part of work
 $S \rightarrow 8 \text{ d} \rightarrow \frac{1}{8}$
 $P \rightarrow 2 \text{ d} \rightarrow \frac{1}{2}$
 let Pitu takes x days
 let x to complete the work.

$\frac{1}{6} + \frac{1}{8} + \frac{1}{2} = 1$ (in 1 day)
 But given that they complete the task in 3 days
 $\therefore$ in 1 day they do $\frac{1}{3}$ part of work
 $\therefore$ we can say

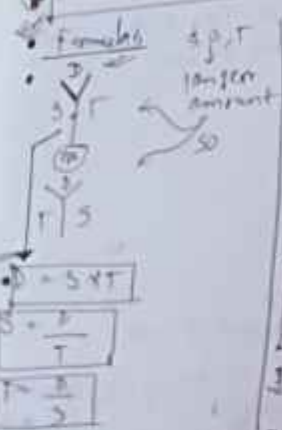
$\frac{1}{6} + \frac{1}{8} + \frac{1}{2} = \frac{1}{3}$
 $\frac{1}{6} = \frac{1}{3} - \frac{1}{8} - \frac{1}{2}$
 $\frac{1}{6} = \frac{2}{24} - \frac{3}{24} - \frac{12}{24}$
 $\frac{1}{6} = -\frac{13}{24}$

$\therefore R = 24$ days (if working alone)
 Ratio of work done by R, S, P is $= \frac{1}{24} : \frac{1}{8} : \frac{1}{2}$ (in 1 day)
 $= 1 : 3 : 12$

$\therefore$ If Rs. 800 given for the task with 800 RS. But with help of Pitu they finish the task in 3 days. i.e. now R will also be paid from that 800 RS.
 $\therefore$ The R will get $= \frac{1}{13} \times 800 = 100 \text{ RS}$

NAME - Rajasree Latha
 CLASS - XII
 Roll - 23 Sec - A
 REGISTRATION No. - 4192315253
 UB - Physics
 Host - Balamangal school's 11th 12th

Time, distance, speed (m/s hrs) 26/02/24



Types of Sum (4)

- S same
- S same
- T same
- Avg speed

Avg S = $\frac{\text{Total distance (D)}}{\text{Total time (T)}}$

Is $\frac{D_1}{T_1} = \frac{D_2}{T_2} = \frac{D_3}{T_3}$ to

$T_1 D_2 = T_2 D_1$ $T_1 D_3 = T_3 D_1$ $T_2 D_3 = T_3 D_2$

$D = D_1 + D_2 + D_3 = 12m$

Q + by increases his speed to 5/5 times of his speed. By this he reaches his school 30 mins before the usual time. How much time does he takes usually?

Ans, D is same

$D = D$

$S_1 T_1 = S_2 T_2$ [Let S_1 (usual speed) = 5]

$5 T_1 = \frac{25}{5} (T_1 - 30)$ $T_1 = T_2$

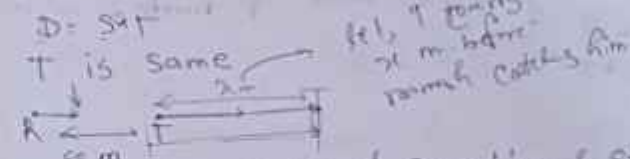
$5 T_1 = 5 T_1 - 240$ $S_2 = \frac{25}{5}$

$5 T_1 = 240$

$T_1 = 48 \text{ mins}$

$T_2 = (T_1 - 30)$ (as he takes 30 min less than usual time)

Q Ramesh sees a thief at a distance of 80m. Ramesh starts chasing the thief who is running at a speed of 5 m/s. Ramesh is chasing him with a speed of 7 m/s. How much distance does the thief cover before Ramesh catches him?



Both of them started at same time & stopped at the same time, i.e. the time taken by both is same. (Both of them were running for the same amt of time)

$T_1 = T_2$

$\frac{D_1}{S_1} = \frac{D_2}{S_2}$

$\frac{80}{7} = \frac{x}{5}$

$\frac{D_2}{S_2} = \frac{D_1}{S_1}$

$\frac{x}{5} = \frac{80}{7}$

[∴ Ramesh covers 80m to catch the thief]

$7x = 400 = 7m$

$x = 400$

∴ he 200m

NAME - Rajasree Laha
CLASS - XII
Roll - 23 Sec - A
REGISTRATION No. - 41923158
UB - Physics
Host - Balarama girls' High

Honk - Bismarck girl's high

⑤ Surendra travels from home to office by car with an avg speed of 50 km/hr. He is late by 30 mins. But when he comes with a speed of 60 km/hr, he reaches his office 10 mins earlier. How far is his office from his home?

2 ways

Easy way
 $D = S \times T$



Diff. b/w this 2 speeds of clock
 $= 30 + 10 = 40 \text{ min}$

this is the diff. of time

$$T_1 - T_2 = 40 \text{ min}$$

$$\frac{D}{S_1} - \frac{D}{S_2} = \frac{40}{60} \text{ hrs}$$

or

$$\frac{x}{50} - \frac{x}{60} = \frac{40}{60}$$

$$\Rightarrow x \left(\frac{1}{50} - \frac{1}{60} \right) = \frac{2}{3}$$

$$\Rightarrow x + \frac{10}{50} = \frac{2}{3}$$

$$\Rightarrow x = \frac{2}{3} \text{ km}$$

[Here, S & T are different but D is same]
4th way

2) Regular method

S, T not same

$D = \text{same}$

$$D = D$$

$$\Rightarrow S_1 \times T_1 = S_2 \times T_2$$

$$\Rightarrow 50 \left(\frac{x}{50} \right) = 60 \left(\frac{x}{60} \right) \text{ [let, regularly S reaches in office with } x \text{ min]} \\ \Rightarrow x = \frac{3}{2} \text{ hrs}$$

$$\therefore D = S \times T = \frac{50 \times \frac{3}{2}}{1} = 75 \text{ km}$$

⑥ Rohit drives from his home at a speed of 30 km/hr & reaches his bank 20 mins late. Then the next day he increases his speed by 15 km/hr but still he is late by 8 mins. How far is his bank from his home?

$$\text{Diff.} = 20 - 8 = 12 \text{ mins}$$

$$T_1 - T_2 = 12 \text{ min}$$

$$\Rightarrow \frac{x}{30} - \frac{x}{45} = \frac{12}{60}$$

$$\Rightarrow x \left(\frac{1}{30} - \frac{1}{45} \right) = \frac{12}{60}$$

$$\Rightarrow x = \frac{18 \times 3}{30 + 45} = \frac{18 \times 3}{75}$$

$$\Rightarrow x = 18 \text{ km}$$

NAME - Pooja Sree Isha

CLASS - XII

Roll - 23 Sec - A

REGISTRATION NO. - 4152515855

UB - Physics

School - Bahurampur girls' High School

① Pratik travels some distance at a speed of 16 km/hr using a bike, 124 km at 31 km/hr by car & another 105 km at 7 km/hr in horse cart. Then find his avg speed for the entire distance travelled?

$$D = S \cdot T$$

$$S = \frac{D}{T}$$

$$\text{Avg } S = \frac{T \cdot D}{T \cdot T}$$

$$\text{Avg } S = \frac{16 \cdot 96 + 31 \cdot 124 + 7 \cdot 105}{96 + 124 + 105} \text{ km}$$

$$T = T_1 + T_2 + T_3$$

$$= \frac{D_1}{S_1} + \frac{D_2}{S_2} + \frac{D_3}{S_3}$$

$$= \frac{96}{16} + \frac{124}{31} + \frac{105}{7}$$

$$= 6 + 4 + 15 = 25 \text{ hrs}$$

② Rohit covers one-fourth ($\frac{1}{4}$) of the total distance at 20 km/hr, one-fourth at 10 km/hr & rest of his journey at 30 km/hr. Find Rohit's avg speed for the whole distance?

$$D = S \cdot T$$

$$S = \frac{D}{T}$$

$$\text{Avg } S = \frac{T \cdot D}{T \cdot T}$$

$$\text{Let total } D = x \text{ km}$$

$$x = 20 \text{ km/hr}$$

$$x = 10 \text{ km/hr}$$

$$x = \frac{20 \cdot x}{4} + \frac{10 \cdot x}{4} + \frac{30 \cdot x}{4}$$

$$T = T_1 + T_2 + T_3$$

$$= \frac{x}{20} + \frac{x}{10} + \frac{x}{30}$$

$$= \frac{x}{20} \left(\frac{1}{2} + \frac{1}{2} + \frac{1}{3} \right)$$

$$= \frac{x}{20} \left(\frac{1}{2} + \frac{1}{2} + \frac{1}{3} \right)$$

$$= \frac{x}{20} \left(\frac{16 + 3 + 8}{24} \right)$$

$$= \frac{x}{20} \cdot \frac{27}{8}$$

$$= \frac{27x}{160}$$

$$\therefore \text{Avg } S = \frac{x \cdot 160}{27x}$$

$$= \frac{160}{27} \text{ km/hr}$$

③ A walks from Jammu to Delhi & at the same time B starts walking from D to J. After passing each other, they complete their journeys in 36 hours & 288 hours, respectively. Find the ratio of speed of A to that of B?

$$S: B_s = \sqrt{J} : \sqrt{X} \leftarrow \text{method}$$

we know, $X = 361$
 $J = 289$

$$A_s: B_s = \sqrt{289} : \sqrt{361} \\ = 17 : 19$$

17) A car travelling with 5/7th of its actual speed covers 42 km in 1 hr 40 mins 48 secs. Find the actual speed of the car.

D.S.T

$$D.S.T = \frac{D}{T}$$

Let S of the Car = x
 Car is currently travelling 5/7th of its actual speed $\rightarrow \frac{5x}{7}$

$$T = 1 \text{ hr } 40 \text{ min } 48 \text{ s} \\ = 60 \times 6 + 40 \times 60 + 48 \\ = 6048 \text{ Sec}$$

$$D = 42 \text{ km} = 42000 \text{ m}$$

$$S = \frac{D}{T}$$

$$\frac{5x}{7} = \frac{42000}{6048}$$

$$\Rightarrow x = \frac{350 \times 18}{36 \times 5} = \frac{350 \times 18}{180} = 35 \text{ km/hr}$$

$$\begin{array}{l} \text{km/hr} \rightarrow \frac{5}{7} \text{ m/s} \\ \text{m/s} \rightarrow \frac{18}{5} \text{ km/hr} \end{array}$$

Profit & Loss (2-3)

Formula



Let,

RS. 200 $\rightarrow$ C.P

(Cost Price) $\rightarrow$ Amount we spend to create the book

If you want to sell the book with 10% profit, after selling the book, then we'll sell the book at price = $200 + 100 = 300 \text{ RS.}$
 $\rightarrow$ selling price of the book = 300

$$\therefore \boxed{SP = CP + \text{Profit in RS.}}$$

If we want to express the profit into %, then

$$\therefore \boxed{\% \text{ Profit} = \frac{\text{Profit in RS.}}{C.P.} \times 100}$$

As Profit is always calculated on the C.P.

~~% Loss~~

Let, C.P = RS. 200

S.P = RS. 150

$\therefore$ Loss is = 50 RS.

$$\boxed{SP = C.P - \text{Loss in RS.}}$$

$$\boxed{SP = CP + \text{Loss in RS.}}$$

$$\therefore \boxed{\% \text{ Loss} = \frac{\text{Loss in RS.}}{C.P.} \times 100}$$

As Loss is always calculated on C.P.

NAME - Rajasree Laha

CLASS - 8th

Roll - 23 SEC - A

REGISTRATION No. - 4102315253

UR - Physics

Hostel - Balaram girls High Sch

For Profit

Box $\rightarrow$ Rs 500 = CP
 Let Profit = 10%, then Profit is %?

$$SP = CP + \text{Profit} \text{ Rs. } \frac{CP \times \text{Profit}}{100}$$

$$SP = (100 + 10) \times \frac{500}{100}$$

[As, whole thing is taken as 100%]

Hence,

$$SP = (100 + 10) \times \frac{500}{100}$$

$$SP = \frac{55000}{100} = 550/-$$

For Loss

Box $\rightarrow$ CP = 500 Rs.
 Let, sold to book in 20% loss, then SP?

$$SP = (100 - m) \times \frac{CP}{100}$$

Hence,

$$SP = (100 - 20) \times \frac{500}{100}$$

$$= \frac{400}{100} \times 500 = 400 \text{ Rs}$$

For Profit	Profit %	Profit Rs.
For Loss	Loss %	Loss Rs.
For Profit	Profit %	Profit Rs.
For Loss	Loss %	Loss Rs.
For Profit	Profit %	Profit Rs.
For Loss	Loss %	Loss Rs.

[As, whole thing is taken as 100%]

Loss % $\rightarrow$ Loss %

① Ajay incurred loss of 20% by selling a vase for Rs 3800. To get a profit of 20% at what price should he sell the vase?

According to the Qn,

$$SP = (100 - \text{Profit}) \times \frac{CP}{100}$$

$$3800 = (100 - 20) \times \frac{CP}{100}$$

$$\Rightarrow 3800 = \frac{80}{100} \times CP$$

$$\Rightarrow CP = \frac{3800 \times 100}{80} = 4750$$

Now, CP = 4750
 SP = ?
 Profit = 20%

$$SP = (100 + 20) \times \frac{CP}{100}$$

$$= \frac{120}{100} \times 4750$$

$$= 5700$$

② Rambhadr sells paper planes at the rate of 20 planes for Re 1. If he gets profit of 20%, how many planes did he buy in 1 rupee?

$$SP = \text{Rs. } 1 = (100 + \text{Profit}) \times \frac{CP}{100} = (100 + 20) \times \frac{CP}{100}$$

$$\Rightarrow 1 = \frac{120}{100} \times CP$$

$$\Rightarrow CP = \frac{100}{120} \text{ Rs}$$

20 $\rightarrow$ 10/120 $\rightarrow$ 20 $\times$ 12 = 240
 10? $\rightarrow$ 1 $\rightarrow$ 24 planes

$$\frac{1}{P} = \frac{P_{100}}{CP}$$

$$\frac{1}{L} = \frac{L_{100}}{CP}$$

$$SP = (100 + P) \times \frac{CP}{100}$$

$$SP = (100 - L) \times \frac{CP}{100}$$

NAME - Rajat Singh
 CLASS - 10
 Roll - 23 Sec - A
 REGISTRATION No. - 415231203
 UB - Physics
 School - Budhwaran Girls High School

Q) Uma wants to gain 15% profit on her sale of sugar. She buys 1kg of sugar at RS. 24 per kg to mix with 1kg of sugar bought at RS. 28 per kg. She sells the sugar mix at ? — (with profit?)

$$SP = (100 + 15)\% \times CP \\ = \frac{115}{100} \times CP$$

CP
1kg in 24 RS/kg

She spends = $24 \times 120 = 2880$ RS.

1kg in 28 RS/kg

She spends = $28 \times 120 = 3360$ RS.

She mixes RS. two

total expenditure / total amt. spent

$$= \frac{2880 + 3360}{2}$$

$$= 3120 \text{ RS}$$

$$CP = 3120$$

$$SP = \frac{115}{100} \times 3120$$

$$= 3588 \text{ RS}$$

Q) Gaddi buys some oranges in a shop at 4 per rupee. She goes to other shop & buys same no. of oranges at 5 per rupee. She then combines them in a basket & sells them at 4 per rupee. Will she get a profit or loss? And how much?

$$\text{if } SP - CP > 1 \rightarrow \text{Profit} \quad \text{if } SP - CP < 1 \rightarrow \text{loss}$$

CP of orange
1st shop

4 oranges in 1 RS.

$\therefore$ Price of 1 orange $\rightarrow \frac{1}{4}$ RS

2nd shop

5 oranges in RS. 1

$\therefore$ Price of 1 orange $\rightarrow \frac{1}{5}$ RS.

if she mixes 1 orange from the 1st shop with 1 orange of the 2nd shop, i.e. 2 oranges

$$\text{The cost CP of this 2 oranges} = \frac{1}{4} + \frac{1}{5} \\ = \frac{5+4}{20} \\ = \frac{9}{20} \text{ RS.}$$

2 oranges (mixed) $\rightarrow \frac{9}{20}$ RS.

1 orange (average) $\rightarrow \frac{9}{40}$ RS.

SP of oranges = $\frac{4}{1} \text{ RS. (as in question)}$

$\therefore$ CP of 1 orange = $\frac{9}{40}$ RS.

NAME - Pooja Sree Laha

CLASS - XII

Roll - 23 Sec - A

Registration No. - 4192315233

UB - Physics

Hostel - Baluranga Girls' High School

SP of mango

4 mangoes in 1 RS

1 mango $\frac{1}{4}$ RS

SP of 1 mango = $\frac{1}{4}$ RS

now

$$SP - CP = \frac{1}{4} - \frac{2}{10}$$

$$= \frac{10 - 8}{40} = \frac{2}{40} = \frac{1}{20} \text{ (ve)}$$

$\therefore$ she gets profit

$$\% \text{ profit} = \frac{1 \times \frac{1}{20}}{\frac{2}{10}} = \frac{100}{20} = 5\%$$

$$= \frac{100\%}{2} \text{ or } 11\frac{1}{2}\% \text{ or } 11.11\%$$

Ques

⑤ Kamlesh gets a profit of 20% in one trade & suffers a loss of 20% in the 2nd when he sells 2 cycles for RS 4000 each. What is his profit or loss in the entire trade?

2 trades $\rightarrow$ 1st cycle
 $\rightarrow$ 2nd cycle

$$+20\% - 20\%$$

SP = same = 4000 RS/cycle

Therefore, there will be this kind of cases, where

$\frac{1}{n} \text{ RS}$ & also $\frac{1}{n} \text{ RS}$ are also same, for

2-trades are something then the answer will always be a **Loss** as the **loss**

Percentage is $\frac{1}{n} \text{ RS} \times \frac{1}{n} \text{ RS}$ (let there be no of

$$\left[\frac{1}{n} \text{ RS} \times \frac{1}{n} \text{ RS} \right] \times \frac{100}{100} \text{ (not } \frac{100}{100} \text{)}$$

$$\left[\frac{1}{n} \text{ RS} \times \frac{1}{n} \text{ RS} \right] \times \frac{100}{100}$$

$$\therefore \text{here } \frac{1}{2} \text{ RS} \times \frac{1}{2} \text{ RS} = \frac{4000 \times 4000}{100} = 4\%$$

$\therefore$ 4% loss in the entire trade

⑥ Simran bought pet food worth RS 5600. She then sold $\frac{1}{3}$ rd of it incurring a loss of 40%. What profit she must earn on rest of the supplies to nullify this loss?

$$\begin{matrix} (1) & (2) \\ 2 & 3 \\ 20\% & 140\% \end{matrix}$$

$\therefore$ she must earn 140% profit on the rest of the supplies to nullify this loss.

NAME - Rajasree Laha

CLASS - 11

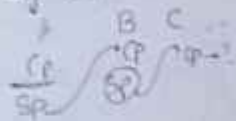
Roll - 23 Sec - A

REGISTRATION No. - 4152315833

UB - Physics

School - Bhatnagar Girls' High School

Q) A sold a car to B at a profit of 25%. B incurred loss of 15% while selling the same car to C. A spent Rs. 50,000 for this car. At what price did C buy it?



A,
 $CP = 50,000$
 $SP = (100 + 25)\% \times CP$
 $= \frac{125}{100} \times 50,000$
 $= 62,500 \text{ RS}$

B
 $CP = 62,500$
 $SP = (100 - 15)\% \times 62,500$
 $= \frac{85}{100} \times 62,500$
 $= 53,125 \text{ RS}$

C
 $CP = SP \text{ of B} = 53,125 \text{ RS}$
 Q) A cheater manipulated his weighing machine so that it shows 1 kg for 970 grams. How much profit % does he get? if we consider the actual wt = $CP = 970 \text{ g}$ & the fake wt = $SP = 1 \text{ Kg}$
 $\therefore \frac{1000 - 970}{970} \times 100 = 3.09\%$

$\therefore SP = CP + \text{Profit}$
 $\Rightarrow 1000 = 970 + \text{Profit}$
 $\therefore \text{Profit} = 30 \text{ RS}$ [for every kg he gets 30 RS profit]

this is Profit in grams

$\therefore \text{Profit} = \frac{30}{970} \times 100$
 $= \frac{300}{97} \% \approx 3 \frac{2}{97} \%$

Q) Rohit got Profit of $11\frac{1}{2}\%$ by selling his old car. However he realized that he sold it for Rs. 8100 more, his profit would be 38.5%. At what price did he buy the car? (1.5% Profit)

if, 8100 RS $\rightarrow$ 38.5% (Profit)
 the diff. b/w this 2 profits = $38.5 - 11.5\%$

$\therefore$ If he sold his car for 8100 RS more, the profit would be 27% more.

$\therefore 27\% \times CP = 8100$
 $\Rightarrow \frac{27}{100} \times CP = 8100$

$\therefore CP = 30,000 \text{ RS}$

Q) Chaman sells 40 fans at 10% profit. He wants a total of 20% profit on the entire sale. Since he got 160 fans at rate of Rs. 100

NAME - Raju Suresh
 CLASS - 10
 Roll - 23 SEC - A
 REGISTRATION No. - 4152185
 DR - Physics
 HOD - Baburam 10/1/23 11:30

each, what profit must he sell the remaining fans?

100 fans → Rs 100 for each

100 fans → 100 × 10 = 1000

100 fans → ? } 20%

Total Rs = 1000

100 fans → 1000 Rs (for 100 fans) + 100 fans → 1000 Rs (for 100 fans)

$\frac{1000 \times 100}{100} = 1000$ $\frac{1000 \times 100}{100} = 1000$ $\frac{1000 \times 100}{100} = 1000$

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Q) Suman buys 100 chocolates for Rs 1000. She wanted to earn 30% profit by selling them. But Ramesh visited her & she gave him 75% of those chocolates at the cost price itself. But even after doing this, she earned a profit of 30%. How much did she sell each chocolate?

1000 → 1000 Rs

100 → 1000

1000 → 1000

1000 → 1000

1000 → 1000

Rs 1000

30% profit is wanted by her

$\therefore \frac{1000 \times 100}{100} = 1000$

$\frac{1000 \times 100}{100} = 1000$

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Rs 1000

30% profit is wanted by her

$\therefore \frac{1000 \times 100}{100} = 1000$

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NAME - Yashwantrao Jadhav

CLASS - 10

Roll - 25 SEC - B

DATE - 25/05/2020

UB - Physics

Topic - Simple Interest

① A shopkeeper bought a profit of 15% after selling a box at 20% discount on the marked price. The ratio of the cost price & marked price of the box is?

Let marked price = x Rs.

$$SP = (100 + 15)\% \times CP \quad | \quad SP = (100 - 20)\% \times MP \times 2$$

$$1.15 CP = 1.6 CP \quad | \quad SP = 80\% \times x$$

$$1.15 CP = 1.6 CP \quad | \quad SP = \frac{80}{100} \times x$$

$$\frac{1.15}{1.6} + CP = \frac{80}{100} \times x$$

$$\frac{1.15}{1.6} + CP = \frac{80}{100} \times x \quad \therefore CP : x = 16 : 23$$

② The ratio of cost price & selling price is 5 : 6. The profit percent is?

$$CP : SP = 5 : 6 \quad \text{Let } CP = 5 \text{ Rs (let } x) \quad SP = 6 \text{ Rs}$$

$$P = SP - CP = 6 - 5 = 1 \text{ Rs} \quad \therefore \text{Profit} = 1 \text{ Rs}$$

$$\therefore \text{Profit} = \frac{1}{5} \times 100 = 20\%$$

③ If selling price of 40 articles is equal to cost price of 50 articles, the profit percent is?

$$SP_{40} = CP_{50}$$

$$40 \times SP = 50 \times CP$$

$$\frac{SP}{CP} = \frac{50}{40} = \frac{5}{4}$$

$$\therefore \text{Profit} = \frac{5 - 4}{4} \times 100 = 25\%$$

④ A fruit seller buys 10 mangoes at 2 for a rupee & sells them at 3 for a rupee. His gain percent is?

$$\text{CP} = 10 \times \frac{1}{2} = 5 \text{ Rs} \quad \text{SP} = 10 \times \frac{2}{3} = \frac{20}{3} \text{ Rs}$$

$$\text{Profit} = \frac{20}{3} - 5 = \frac{5}{3} \text{ Rs}$$

$$\therefore \text{Profit} = \frac{5}{5} \times 100 = 100\%$$

① Boats & Streams (1-2 hrs)

Formula

B & S

Speed River & Speed Boat

$$P_s = x + y \quad \rightarrow \text{Downstream}$$

$$U_s = x - y \quad \rightarrow \text{Upstream}$$

① What will be the boat's speed in still water & speed of river, if the boat takes 2 hrs to row 48 km upstream & 3 hrs to row the same distance downstream?

Sol: Given, x = Speed in still water (In general, we consider the speed of boat in still water not in running water)

$$D = S \times T \quad \text{Upstream} \quad \text{Downstream}$$

$$48 = (x - y) \times 2 \quad 48 = (x + y) \times 3$$

$$24 = x - y \quad 48 = x + y$$

$$x - y = 24 \quad \text{--- (1)}$$

$$x + y = 48 \quad \text{--- (2)}$$

$$2x = 72 \quad \therefore x = 36 \text{ km/hr}$$

$$y = 12 \text{ km/hr}$$

NAME - Yashraj Singh

CLASS - XII

ROLL - 23 Sec - A

REGISTRATION NO. - 419231

JB - Physics

Host - Balamangal school

Q. Shipman takes twice as long to swim up as to swim down the river & has a speed of 12 km/hr in still water. What is river's speed?

$x = 12 \text{ m/hr}$ $y = ?$ as long = time

$$\begin{aligned} v_s &= 12 \\ v_s + y &= 2(v_s - y) \\ 12 + y &= 2(12 - y) \\ 12 + y &= 24 - 2y \\ 3y &= 12 \\ y &= 4 \end{aligned}$$

$t_u = 2 \times t_d$
 $\frac{d}{v_s - y} = 2 \times \frac{d}{v_s + y}$
 $\frac{1}{12 - y} = 2 \times \frac{1}{12 + y}$
 $12 + y = 2(12 - y)$
 $12 + y = 24 - 2y$
 $3y = 12$
 $y = 4$

[Hence, we always have to consider that $v_u = 2 \times v_d$ (as the distance of river is same)]

Q. It takes P 1 hr to row a place & come back. If the river is running at 2.4 m/hr & P has a speed of 12 m/hr in still water, what distance is the place from P's starting point?

$x = 12$, $y = 2.4$

Let the distance = d

Up: $P \rightarrow \text{Place}$
 Down: $\text{Place} \rightarrow P$

$t_u + t_d = 1 \text{ hr}$
 $\frac{d}{v_s - y} + \frac{d}{v_s + y} = 1$

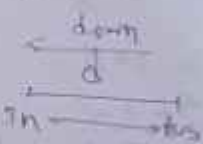
$$d \left[\frac{1}{v_s - y} + \frac{1}{v_s + y} \right] = 1$$

$\Rightarrow d \times \frac{2 \times 12}{12^2 - 2.4^2} = 1$
 $\Rightarrow d = 5.76 \text{ km}$

Q. An ocean current flows at a rate of 1.5 m/hr. A shark can swim in still water at the rate 4.5 m/hr. What is the average speed for the entire distance travelled if the shark swims from India to Australia & comes back?

Avg $s = \frac{\text{Total } d}{\text{total } t}$
 $\frac{d + d}{t_u + t_d}$

$t_u + t_d = \frac{d}{v_s - y} + \frac{d}{v_s + y}$
 $= d \left(\frac{1}{4.5 - 1.5} + \frac{1}{4.5 + 1.5} \right)$
 $= d + \frac{2 \times 4.5}{(4.5)^2 - (1.5)^2}$
 $= \frac{d}{2}$



$\frac{1}{4.5} + \frac{1}{2 \times 4.5} = \frac{1}{2 \times 4.5}$

From (1),
 Avg $s = \frac{2d}{d + 2} = 4 \text{ m/hr}$

NAME - Rajasree Isha
 CLASS - 10
 Roll - 25 Sec - A
 REGISTRATION No. - 1152315233
 28 - Physics
 Host - Balamangal High School

⑤ Raj takes 4 hrs more while swimming upstream than downstream. His speed in still water is 10 km/hr. The speed of stream is 2 km/hr. What is the distance?

$x = 10 \text{ km/hr}$, $y = 2 \text{ km/hr}$

$T_u = 4T_d$ [let dist = d]

$\frac{d}{x-y} = 4 + \frac{d}{x+y}$

$\frac{d}{10-2} = 4 + \frac{d}{10+2}$

$\frac{d}{8} = 4 + \frac{d}{12}$

$\frac{d}{8} - \frac{d}{12} = 4$

$\frac{3d - 2d}{24} = 4$

$\frac{d}{24} = 4$

$d = 96 \text{ km}$

⑥ Raj swims 20 km downstream in same time as 14 km upstream. What is his speed in still water if speed of stream is 3 km/hr?

$x = ?$, $y = 3 \text{ km/hr}$

$T_d = T_u$

$\frac{20}{x+3} = \frac{14}{x-3}$

$13x - 33 = 7x + 21$

$6x = 54$
 $x = 9 \text{ km/hr}$

⑦ Ratio of Gudi's swimming speed in still water to the speed of river is 7:1. She swims 4.2 km up the river in just 14 min. How much time will Gudi take to swim 18.4 km down the river?

$x:y = 7:1$

$T_u = 14 \text{ min}$, $d_u = 4.2 \text{ km}$

$T_d = ?$, $d_d = 18.4 \text{ km}$

$S_U = \frac{4.2}{14}$
 $= 0.3 \text{ km/min}$

$x - y = 0.3$

$7y - y = 0.3$

$6y = 0.3$

$y = \frac{0.3}{6}$

$y = 0.05 \text{ km/min}$

⑧ Find the ratio of swimming speed of Raj in still water to speed of river, if ratio of time taken to go 10 km upstream to time taken to go 10 km downstream is 11:5?

Why? $T_u : T_d = 11:5$ [$d = d_u = d_d = 10 \text{ km}$]

$\frac{T_u}{T_d} = \frac{11}{5}$

NAME - Rajat Singh
CLASS - 10
ROLL NO - 25 SEC - B
REGISTRATION NO. - 1155
18 - Physics
MOB - 9876543210

$$\frac{10x + 12y}{10x + 12y} = \frac{13}{2}$$

$$\frac{10x + 12y}{10x + 12y} = \frac{13}{2}$$

$$\frac{10x + 12y}{10x + 12y} = \frac{13}{2}$$

$$\frac{10x + 12y}{10x + 12y} = \frac{13}{2}$$

$$\frac{10x + 12y}{10x + 12y} = \frac{13}{2}$$

Q. A boat goes 11 km/hr with the stream & 5 km/hr against the stream. The speed of the boat in still water is 8 km/hr. What is the speed of the stream?

$$x + y = 11$$

$$x - y = 5$$

$$2x = 16$$

$$x = 8$$

$$y = 3$$

$$\frac{10x + 12y}{10x + 12y} = \frac{13}{2}$$

$$\frac{10x + 12y}{10x + 12y} = \frac{13}{2}$$

$$\frac{10x + 12y}{10x + 12y} = \frac{13}{2}$$

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$$\frac{10x + 12y}{10x + 12y} = \frac{13}{2}$$

$$x+y=12 \text{ --- (vi)}$$

It is, let's say, in eq (i)

$$\frac{-4x}{5} + \frac{+12}{5} = 13$$

$$\frac{-4x}{5} + \frac{12}{5} = 13$$

$$\frac{-4x}{5} = 8$$

$$-4x = 40$$

$$x = -10$$

$$x+y=12$$

$$-10+y=8$$

$$y=18$$

$$y=18$$

$$y=18$$

$$y=18$$

$$y=18$$

$$y=18$$

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$$y=18$$

$$y=18$$

$$y = 2.8 \text{ m/s}$$

$$y = 2.8 \text{ m/s}$$

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$$y = 2.8 \text{ m/s}$$

Exercise (4-5) (ms)

Remainders & factors

(marked)

$(5x^3 + 39x^2 + 17x + 42) \div (x+5)$, Rem?

$f(x) = (5x^3 + 39x^2 + 17x + 42) \div (x+5)$

$f(x) = 5x^2 + 29x + 17$

$5x^2 + 29x + 17$

$5x^2 + 29x + 17$

$5x^2 + 29x + 17$

$5x^2 + 29x + 17$

$5x^2 + 29x + 17$

Let's use the ans. is the here.

If absolutely, completely exactly these are mentioned, i.e. the remainder said that the dividend completely / exactly / absolutely divides the dividend, that means, Rem=0, otherwise Rem≠0.

(1) $(4x^3 + 13x^2 - 5x + 1) \div (x+4)$. Find the remainder.

Let $x = -4$

$4(-4)^3 + 13(-4)^2 - 5(-4) + 1$

$= -256 + 208 + 20 + 1$

$= -27 = \text{Rem}$

(2) $(5x^3 - 3x^2 + 17x + 42) \div (x+5)$. Find the value of 'c' if the remainder is 5.

$f(x) = 5x^3 - 3x^2 + 17x + 42$

Given, $x = -5$

$5(-5)^3 - 3(-5)^2 + 17(-5) + 42 = 5$

$-125 - 75 - 85 + 42 = 5$

$-243 = 5$

$-243 = 5$

$-243 = 5$

$-243 = 5$

$-243 = 5$

$-243 = 5$

$-243 = 5$

$-243 = 5$

NAME - Rajasree Latha

CLASS - XII

Roll - 123 Sec - A

Registration No - 419231582

18 - Physics

Host - Bahangan and's High

Q. The 9th term of the series $(x^{500} - 1) \div (x + 1)$ is then

$$\begin{array}{l} x^0 = 1 \\ x^1 = -1 \end{array} \quad \left| \begin{array}{l} x^{500} = 1 \\ x^{501} = -1 \end{array} \right. \quad \begin{array}{l} \text{Odd Powers} \\ \text{So } x^{500} = 1 \\ \text{Even Powers} \\ \text{then } + \end{array}$$

Q. $(3x^3 + 6x^2 + 4x) \div (x^2 - 4)$ leaves out remainder $2x$. Find the value of x in the expression

$$\begin{array}{l} x^2 - 4 = 0 \\ x^2 = 4 \\ 3x^3 + 6x^2 + 4x = 12x \quad (\text{given}) \\ 3x(4) + 6(4) + 4x = 12x \\ 12x + 24 + 4x = 12x \\ 16x + 24 = 0 \\ 16x = -24 \\ x = -\frac{3}{2} \end{array}$$

Q. $(x^{1001} + 5x^{999} + a)$ is completely divisible by $(x+1)$. Find the value of a .

$$\begin{array}{l} x+1=0 \\ x=-1 \end{array} \quad \left| \begin{array}{l} \text{Completely divisible, means remainder} \\ x^{1001} + 5x^{999} + a = 0 \\ (-1)^{1001} + 5(-1)^{999} + a = 0 \\ -1 - 5 + a = 0 \\ a = 6 \end{array} \right.$$

Q. If $5^x - 5^{x-1} = 500$. Find the value of x^x .

$$\begin{array}{l} 5^x - 5^{x-1} = 500 \\ 5^{x-1}(5 - 1) = 500 \\ 4 \cdot 5^{x-1} = 500 \\ 5^{x-1} = 125 \\ x-1 = 3 \\ x = 4 \\ x^x = 4^4 = 256 \end{array}$$

Algebra tips - the power of substitution

- Find the val. of lengthy algebraic expression
- Which 2 algebraic expressions are same?
- Insufficient no. of equations are given

Method: Substitution method

Assume the value of the variable $\rightarrow$ Substitute $\rightarrow$ Find answers.
(smaller val.) $\rightarrow$ (Larger val.)

- Assume such values that don't make the denominator zero

Eg. $\frac{1}{x-1} > 0$ we'll try to assume something that is smaller but not 1 (this will make $x > 0$)

NAME - Pooja Sree Isha
CLASS - XII
Roll - 23 Sec - A
Registration No. - 4192315230
18 - Physics
School - Baburam Vohra's High School

Q) If $x+y+z=0$, find the value of $x^3+y^3+z^3+3xyz$

$$x+y+z=0$$

$$\text{Let } y=0$$

$$x+0=0$$

$$x=-3$$

$$x^3+y^3+z^3+3xyz$$

$$= (-3)^3 + 0 + 0$$

$$= -27 + 27$$

$$= 0$$

Q) If $a+b+c=25$ & $a^2+b^2+c^2=225$, find the value of $4a^2b^2+c(a-b)$

$$a+b+c=25 \quad \text{--- (i)}$$

$$a^2+b^2+c^2=225 \quad \text{--- (ii)}$$

$$\therefore \text{from (i) \& (ii)}$$

$$a+b=25 \quad \text{--- (iii)}$$

$$a^2+b^2=225 \quad \text{--- (iv)}$$

$$\text{from (iii)}$$

$$(a+b)^2 = (a^2+b^2)+2ab$$

$$\Rightarrow 625 = (225) + 2ab$$

$$\Rightarrow 2ab = 400$$

$$\Rightarrow ab = 200$$

$$c=0 \text{ (def)}$$

$$4a^2b^2+c(a-b)$$

$$= 4a^2b^2$$

$$= (2ab)^2$$

$$= (400)^2$$

$$= 160000$$

Q) If $(x+2/a)^3 = (2a-2/a)^3$, find $(x^2+y^2+z^2)/(x^2+y^2)$

$$x = a + \frac{2}{a} \quad \text{--- (i)}$$

$$y = 2a - \frac{2}{a} \quad \text{--- (ii)}$$

3 variables, 1 extra

var = 0 (also, a is common)

let, $a=0$ (in both eqs)

$$\text{from (i), } x=3$$

$$\text{from (ii), } y=0$$

same algebra tips

$$\text{let } x + \frac{1}{x} = a, \text{ then } x^2 + \frac{1}{x^2} = ?$$

$$(x + \frac{1}{x})^2 = a^2$$

$$\Rightarrow x^2 + \frac{1}{x^2} + 2 = a^2$$

$$\Rightarrow x^2 + \frac{1}{x^2} = a^2 - 2$$

whenever will be the value of x^2 , just substitute

let, $x + \frac{1}{x} = 7$ (i.e. a=7)

$$x^2 + \frac{1}{x^2} = 49 - 2 = 47$$

$$\frac{(x^2+y^2)(y^2+z^2)}{(x^2+y^2)}$$

$$(x^2+y^2)$$

$$= -9 + 3/$$

$$(5-6)$$

$$= -1$$

$$\text{let } x - \frac{1}{x} = a, \text{ then } x^2 + \frac{1}{x^2} = ?$$

$$(x - \frac{1}{x})^2 = a^2$$

$$\Rightarrow x^2 + \frac{1}{x^2} - 2 = a^2$$

$$\Rightarrow x^2 + \frac{1}{x^2} = a^2 + 2$$

$$\text{let } a=7$$

$$\Rightarrow x^2 + \frac{1}{x^2} = 49 + 2 = 51$$

square both sides
to get the value
Greater Power (Retained)
= Square

NAME - Rajasree Laha
CLASS - 10
Roll - 23 SEC - A
REGISTRATION NO. - 41023152
18 - Physics
Hool - Baboragan girls' High

$$\Rightarrow x^2 + \frac{1}{x} = a^2, \quad x + \frac{1}{x} = ?$$

$$x^2 + \frac{1}{x} = a^2$$

$$\Rightarrow (x + \frac{1}{x})^2 = a^2$$

$$\Rightarrow (x + \frac{1}{x})^2 = 2 + \frac{1}{x^2} + x^2$$

$$\Rightarrow (x + \frac{1}{x})^2 = a^2 + 2$$

$$\Rightarrow x + \frac{1}{x} = \sqrt{a^2 + 2}$$

Whatever be the value of a , put it add it with 2 & root the result.

$$\text{let, } a^2 = 25$$

$$\Rightarrow x + \frac{1}{x} = \sqrt{27} = 3\sqrt{3}$$

(13) a) Find $x^2 + \frac{1}{x}$; if $x + \frac{1}{x} = \sqrt{47}$

$$(x + \frac{1}{x})^2 = 2 + \frac{1}{x^2} + x^2 = 47$$

$$\Rightarrow (x + \frac{1}{x})^2 = 47$$

$$\Rightarrow x^2 + \frac{1}{x^2} + 2 = 47$$

$$\Rightarrow x^2 + \frac{1}{x^2} = 45$$

$$\Rightarrow x^2 + \frac{1}{x} = a^2, \quad x + \frac{1}{x} = ?$$

$$(x + \frac{1}{x})^2 + 2 + \frac{1}{x^2} = a^2$$

$$\Rightarrow (x + \frac{1}{x})^2 = a^2 - 2$$

$$\Rightarrow x + \frac{1}{x} = \sqrt{a^2 - 2}$$

$$\text{let, } a^2 = 4$$

$$\Rightarrow x + \frac{1}{x} = \sqrt{4 - 2} = \sqrt{2}$$

Simplify the LHS & find the value (after simplifying it will be in form of $\sqrt{\text{something}}$)
Less than (not) = 0 equality

b) Find, $x^8 + \frac{1}{x^8}$ if $x + \frac{1}{x} = \sqrt{6}$

$$x + \frac{1}{x} = \sqrt{6}$$

$$x + \frac{1}{x} = \sqrt{6}$$

$$\Rightarrow x^2 + \frac{1}{x^2} + 2 = 6$$

$$\Rightarrow x^2 + \frac{1}{x^2} = 4$$

$$\Rightarrow (x^2 + \frac{1}{x^2})^2 = 16$$

$$\Rightarrow x^4 + \frac{1}{x^4} + 2 = 16$$

$$\Rightarrow x^4 + \frac{1}{x^4} = 14$$

$$\Rightarrow (x^2 + \frac{1}{x^2})^2 = (14)^2$$

$$\Rightarrow x^4 + \frac{1}{x^4} + 2 = 196$$

$$\Rightarrow x^4 + \frac{1}{x^4} = 194$$

b) $x^2 + \frac{1}{x^2} = 23$, find $x + \frac{1}{x}$

$$x^2 + \frac{1}{x^2} = 23$$

$$\Rightarrow (x + \frac{1}{x})^2 + 2 = 23$$

$$\Rightarrow (x + \frac{1}{x})^2 = 21$$

$$\Rightarrow (x + \frac{1}{x})^2 = 21$$

$$\Rightarrow x + \frac{1}{x} = \sqrt{21}$$

$$\Rightarrow (x + \frac{1}{x})^2 = 21$$

$$\Rightarrow x + \frac{1}{x} = \sqrt{21}$$

$$\Rightarrow x + \frac{1}{x} = \sqrt{21}$$

$$\Rightarrow (x + \frac{1}{x})^2 = 21$$

$$\Rightarrow x + \frac{1}{x} = \sqrt{21}$$

$$\Rightarrow (x + \frac{1}{x})^2 = 21$$

$$\Rightarrow x + \frac{1}{x} = \sqrt{21}$$

$$\Rightarrow x + \frac{1}{x} = \sqrt{21}$$

NAME - Rajuwar Loh
CLASS - 10
Roll - 25 Sec - A
REGISTRATION No. - 41923152
10 - Physics
Host - Bulungwa Girl's H.S.

Q20 If $u+y^2$, find $u^5 y^5$ in $2x^4 y^4 + 4y^3 - 2y$

$$u+y^2 = 1 \quad (1+1^2 = 2 \text{ var})$$

$$\text{Let } y \text{ (extra var)} = 0$$

$$n=1$$

$$= u^5 y^5 - 2x^4 + y^4 + x^3 y^3 - 2x^2 y + -2y$$

$$= 1 - 2$$

$$= -1$$

Q21 If $3x + 2y + 5\sqrt{7}$, find $1/x$

$$\frac{1}{x} = \frac{27 - 5\sqrt{7}}{(27)^2 - (5\sqrt{7})^2}$$

$$= \frac{27 - 5\sqrt{7}}{557}$$

Q14 Permutation & combination
(1-2 hrs)
Formulae (case)

• Factorial, $5! = 5 \times 4 \times 3 \times 2 \times 1$

$$n! = n \times (n-1) \times (n-2) \times \dots \times 3 \times 2 \times 1$$

• Let there are 7 things, & we want to arrange them in a single line, so in how many ways we can arrange this 7 things?

$$1111111 \rightarrow 7! \text{ ways}$$

for arranging 'n' things, we have $n!$ ways possible

• If we have 7 things, & we want to arrange them in a circular manner, then we have $(7-1)! = 6!$ ways possible.

• For arranging 'n' things in circular fashion, there are $(n-1)!$ ways possible

Formula (P & C)

2 concepts / 2 things
Select & arrange

Combination means Select

Permutation = Select & arrange

• 'n' things in a box we have to select 'r' things

$$\boxed{n}^r \Rightarrow {}^n C_r \text{ (combination / select)}$$

$${}^n C_r = \frac{n!}{r! (n-r)!} \quad [{}^n P_r = n]$$

[C = combination]

NAME - Rajasree Laha
CLASS - 10
Roll - 23 Sec - A
REGISTRATION NO. - 41525152
1B - Physics
Host - Balamangal girls High

• we have n things in a box.
we have to select r things i.e., there have to
arrange them.

$$\boxed{r} \text{ things} \Rightarrow {}^n P_r \text{ (permutation/arrange)}$$

$${}^n P_r = \frac{n!}{n!(n-r)!} \times r!$$

(count) (arranging r things in $r!$ ways)

$$\boxed{{}^n P_r = \frac{n!}{(n-r)!}} \text{ [Permutation]}$$

① In Jaya's bag there are 3 boxes of history
4 boxes of science & 2 boxes of maths.
For how many ways can Jaya arrange the
books so that all the books of same
subject are together?

select (list them)
arrange (is there)

∴ permutation

but it's not select & arrange, just arrange.
So instead of using P , we use normal '!'

(3H) (4S) (2M) 3 boxes
total = 3 boxes

3 boxes can be arranged in $3!$ ways (by doing

this, we will move each group, the individual
books will not be moved. By that we can keep
the books of same subject together.

now, in each group there are multiple no. of
books, they can be arranged as—

$$\begin{aligned} 3H &\rightarrow 3! \text{ ways} \\ 4S &\rightarrow 4! \\ 2M &\rightarrow 2! \end{aligned}$$

∴ total ways to arrange the books
so that books of same subj. remain together

$$TS = 3! \times 4! \times 2! \quad [\because 3! \text{ total ways \& in 4H 3 groups } 3! 4! 2!, \text{ so } 3!]$$

$$= 1728 \text{ ways}$$

② In how many ways can we arrange
the word ~~FUZ~~ 'FUZZTONE' so that all
the vowels come together?

F U Z Z T O N E
Vow (3)

total words = 8

word ③ → 1 imp → 3!

left alphabets = 8 - 3 = 5

F, Z, Z, T, N

5 alphabets + 1 imp = 6 → 6!

NAME - Vinay Kumar Jaha

CLASS - XII

ROLL - 23 SEC - A

ORISRATION No - 4152315

JB - Physics

Host - Bhubaneswar 751001

$\therefore$ 8 not should be $= 6! \times 2!$ but $\times$
 $\therefore$ 2 gets repeated twice

1, 2, 3, 4, 5

duplication (2 times)

total no. of ways to arrange that word so that all vowels come together

$$= \frac{6! 3!}{2!} = 2160 \text{ ways}$$

③ If suraj doesn't want vowels together then in how many can he arrange letters of the word 'MARKER'?

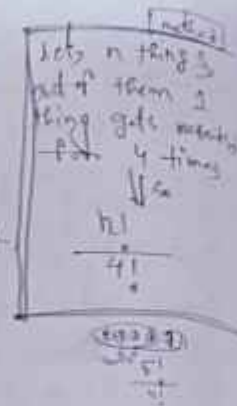
MARKER
 vowels (2)

total words = 6

vowels = 2 $\rightarrow$ 1st $\rightarrow$ 2!

1st 4 letters = 6 - 2 = 4 m, r, k, e

to arrange a way such that vowels come together 2!
 doesn't



total ways = vowels together + vowels not together

In general total ways = $n!$ but here the statement is such a way where vowels are together, so then the total ways are 15 = together + not together $\rightarrow$ 15

$\therefore$ vowels not together = ~~total~~ total ways - vowels together

$$= 6! (\text{total words}) - \text{vowels together}$$

$$2! (\text{duplication})$$

now, the no. of ways, by which the vowels remain together is,

$$4 + 1 \text{st} = 5 \rightarrow 5! \text{ ways}$$

$\therefore$ 2 vowels can be arranged in 2! ways

$$\therefore \frac{5! \times 2!}{2!} (\text{for repetition of e})$$

$$= 5! = 120 \text{ ways}$$

$\therefore$ no. of ways vowels are not together

$$= \frac{6!}{2!} - \frac{5! \times 2!}{2!}$$

$$= \frac{6! - 5! \times 2!}{2!}$$

$$= \frac{6!}{2!} - 120$$

$$= 360 - 120 = 240 \text{ ways}$$

NAME - Rajat Singh

CLASS - XII

Roll - 23 Sec - A

Registration No. - 415231583

18 - Physics

Hostel - Bahadurganj Dist. 3, Noida

④ without repetition, using digits 2, 3, 4, 5, 6, 7, 8, 9, how many no.s can be made which lie b/w 500 & 1000?

2 methods

(1) way
same logic

5 — (500-1000)

1, 2, 3, 4, 5, 6, 7, 8, 9

Now, no repetition,

5, 3, 4, 5, 6, 7, 8, 9

7 no.s

in b/w

5 is already used

∴ total no.s left now = 6

3 6 5 (possible choices)

same way

∴ $3 \times 6 \times 3 = 90$ no.s can be made in b/w

500 - 1000.

[If repetition will be allowed,
 $\frac{3}{5, 6, 7} \times \frac{1}{1} \times \frac{1}{1} = 147$]

way
formula based

eg ${}^9C_1 \times {}^8C_1 = 90$ no.s possible

6, 8

6 → select

When mentioned, 'weightage digits', select this digits etc, we'll understand it is combination.

⑤ 4 members form a group out of total 8 members. i) In how many ways it is possible to make the group if 2 particular members must be included. ii) In how many ways it is possible to make the group if 2 particular members must not be included?

total 8 members a b c d e f g h

i) must members = 2

∴ left members = 6 (∵ In 1 group 4 members)

now from 8 mems, 2 are selected

rem = 8 - 2 = 6 → from these we have to

select 2 mem members for the group

2	6
2	2
3	6
3	2

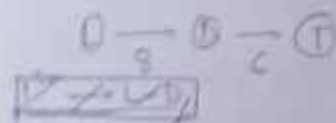
2 particular members can be selected by 1 way from rest of 6 mems, we are choosing 2 mems.

total no. of ways possible = $1 \times 6 \times 2$ (ways)
 = 12 ways

2 ways must not be included,
 as 2 = 0 $\rightarrow$ have to choose
 not 4 more (1 way)
 then

total no. of ways possible
 = 12
 = 12 ways

⑥ There are 8 routes from London to Delhi. And there are 2 routes from Delhi to Tokyo. In how many different ways can Raj travel from London to Tokyo via Delhi?



1st way L-D $\rightarrow$ ① choices from D-T
 and L-D $\rightarrow$ ② . . . D-T

etc. L-D $\rightarrow$ ⑧ . . . D-T
 = 16 choices

⑦ There are 35 people in a group. There are 12 school girls, 10 school boys, 5 senior citizens & 8 babies in the group. The organizer of the group wants to select a school girl or school boy as leaders of the group. In how many ways can he do so?

ways
 from a
 1 leader
 from 12 school girls / 10 school boys
 $12 + 10$

$\therefore$ total ways = $22 \times 1 = 22$

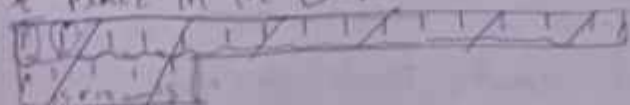
Logic

$12 + 10 = 22$
 43 is 6

1 leader
 out of them

to select 1 leader from
 22 people the organizer
 has 22 choices

⑧ In a class, there are 15 students. During a Christmas party all of them shook hands with elo only once. How many handshakes took place in the class?



NAME - Rajat Singh
 CLASS - 10

Roll No. - 110231823
 10 - Physics
 Host - Balamangal 10/11/2020

15 students in (4th standard)

② for a handshake

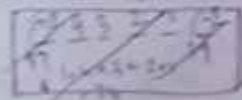
$$n = 15 + 7 = 105 \text{ handshakes } ({}^n_2)$$

[ok, there are 2 teams & they are playing match with each other only once, then, 2 (for a match 2 teams are must)]

• If 'only once' not given,

$$n = 15 \times 14 = 210 \text{ (not that imp)}$$

③ A bank has 6 digit account no. with no repetition of digits within an account no. The first & last digit of the account no. are fixed to be 4 & 7 - how many such account nos are possible?



0, 1, 2, 3, 5, 6, 8, 9

A B C D E F

$$\text{no. of accounts possible} = 8 \times 7 \times 6 \times 5 \times 4 \times 3 = 1680$$

④ A trekking group is to be formed having 6 members. They are to be selected from 3 girls, 4 boys & 5 teachers. In how many ways can the group be formed so that there are 3 teachers & 3 boys or 2 girls & 4 teachers?

$$(3t \& 3b) \quad 3t = 4b, 5t$$



(3t and 3b) OR (2g and 4t)

$$\rightarrow ({}^3_3 \times {}^4_3) + ({}^3_2 \times {}^5_4)$$

$$= (10 \times 4) + (3 \times 5)$$

$$= 40 + 15$$

$$= 55 \text{ ways}$$

⑤ In a railway line there are 20 stops. A ticket is needed to travel 1 or 2 stops. How many different tickets would the need to prepare to cater to all possibilities?



$$\text{no. of stops} = 20 (n)$$

$$\text{ticket is 1 or 2 stops (n)}$$

$$\text{no. of tickets needed} = {}^{20}_2 = 190$$

$$\text{traveling from A side} = 190 + ({}^{20}_2 \times 2) = 190 + 190 = 380$$

total tickets = 380 (more) tickets.
 (12) 17 students are present in a class. In how many ways can they be made to stand in 2 circles of 8 & 9 students?

17 students
 8 9

1st → select 9 students & arrange them

select = ${}^{17}C_9$ & arrange = $(9-1)! = 8!$

2nd → arrange the rest of students

Rest = $17-9 = 8$ S

arrange = $(8-1)7!$

total no. of ways = ${}^{17}C_9 \times 8! \times 7!$

By (2nd way)

$$= \frac{17!}{9!} \times 7! = 17! \times \frac{7!}{9!}$$

1st → select 8 students & arrange them

${}^{17}C_8 = \frac{17!}{9!} \times 8! \times 7!$

2nd → arrange rest of the students

Rest = $17-8 = 9$ S

arrange = $(9-1)8!$

total no. of ways = $\frac{17!}{9!} \times 7!$ — (1st way)

total no. of ways = (1st way) OR (2nd way)

$$= \frac{17!}{9!} \times 7! + \frac{17!}{9!} \times 8!$$

$$= \frac{17!}{9!} (7! + 8!)$$

[1st, linear fashion is mentioned]

$17S \rightarrow 8 \quad 9$

1st → select 9 S & arrange = ${}^{17}P_9 = \frac{17!}{8!}$

2nd → arrange rest of 8 S = $8!$

total ways = $\frac{17!}{8!} \times 8! = 17!$

1st → select 8 S & arrange = ${}^{17}P_8 = \frac{17!}{9!}$

2nd → arrange rest 9 S = $9!$

total ways = $\frac{17!}{9!} \times 9! = 17!$

(13) A locker in bank has 3 digit lock. Mahesh forgot his password & was trying all possible combinations. He took 6 seconds for each try. The problem was that each digit can be from 0-9 how much time will be needed to try all of the combinations?

NAME - Rajasree Isha
 ASS - 10
 ALL - 23 SEC - A
 ILLUSTRATION No. - 4152315233
 UB - Physics
 Host - Balamangal High School

$$\left[\begin{array}{c} 10 \\ 10 \\ 10 \end{array} \right] \rightarrow 0-9$$

1000 possible ways

$$\frac{10}{1} \cdot \frac{10}{1} \cdot \frac{10}{1} = 10 \cdot 10 \cdot 10 = 1000$$

1000 possible ways

$$1 \text{ try (1 way)} \rightarrow 65$$

$$1000 \times (1 \text{ way}) = 65 \times 1000 = 65000$$

$$\frac{65000}{65} = 1000 \text{ to try all combination}$$

14) In a room there are 2 green chairs, 3 yellow chairs & 4 blue chairs. In how many ways can Raj choose 3 chairs so that atleast 1 yellow chair is included?

2G, 3Y, 4B

1 Y chair	2 Y chair	3 Y chair
$1 \text{ Y} \cdot 2 \text{ others} = {}^3C_1 \cdot {}^6C_2 = 3 \cdot 15 = 45$	$2 \text{ Y} \cdot 1 \text{ other} = {}^3C_2 \cdot {}^6C_1 = 3 \cdot 6 = 18$	$3 \text{ Y} = {}^3C_3 = 1$

$$\therefore \text{no. of ways possible} = 1 \text{ Y on 2 Y on 3 Y} \\ = 45 + 18 + 1 = 64 \text{ ways}$$

① total ways = choosing atleast 1 chain + choosing no. of chain

$$\begin{aligned} \text{choosing atleast 1 chain} &= \text{total ways} - \text{choosing no. of chain} \\ &= {}^9C_3 - {}^6C_3 \\ &= 84 - 20 \\ &= 64 \text{ ways} \end{aligned}$$

② How many combinations are possible while selecting 4 letters from the word 'smokejack' with the condition that J must appear in it? smokejack

1 will be J must (no choice)

(last arranging isn't given, given is selecting)

$$\begin{aligned} &= {}^6C_1 + {}^5C_2 + {}^4C_3 \\ &= 6 + 15 + 4 = 25 \end{aligned}$$

NAME - Rajasree Laha
ASS - 311
ALL - 23 SEC - A
REGISTRATION No. - 41923158
VA - Physics
HOD - Balaraman sir's 11/5/20

(15) Averages (At least 290)

Things

• Equitable distribution (eg. 100 RS & 5 children if you want equal distribution then
 $\rightarrow 20/-$ $4 \rightarrow 20/-$
 $3 \rightarrow 20/-$ $5 \rightarrow 20/-$
 $\therefore$ Avg amt ($\frac{100}{5}$ RS) with each ($= 20/-$)

eg

1) D.M. & D.M.
 1st 100/- 2nd 25/- 3rd 75/-

$$\text{Avg} = \frac{100 + 25 + 75}{3} = \frac{200}{3} = 66.67$$

100/- each 100/- 100/- (Avg amt with each person)

Avg = $\frac{\text{Sum of all the observations}}{\text{total observations}}$	$\frac{\text{Sum}}{\text{total}}$
--	-----------------------------------

ii) Batsman $\rightarrow 20$ m (total)

total score $\rightarrow 750$ (total sum)

$$\therefore \text{avg} = \frac{750}{20} = 37.5 \text{ per match}$$

• Total sum or sum

$$\text{F.D. or Avg} = \frac{\text{Sum}}{\text{Total no.}}$$

$$\Rightarrow \text{Sum} = \text{avg} \times \text{total no.}$$

eg
 1 2 3 4 5
 150 100 50 20 1000

total sum (adding) here
 $150 + 100 + 50 + 20 + 1000 = 1320$

eg 4325

1 2 3
 10 15 20

total age = 15 + 25 + 40 = 80 years

• total sum = nothing but addition

Q knowing that Vijay's expenditure for first 3 days is RS. 100, RS. 125 & RS. 85, what is his 4th day expenditure as his 4 days avg expenditure RS. 90?

$$\text{Avg} = 90$$

$$\therefore \text{Avg} = \frac{\text{Sum}}{\text{total no.}}$$

$$\rightarrow 90 = \frac{100 + 125 + 85 + x}{4}$$

$$\therefore x = 50/-$$

Q what will be average price of all the goods bought, if Ajay buys 30 crasens for RS. 3 each, 35 chocolates for RS. 10 each & RS. 4 clips at the rate of RS. 4 each?

$$1 \text{ craser} \rightarrow 3/-$$

$$30 \text{ } \rightarrow 3 \times 30 = 90/-$$

$$1 \text{ chocolate} \rightarrow 10/-$$

$$35 \text{ } \rightarrow 10 \times 35 = 350/-$$

$$S_1 = 41$$

$$S_2 = 42.5$$

$$Avg = \frac{41 + 42.5}{2} = 41.75$$

$$\frac{41 + 42.5}{2} = 41.75$$

$$Avg = 41.75$$

③ Of the 20 cycles sold by Ajay, average cost of 12 cycles is Rs. 18000. In total he earned Rs. 3,00,000. What was the average cost of remaining cycles?

$$S_1 = 18000$$

$$S_2 = 300000$$

total earned 3,00,000

$$18000 \times 12 = 216000$$

$$S = \frac{216000}{12} = 18000$$

From 12 cycles he earned 2,16,000/-
Rest amt he earned from 8 cycles = 3,00,000 - 2,16,000 = 84,000/-

$$S_1 = 84000$$

$$S_2 = 84000$$

$$T = 8$$

$$Avg \text{ cost of 8 cycles} = \frac{84000}{8} = 10500/-$$

④ Without considering the salary of the boss, the average salary received by Rs. 1000. What will be salary of boss if avg salary of 11 employees & boss is Rs. 18,000?

$$S_1 = 18000$$

$$T = 11 \text{ emp \& boss}$$

$$Avg = \frac{S}{T}$$

$$Avg \text{ of 12 people} = 18000$$

$$S = \frac{S}{T}$$

$$18000 = \frac{S}{12}$$

$$S = 216000$$

$$\therefore \text{Boss's sal} = S_2 - S_1$$

$$= 216000 - 198000$$

$$= 18000/-$$

Excluding boss's sal
Avg of 11 people = 17,000

$$\frac{S}{11} = 17000$$

$$S = 17000 \times 11$$

$$S = 187000$$

⑤ Avg age of 5 people is 42 years. Another group has 3 people who have avg age of 81 years. When both groups are mixed, what is the avg age of all people?

$$\begin{aligned} \text{Avg}_5 &= \frac{S_1}{5} & \text{Avg}_3 &= \frac{S_2}{3} \\ \Rightarrow 42 &= \frac{S_1}{5} & \Rightarrow 81 &= \frac{S_2}{3} \\ \Rightarrow S_1 &= 210 \text{ yrs} & \Rightarrow S_2 &= 243 \text{ yrs} \end{aligned}$$

Both groups are mixed,

$$\begin{aligned} \text{New avg mixed} &= \frac{210 + 243}{8} \\ &= \frac{453}{8} \\ &= 56.625 \end{aligned}$$

⑥ 3 boxes have some avg weight, when one box which weights 80 kg is replaced by another box, the avg weight increase by 5 kg, how much the new box weight?

$$\begin{aligned} \text{Avg}_3 &= \frac{S_1}{3} & \text{Avg}_{\text{new}} &= \frac{S_2}{3} \\ \Rightarrow \text{Avg}_3 &= \frac{80}{3} & \Rightarrow \text{Avg}_3 + 5 &= \frac{S_2}{3} \\ \Rightarrow S_1 &= 80 + 10 & \Rightarrow \frac{80 + 10}{3} + 5 &= \frac{S_2}{3} \\ & & \Rightarrow 80 + 10 + 15 &= S_2 + 15 \\ & & \Rightarrow S_2 &= 104 \text{ kg} \end{aligned}$$

Formula: new avg = $\frac{\text{existing avg} \times \text{no. of people} + \text{new avg} \times \text{no. of people}}{\text{total no. of people}}$

Example: If avg age of 5 people is 42 and avg age of 3 people is 81, then the avg age of 8 people is 56.625.

⑦ How old will be if ratio of his age to his son's is 11:2 & avg age of his & his both grandsons is 50 years?

Raj's age : His grandsons age = 11:2

$$\begin{aligned} \text{R's age} &= 11x \\ \text{His age} &= 2x \end{aligned}$$

$$\begin{aligned} \text{Avg A of his &\& \text{his both grandsons} &= \frac{S_1 + S_2 + S_3}{3} \\ \Rightarrow 50 &= \frac{11x + 2x + 2x}{3} \end{aligned}$$

$$\begin{aligned} \Rightarrow 150 &= 15x \\ \Rightarrow x &= 10 \text{ yrs} \end{aligned}$$

$$\begin{aligned} \therefore \text{R's age} &= 11 \times 10 \\ &= 110 \text{ yrs} \end{aligned}$$

NAME - Rajasree Lohar
CLASS - XII
ROLL - 23 SEC - A
REGISTRATION No. - 419231523
SUB - Physics
School - Baburagan girls' High

Q Had try scored 18 runs more in his 3rd innings & 4 runs more in his 7th innings, his avg would have become 66 runs. But it is 64 runs. How many runs did he play?

Let total innings = x

$$\begin{aligned} \text{Avg} &= 64 \\ \therefore 64 &= \frac{S}{x} \end{aligned} \quad \text{--- (1)}$$

In 3rd = 18 runs
7th = 4 runs

$$\begin{aligned} \text{Avg} &= 66 \\ \therefore 66 &= \frac{S + 22}{x} \end{aligned}$$

$$\begin{aligned} \therefore 66 &= \frac{64x + 22}{x} \\ \therefore 66x + 22 &= 64x \\ \therefore 2x &= -22 \\ \therefore x &= 11 \text{ runs} \end{aligned}$$

Q In a group of people, the oldest & the youngest have an age difference of 100 years. If these 2 are left out of counting, then the avg age of the remaining 40 people is 28. The avg age of entire group being 30, how old is the eldest person?

$$\begin{aligned} \text{Let oldest} &= x, \text{ youngest} = y \text{ years} \\ x - y &= 100 \Rightarrow y = x - 100 \text{ --- (1)} \\ \text{Total age of 40 people} &= 40 \times 28 = 1120 \\ \text{Total age of 42 people} &= 42 \times 30 = 1260 \end{aligned}$$

Let x be the age of the oldest person.

$$30 = \frac{S + x + y}{42}$$

$$30 \times 42 = 1120 + x + y \quad \text{(from (1))}$$

$$1260 = 1120 + x + y - 100 \quad \text{(from (1))}$$

$$1260 - 1120 = 2x - 100$$

$$140 = 2x - 100$$

$$220 = 2x$$

$$x = 110 \text{ years}$$

Q A batsman played 11 innings & has a certain average. This avg increases by 2 runs when his innings of 32 runs, 33 runs & 34 runs are replaced by 3 innings. Find the average of these 3 runs.

$$\begin{aligned} \text{Avg} &= \frac{32 + S + 33 + 34}{11} \\ \text{Avg} + 2 &= \frac{S + 99}{9} \end{aligned}$$

$$\begin{aligned} \therefore \text{Avg} &= \frac{S + 99}{9} \\ \therefore \text{Avg} + 2 &= \frac{S + 99}{9} + 2 \\ \therefore \text{Avg} + 2 &= \frac{S + 117}{9} \end{aligned}$$

NAME - Rajasree Latha
CLASS - 10
ROLL - 23 SEC - A
REGISTRATION No. - 41923/523
SUB - Physics
SCHOOL - Balaraman Girls High

④ Series (2-3 hrs)

Types & tips

- numeric
- letter/alphabet
- Alpha-numeric

Ans types

- next no/letter/group of nos/2 or of alphabet
- missing no/letters/... etc.
- odd mark out (And the odd no/letters like that)

Tips

- ① Find the relation b/w the given elem.
- ② solve the relation/m & get the ans.

1 2 3 4 5 6
1st 2nd 3rd 4th 5th 6th

Pattern → Using it we can solve the m.

eg 1 2 3 4 5 6

1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20 21 22 23 24 25 26 27 28 29 30 31 32 33 34 35 36 37 38 39 40 41 42 43 44 45 46 47 48 49 50 51 52 53 54 55 56 57 58 59 60 61 62 63 64 65 66 67 68 69 70 71 72 73 74 75 76 77 78 79 80 81 82 83 84 85 86 87 88 89 90 91 92 93 94 95 96 97 98 99 100

The series (s) must be related to sq of no, cube of numbers, multiplication etc.

eg 1, 3, 7, 10, 15 (increasing but slowly increasing)

The series must be related to addition, subtraction, mul, div. The series must be of prime nos. Fibonacci series.

1st type

Numeric series

i) Addition series

eg 10, 25, 32, 40 ?
25-10 = 15
32-25 = 7
40-32 = 8
15+7 = 22
22+8 = 30
30+10 = 40
40+15 = 55
55+22 = 77
77+30 = 107
107+40 = 147
147+55 = 202
202+77 = 279
279+107 = 386
386+147 = 533
533+202 = 735
735+279 = 1014
1014+386 = 1400
1400+533 = 1933
1933+735 = 2668
2668+1014 = 3682
3682+1400 = 5082
5082+1933 = 7015
7015+2668 = 9683
9683+3682 = 13365
13365+5082 = 18447
18447+7015 = 25462
25462+9683 = 35145
35145+13365 = 48510
48510+18447 = 66957
66957+25462 = 92419
92419+35145 = 127564
127564+48510 = 176074
176074+66957 = 243031
243031+92419 = 335450
335450+127564 = 463014
463014+176074 = 639088
639088+243031 = 882119
882119+335450 = 1217569
1217569+463014 = 1680583
1680583+639088 = 2319671
2319671+882119 = 3201790
3201790+1217569 = 4419359
4419359+1680583 = 6100942
6100942+2319671 = 8420613
8420613+3201790 = 11622403
11622403+4419359 = 16041762
16041762+6100942 = 22142704
22142704+8420613 = 30563317
30563317+11622403 = 42185720
42185720+16041762 = 58227482
58227482+22142704 = 80370186
80370186+30563317 = 110933503
110933503+42185720 = 153119223
153119223+58227482 = 211346705
211346705+80370186 = 291716891
291716891+110933503 = 402650394
402650394+153119223 = 555769617
555769617+211346705 = 767116322
767116322+291716891 = 1058833213
1058833213+402650394 = 1461483607
1461483607+555769617 = 2017253224
2017253224+767116322 = 2784369546
2784369546+1058833213 = 3843202759
3843202759+1461483607 = 5304686366
5304686366+2017253224 = 7321939590
7321939590+2784369546 = 10106309136
10106309136+3843202759 = 13949511895
13949511895+5304686366 = 19254198261
19254198261+7321939590 = 26576137851
26576137851+10106309136 = 36682446987
36682446987+13949511895 = 50631958882
50631958882+19254198261 = 69886157143
69886157143+26576137851 = 96462295094
96462295094+36682446987 = 133144742081
133144742081+50631958882 = 183776700963
183776700963+69886157143 = 253662858106
253662858106+96462295094 = 350125153200
350125153200+133144742081 = 483269895281
483269895281+183776700963 = 667046596244
667046596244+253662858106 = 920709454350
920709454350+350125153200 = 1270834607550
1270834607550+483269895281 = 1754104502831
1754104502831+667046596244 = 2421151099075
2421151099075+920709454350 = 3341860553425
3341860553425+1270834607550 = 4612695160975
4612695160975+1754104502831 = 6366799663806
6366799663806+2421151099075 = 8787950762881
8787950762881+3341860553425 = 12129811316306
12129811316306+4612695160975 = 16742506477281
16742506477281+6366799663806 = 23109306141087
23109306141087+8787950762881 = 31897256903968
31897256903968+12129811316306 = 44027068220274
44027068220274+16742506477281 = 60769574697555
60769574697555+23109306141087 = 83878880838642
83878880838642+31897256903968 = 115776137742610
115776137742610+44027068220274 = 159803205962884
159803205962884+60769574697555 = 220572780660439
220572780660439+83878880838642 = 304451661499081
304451661499081+115776137742610 = 420227799241691
420227799241691+159803205962884 = 580031005204575
580031005204575+220572780660439 = 800603785865014
800603785865014+304451661499081 = 1105055447364095
1105055447364095+420227799241691 = 1525283246605786
1525283246605786+580031005204575 = 2105314251810361
2105314251810361+800603785865014 = 2905918037675375
2905918037675375+1105055447364095 = 4010973485039470
4010973485039470+1525283246605786 = 5536256731645256
5536256731645256+2105314251810361 = 7641570983455617
7641570983455617+2905918037675375 = 10547489021130992
10547489021130992+4010973485039470 = 14558462506170462
14558462506170462+5536256731645256 = 20094719237815718
20094719237815718+7641570983455617 = 27736290221271335
27736290221271335+10547489021130992 = 38283779242402327
38283779242402327+14558462506170462 = 52842241748572789
52842241748572789+20094719237815718 = 72936960986388507
72936960986388507+27736290221271335 = 100673251207659842
100673251207659842+38283779242402327 = 138957030450062169
138957030450062169+52842241748572789 = 191799272198634958
191799272198634958+72936960986388507 = 264736233185023465
264736233185023465+100673251207659842 = 365409484392683307
365409484392683307+138957030450062169 = 484366514842745476
484366514842745476+191799272198634958 = 676165787041380434
676165787041380434+264736233185023465 = 940902020226403900
940902020226403900+365409484392683307 = 1306311504619087207
1306311504619087207+484366514842745476 = 1790678019461832683
1790678019461832683+676165787041380434 = 2466843806503213117
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3407745826729617017+1306311504619087207 = 4714057331348704224
4714057331348704224+1790678019461832683 = 6504735350810536907
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12379324984043367041+4714057331348704224 = 17093382315392071265
17093382315392071265+6504735350810536907 = 23598117666202608172
23598117666202608172+8971579157313750024 = 32569696823516358196
32569696823516358196+12379324984043367041 = 44949021807559725237
44949021807559725237+17093382315392071265 = 62042404122951806502
62042404122951806502+23598117666202608172 = 85640521789154414674
85640521789154414674+32569696823516358196 = 118210218612670772870
118210218612670772870+44949021807559725237 = 163159240420230498107
163159240420230498107+62042404122951806502 = 225201644543182304609
225201644543182304609+85640521789154414674 = 310842166332336719283
310842166332336719283+118210218612670772870 = 429052384945007492153
429052384945007492153+163159240420230498107 = 592211625365237990260
592211625365237990260+225201644543182304609 = 817413269908420294869
817413269908420294869+310842166332336719283 = 1128255436240656914152
1128255436240656914152+429052384945007492153 = 1557307821185664406305
1557307821185664406305+592211625365237990260 = 2149519446550902396565
2149519446550902396565+817413269908420294869 = 2966932716459322691434
2966932716459322691434+1128255436240656914152 = 4095188152700079605586
4095188152700079605586+1557307821185664406305 = 5652495973885744011891
5652495973885744011891+2149519446550902396565 = 7802015420436646408456
7802015420436646408456+2966932716459322691434 = 10768948136895969100890
10768948136895969100890+4095188152700079605586 = 14864136289595048706476
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28318647683917439126823+10768948136895969100890 = 39087595820813408227713
39087595820813408227713+14864136289595048706476 = 53951732110408456934189
53951732110408456934189+20516632263480792718367 = 74468364373889249652556
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102787012057806688779379+39087595820813408227713 = 141874607878620097007092
141874607878620097007092+53951732110408456934189 = 195826339989028553941281
195826339989028553941281+74468364373889249652556 = 270294704362917803593837
270294704362917803593837+102787012057806688779379 = 373081716420724492373216
373081716420724492373216+141874607878620097007092 = 514956324299344589380308
514956324299344589380308+195826339989028553941281 = 710782664288373143321589
710782664288373143321589+270294704362917803593837 = 981077368651290946915426
981077368651290946915426+373081716420724492373216 = 1354159085072015439288642
1354159085072015439288642+514956324299344589380308 = 1869115409371359928668950
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6784249936754399386863557+2579898073659733071990539 = 9364148010414132458854096
9364148010414132458854096+3560975442311024018905965 = 12925123452725156477760061
12925123452725156477760061+4915134527383039458194607 = 17840257980108195935954668
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24624507916862595322818225+9364148010414132458854096 = 33988655927276727781672321
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225035017944360667754708028+89378545276972675518205275 = 314413563221333343272913303
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823096762954204901886819905+314413563221333343272913303 = 1137510326175538245209733208
1137510326175538245209733208+437780764425583746572790899 = 1575291090601121991782524107
1575291090601121991782524107+598061745009834334132111877 = 2173352835610956325914635984
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4233959924740699473011189097+1575291090601121991782524107 = 5809251015341821464793713204
5809251015341821464793713204+2173352835610956325914635984 = 7982603850952777790708349188
7982603850952777790708349188+3096449598565161227801455889 = 11079053449518339018509805077
11079053449518339018509805077+4233959924740699473011189097 = 15313013374259038491520994174
15313013374259038491520994174+5809251015341821464793713204 = 21122264389600859956314707378
21122264389600859956314707378+7982603850952777790708349188 = 29104868240553637747023056566
29104868240553637747023056566+11079053449518339018509805077 = 40183921690071976765532861643
40183921690071976765532861643+15313013374259038491520994174 = 55496935064331015257053855817
55496935064331015257053855817+21122264389600859956314707378 = 76619199453931875213368563195
76619199453931875213368563195+29104868240553637747023056566 = 105724067694485512960391619761
105724067694485512960391619761+4233959924740699473011189097 = 148063667942126207690503510758
148063667942126207690503510758+5809251015341821464793713204 = 196156218957468029155297223762
196156218957468029155297223762+7982603850952777790708349188 = 275982257466995807062380715650
275982257466995807062380715650+11079053449518339018509805077 = 386772791912514146247388520727
386772791912514146247388520727+15313013374259038491520994174 = 539904071103765557868927462471
539904071103765557868927462471+21122264389600859956314707378 = 751126334999766457422074536249
751126334999766457422074536249+29104868240553637747023056566 = 1042171203240320095169097591815
1042171203240320095169097591815+4233959924740699473011189097 = 1465567195687727089870216501712
1465567195687727089870216501712+5809251015341821464793713204 = 20464922972218492363495

It is increasing very quickly $\Rightarrow$ It can't be an A.P.

1 3 6 10 15 ? = 210

? = 110

1 3 6 10 15 21 28 36 ? = 192

③ Division series

2 16 72 36 12 ? 2

It is falling very quickly $\Rightarrow$ It might be an D.P.

2 16 72 36 12 ? 2

? = 6

④ Special numeric series (mostly asked)

① Series (upto 30) ② Prime no. series ③ Composite no. series
④ Cubes (upto 10) ⑤ Fibonacci series

⑥ Mean to it

⑦ Square series / no. near to it

1 0 3 8 15 24 35 ? 63

④ 48 [Study branching - not to do]

0 3 8 15 24 35 ? 63
1-1 4-1 9-1 16-1 25-1 36-1 49-1
2-1 3-1 4-1 5-1 6-1 7-1 8-1

2-1 = 4-1 = 1

= 72

= 48

4 (9 of series)

When there will be a series where pattern is not directly (we can't directly recognize the pattern, then we need to apply tricks of square, cube & mean to it. 'mean to it' means if the no. near to it or mean to square or mean to cube be like that).

⑤ Cube Series / no. near to it

6 12 5 216 343

⑥ Prime no. series

2, 3, 5, 7, 11, 13, 17, 19, ... (prime no.)

⑦ Composite no. series $\rightarrow$ the no. that are not prime

4, 6, 8, 9, 10, 12, ... (composite no. series)

Fibonacci series

0, 1, 1, 2, 3, 5, 8, 13, 21, ...
(Fibonacci series)

2, 3, 5, 8, 13, 21, 34, 55, ... (Fibonacci series)

3, 5, 8, 13, 21, 34, 55, ... (not fibo series)

1, 2, 3, 5, 8, 13, 21, 34, 55, 89, 144, ...
(not increasing / decreasing - it is mixed up)

There are 2 series / 2 new writing on a primary are going on

1, 2, 3, 5, 8, 13, 21, 34, 55, 89, 144, ...

2, 3, 5, 8, 13, 21, 34, 55, 89, 144, ...

Alphabet Series

A B C D E F G H I J K L M N O P Q R S T U V W X Y Z

rev. pos. of letters

2 + 20 = 22

- 1

1st rev. pos. of D $\rightarrow 22 - 4 = 18$

Alphabet series (2 types)

words & constants

alphabet partition (eg. D)

(total of 26 letters)
- original pos. of letter

2 types $\rightarrow$ Single alphabet letters

1st 2nd 3rd

1st 2nd 3rd of letters on alphabet

1st 2nd 3rd of letters on alphabet

1st 2nd 3rd

1st 2nd 3rd

1st 2nd 3rd

1st 2nd 3rd

1st 2nd 3rd

1st 2nd 3rd

1st 2nd 3rd

1st 2nd 3rd

1st 2nd 3rd

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1st 2nd 3rd

1st 2nd 3rd

1st 2nd 3rd

1st 2nd 3rd

1st 2nd 3rd

1st 2nd 3rd

1st 2nd 3rd

Alphanumeric series

A B C D E F G H I J K L M N O P Q R S T U V W X Y Z

1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20 21 22 23 24 25 26

Reverse pos. of letters - 26 - original pos. of letter

NAME - Rajasree Laksh

CLASS - 10

Roll - 25 SEC - A

REGISTRATION No - 4152315231

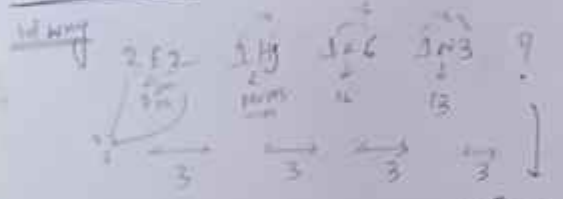
SUB - Physics

School - Balarama vidyalaya High

pld. name
 1993 143 306 183 ?

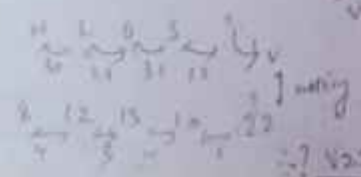


2 wgs / logic



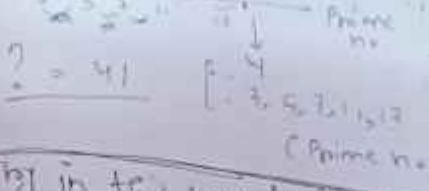
∴ ? = 180

148 112 015 516 ?



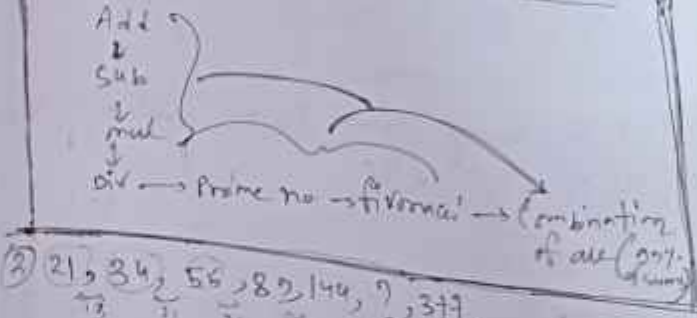
∴ ? = 822

① 2, 5, 10, 17, 28, ?



? = 41

Try in this way to find out the pattern



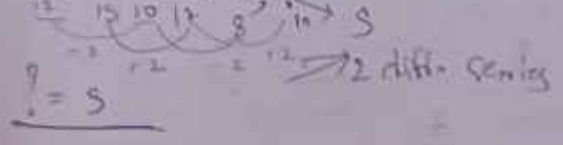
② 21, 34, 56, 82, 144, ? 319

? = 233

③ 21, 34, 56, 82, 144

When this happens then always remember it is a Fibonacci series.

④ 1, 0, 3, 9, 14 ?



? = 5

NAME - Rajasree John
 CLASS - XII
 ROLL - 23 SEC - A
 REGISTRATION NO. - 4192315205
 SUB - Physics
 SCHOOL - Babington Matric High Sch

Check rule
line diagram?

① U, S, P, 2, 6, ?

$$9 = A$$

② U, S, P, 1, 4, 9, 16, 25, 36, 49, 64, 81, 100, 121, 144, 169, 196, 225, 256, 289, 324, 361, 400, 441, 484, 529, 576, 625, 676, 729, 784, 841, 900, 961, 1024, 1089, 1156, 1225, 1296, 1369, 1444, 1521, 1600, 1681, 1764, 1849, 1936, 2025, 2116, 2209, 2304, 2401, 2500, 2601, 2704, 2809, 2916, 3025, 3136, 3249, 3364, 3481, 3600, 3721, 3844, 3969, 4096, 4225, 4356, 4489, 4624, 4761, 4900, 5041, 5184, 5329, 5476, 5625, 5776, 5929, 6084, 6241, 6400, 6561, 6724, 6889, 7056, 7225, 7396, 7569, 7744, 7921, 8100, 8281, 8464, 8649, 8836, 9025, 9216, 9409, 9604, 9801, 10000

③ Find the wrong term in the following series:
10, 14, 28, 40, 64, 88, 130

Wrong term = 34 (as if $28 + 16 = 44$
 $28 + 4 = 32$
4 - doubling)
Correct series = 10, 14, 28, 32, 44, 68, 136

④ Find the wrong term in the following series:
15, 16, 20, 30, 45, 70

Wrong term = 30 (Correct series)
15, 16, 20, 28, 45, 70

⑤ Find the wrong term in the following series:
13, 80, 21, 63, 129

13, 80, 21, 63, 129

Correct series
1, 3, 10, 21, 64, 129

Wrong term = 63

⑥ Find the wrong term in the following series:
3, 2, 8, 12, 18, 24, 18, 32, 23, 42

3, 2, 8, 12, 18, 24, 18, 32, 23, 42

Wrong term = 24

Correct series
3, 2, 8, 12, 18, 24, 32, 42

NAME - Rajasree Laha
CLASS - XII
ROLL - 43 SEC - A
REGISTRATION NO - 419231583
SUB - Physics
School - Baborang Girls High

2. Partnership (2 qns)

Partnership formula

$$\text{Ratio of Investment} \times \text{Time} = \text{Ratio of Profit}$$

Eg -
$$\frac{(A's \text{ investment} \times A's \text{ Time})}{(B's \text{ investment} \times B's \text{ Time})} = \frac{A's \text{ Profit}}{B's \text{ Profit}}$$

Q. P & Q jointly buy a hotel. P invests 6 times the investment of Q. On receiving profit of RS. 84,000 what will be P's share in it?
P & Q → Invested for the same time period

If P's amount ↑ Profit P ↑
Invest

Q → Profit Q ↑

Given, P invests 6 times the inv. of Q
∴ Profit of P = 6 times Profit of Q
let Q = x let P = 6x ✓

$$\frac{P}{Q} = \frac{6}{1} \quad P:Q = 6:1$$

$$P's \text{ share} = \frac{6}{6+1} \times 84,000 \text{ (total profit)}$$

$$= \frac{6}{7} \times 84,000$$

$$B's \text{ share} = \frac{1}{7} \times 84,000$$

$$= 12,000/-$$

$$\frac{1}{7} \times 84,000 = 12,000/-$$

Q. Rishi starts a travel agency by investing Rs. 10,000. After 4 months Raj joins the agency. Raj invests Rs. 5000. What will be profit share of Raj if they earn a total of Rs. 18,700 as profit in the entire year?

$$\text{Formula} = \frac{I \times T}{P}$$

Rishi's part: 10000×12
 Raj's part: 5000×8 [as mentioned, they earned a total of Rs. 18,700 as profit in the entire year]

Ratio of profit share of Raj:
 $\frac{10000 \times 12}{10000 \times 12 + 5000 \times 8} = \frac{120000}{180000} = \frac{4}{3}$

Q. M invests Rs. 400 in teaching classes. After 3 months N comes along with Rs. 800. P joins in with Rs. 20,000. The classes earn a profit of Rs. 11,200 at the end of the year. If profits of M, N & P are in ratio of 6:5:5, for how many months was P involved?

$I \times T = P$

$P_M : P_N : P_P = 6:5:5$

$400 \times 12 : 800 \times 3 : 20000 \times T = 6:5:5$ (given)

actual values

Let, common factor = K

Investment

Time

Profit

$$\text{method values} = 6K, 5K, 5K$$

$$400 \times 12 : 800 \times 3 : 20000 \times T = 6K : 5K : 5K$$

$$48000 : 24000 : 20000T = 6K : 5K : 5K$$

$$48000 = 6K \quad \text{actual val.} \quad \text{actual val.}$$

$$\Rightarrow K = 8000$$

$$20000T = 5 \times 8000$$

$$\Rightarrow T = 2 \text{ months (as rest of units (pro, 100) are in months)}$$

Q. Ratio of profits of P, Q & R are 5:8:3 respectively. What is their investment ratio if their investment time period ratio is 2:5:8 respectively?

$$I \times T = P$$

$$\Rightarrow \frac{I}{T} = \frac{P}{T}$$

$$I = 5 : 8 : 3$$

$$T = 2 : 5 : 8$$

$$I_P : I_Q : I_R$$

$$= \frac{P_P}{T_P} : \frac{P_Q}{T_Q} : \frac{P_R}{T_R}$$

$$= \frac{5}{2} : \frac{8}{5} : \frac{3}{8}$$

$$= \frac{5}{2} : \frac{8}{5} : \frac{3}{8}$$

$$\text{method 1}$$

$$\text{So, LCM}(2, 5, 8) = 40$$

$$= \frac{5}{2} \times 40 : \frac{8}{5} \times 40 : \frac{3}{8} \times 40$$

$$= 100 : 64 : 15$$

$$\text{method-2 (nice)}$$

$$\frac{5}{2} : \frac{8}{5} : \frac{3}{8}$$

$$= \frac{5 \times 40}{2 \times 40} : \frac{8 \times 40}{5 \times 40} : \frac{3 \times 40}{8 \times 40}$$

$$= 100 : 64 : 15$$

$$= 100 : 64 : 15$$

$$\text{fraction but we need whole}$$

$$\text{not fractions}$$

$$\text{is also}$$

$$\text{but it will take more time in mod as HCF is greater than LCM (general case) here} \rightarrow \text{mine}$$

Q. Ajay & Vijay's investment is in ratio 5:2. They get Rs. 3420 as profit after donating 10% of total profit to orphanage. What is Vijay's share?
 $2+T=P$ (As T is not given, some consider that their time periods are same, some need not to consider T)

$$I = P$$

$$I_A = I_V \text{ (given)}$$

$$5:2$$

$$\text{As } I = P, \therefore P_A : P_V = 5:2$$

$$\text{Now, A's share} = \frac{5}{7} \text{ of total profit}$$

$$3420 = \frac{5}{7} \times T \times P$$

$$7P = \frac{3420 \times 7}{5}$$

$$= 4788/-$$

$$\therefore V's \text{ share} = \frac{2}{7} \times 4788 \text{ (Ans)}$$

$$= 684 \times 2$$

$$= 1368/-$$

$$\text{[Overall Profit]} = 4788 + 1368 = 6156 \text{ (Ans)}$$

$$T = \frac{4788}{5} = 957.6$$

$$957.6 \times 5 = 4788$$

$$K = 5320/- \text{ (not needed)}$$

Ans

$$\left[\begin{array}{l} I = P \\ \therefore \frac{I_A}{I_V} = \frac{P_A}{P_V} \end{array} \right] \Rightarrow \frac{5}{2} = \frac{3420}{P_V} \Rightarrow P_V = 1368/-$$



Q. B & P start a bus. B & P invest equally but P invests one-fifth of the total investment amount on getting Profit of Rs. 22,500. How much would be difference between Profit of P & B? No T is given, $\therefore$ same T
 if $I \neq P$

$$I = \frac{1}{5} P$$

P invests $\frac{1}{5}$ of total investment amount

$\therefore$ Profit of P will be also $\frac{1}{5}$ of total profit

$$\therefore P's \text{ Profit} = \frac{1}{5} \times 225000$$

(Profit)

Ans
 As the amount given is the same, profit will be same (in percent)

$$= 45000/-$$

$$I_B = I_P$$

$$P_B = P_A$$

$$\text{Now Profit left} = 225000 - 45000 = 180000/-$$

This will be divided equally among B & P (As they invested same amt, they will get same profit)

$$\therefore P's \text{ Profit} = \frac{180000}{2} = B's \text{ Profit}$$

$$= 90000/-$$

Diff. b/w P_B & P_P

$$P_B - P_P (= P_B - P_P) = 90000 - 45000$$

$$= 45000/-$$

$\therefore P_B - P_P$ (Ans) means P's Profit is 45,000 less than P's Profit

⑦ P & S start a cafe with capital RS 20,000 & RS 40,000 respectively. After a year, out of the profit of RS 15,000, P gets his share of profit. P also some money as his salary. In total, P gets RS 7,000. What is the salary amount he received?

$$I:P = P$$

$$P: S \quad (\text{As in 1 year both P & S is same for both P & S})$$

$$= 20,000 : 40,000$$

$$= 1:2$$

$$= 1:2$$

$$P: S = 1:2$$

then - out of the profit of RS 15,000/-

i.e., total profit = RS 15,000/-

$$P's \text{ share of profit} = \frac{1}{3} \times 15,000$$

$$= 5,000/-$$

then,

P's share + some money = Salary of P = 7,000

$$\rightarrow 5,000 + \text{salary of P} = 7,000$$

$$\rightarrow P's \text{ sal} = 2,000/-$$

- Active Partner = manager, who gets profit along with sal.
- Silent Partner = Investment partner (who put the money in the business)

⑧ RS 1500 were invested by Ram & Shyam together to start a small business. They got a profit of RS 200 at the end of the year. Shyam took his profit share of RS 60. How much did Ram invest?

$$I:P = P$$

$$P_{\text{Ram}} : P_{\text{Shyam}}$$

$$= R \times 12 : S \times 12$$

$$\text{invested} \quad \text{invested}$$

$$R \quad S$$

$$\text{then } P_S = 600/-$$

$$\text{total profit} = 2000/-$$

$$\therefore P_R = 2000 - 600$$

$$= 1400/-$$

$$\therefore P : P_S = 600 : 1400$$

$$= 6 : 14 = 3 : 7$$

$$\therefore P : P_S = 3 : 7$$

$$\therefore P : S = 3 : 7$$

$$\therefore \text{Investment ratio } P : S = 3 : 7$$

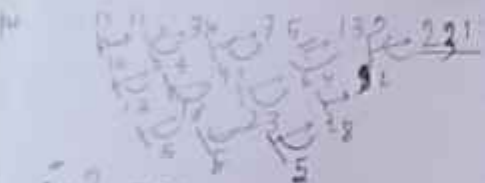
$$\therefore P's \text{ invest} = \frac{3}{10} \times 1500 \quad (\text{total investment})$$

$$= 450/-$$

Here, in profit share, Shyam's share of profit = 600/- (which is his share of investment)

② Imp. (choice)
 1, 1, 34, 75, 130?

• Irregular method



③ 28, 33, 31, ?, 34, 30
 28, 33, 31, ?, 34, 30
 28, 33, 31, ?, 34, 30

∴ ? = 36

④ Probability (2-3 Ans)

P = chance

P = what I want
 Total

eg - 4R & 2G Balls

$$P(\text{Picking a R ball}) = \frac{4}{6} = \frac{2}{3}$$

- MD → + (mul)
- OR → + (add)
- Total P = 1 (Calculus)

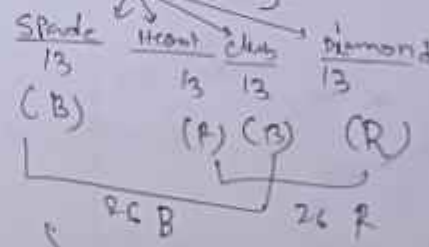
✓ why
 Same as Cwe can only get 2 type of balls from 2-type of balls
 $R + P_G = \frac{4}{6} + \frac{2}{6} = 1$

$$P + \bar{P} = 1$$

Common types

i) coin → H/T 2 possibilities

ii) Cards → 52 cards 52



$$\text{Card no} = 2 - 10$$

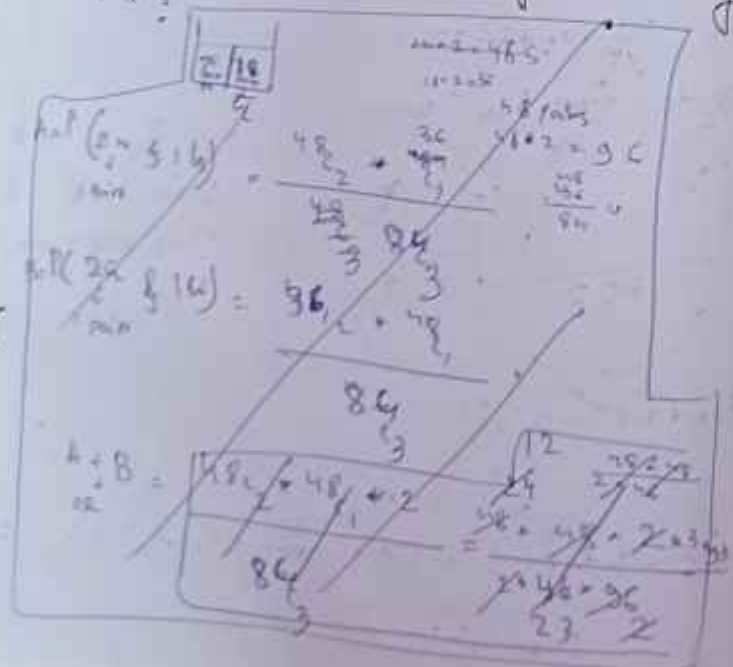
J, K, Q, A

3) dice → 6 faces → each faces has 1, 2, 3, 4, 5, 6
 6 possibilities

G. (1) 50 Socks, balls, marbles, shirts etc. (colours)

P Possibilities depend upon no. of s, b, m, & c
& i.e.
Case by Case.

Q. Suresh keeps all his socks in a single drawer. He has 24 Pairs of white socks & 18 Pairs of Grey Socks. Suresh picks 3 socks randomly. Find the Prob. of Suresh choosing a matching pair?



$$= \frac{24C2 + 18C2}{24C3} = \frac{276 + 153}{2024} = \frac{429}{2024} = \frac{33}{156} = \frac{11}{52}$$

$$A = P(2W1B) = \frac{24C2 \cdot 18C1}{24C3}$$

$$B = P(2G1W) = \frac{18C2 \cdot 24C1}{24C3}$$

$$A+B = \frac{24C2 \cdot 18C1 + 18C2 \cdot 24C1}{24C3} = \frac{40608 + 30240}{2024} = \frac{70848}{2024} = \frac{432}{127}$$

$$\frac{4000}{10000} = 40\% \quad \frac{1}{2} = 50\%$$

What ever
shows he
chooses in
any time, he
always gets
something he

Always forms
(always) (short quantity)

the task is 1

What will be the probability of drawing a blue
on a slide from a well shuffled standard
deck of 52 playing cards?

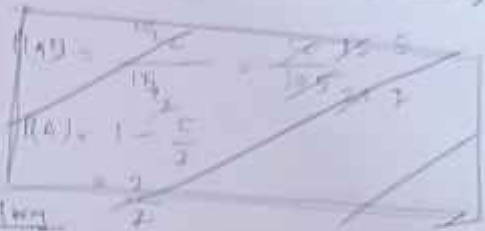
$$\frac{13 \times 4}{52} = \frac{52}{52} = 1$$

(blue & black
are mutually
exclusive)

③ A box has 6 blue, 4 red, 2 white & 3
blue shirts. When 2 shirts are picked
randomly, what is the probability that at least
one is white or both are blue?

$$60, 40, 20, 30$$

$$P(A) = \frac{10}{100} = \frac{1}{10}$$



2 shirts → 1st & 2nd

24 OR 28

$$\left(\frac{24}{100} \times \frac{23}{99} \right) + \left(\frac{28}{100} \times \frac{27}{99} \right)$$

$$= \frac{2}{10 \times 11} + \frac{6}{10 \times 11}$$

$$= \frac{8}{10 \times 11}$$

$$= \frac{4}{55}$$

$$P(A) = \frac{24}{100} + \frac{28}{100}$$

$$= \frac{52}{100}$$

④ A pot has 24, 68, 42 & 8 mm balls.
If one ball is picked randomly from
the pot, what is the prob. of it being less
or greater? 24, 42, 68

$$P(A) = \frac{S_1 + S_2}{S_1 + S_2 + S_3 + S_4}$$

$$= \frac{24 + 42}{24 + 42 + 68 + 8}$$

$$= \frac{66}{142}$$

$$= \frac{33}{71}$$

⑤ There are 2 pots. One pot has 5 red & 3 green
marbles other has 4 red & 2 green marbles.
What is the probability of drawing a red
marble?

1. 1st pot 25
2. 2nd pot 25
3. 3rd pot 25
4. 4th pot 25
5. 5th pot 25
6. 6th pot 25
7. 7th pot 25
8. 8th pot 25
9. 9th pot 25
10. 10th pot 25

$$P(A) = \frac{5}{8} + \frac{4}{10}$$

$$= \frac{5}{8} + \frac{2}{5}$$

$$= \frac{15 + 8}{24}$$

$$= \frac{23}{24}$$

$$P(B) = \frac{1}{2} = \frac{12}{24}$$

But we have to consider
the prob. of choosing
a 1st pot.

Let, we are choosing 1st
& drawing a red marble.

$$P(A) = \frac{1}{2} \times \frac{5}{8}$$

1st pot 25

$$P(A \text{ OR } B) = P(A) + P(B)$$

$$= \frac{1}{2} \times \frac{5}{8} + \frac{1}{2} \times \frac{4}{10}$$

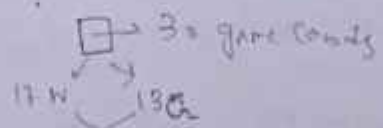
$$= \frac{1}{2} \left(\frac{5}{8} + \frac{2}{5} \right)$$

$$= \frac{1}{2} \times \frac{15 + 8}{24}$$

$$= \frac{1}{2} \times \frac{23}{24}$$

$$= \frac{23}{48}$$

⑥ In a set of 30 game cards, 17 are white
& rest are green. 4 white & 5 green are
marked IMPORTANT. If a card is chosen
randomly from this set, what is the probability
of choosing a green card or an 'important'
card?



IMPORTANT cards = 4W & 5G

A = green card, B = important

$$P(A) = \frac{13}{30}, P(B) = \frac{9}{30}$$

$$P(A \text{ OR } B) = P(A) + P(B) - P(A \cap B)$$

$$= \frac{13}{30} + \frac{9}{30} - \frac{4}{30}$$

$$= \frac{18}{30} = \frac{3}{5}$$

$$\frac{13}{30} + \frac{5}{30} + \frac{4}{30} + \frac{5}{30} - \frac{5}{30}$$

$$A \text{ of } 13 \rightarrow A \cup B \rightarrow A \cap B$$

$$P(A \cup B) = P(A) + P(B) - P(A \cap B) \quad \text{if } A \text{ \& } B \text{ are mutually ex.}$$

$$= \frac{13}{30} + \frac{5}{30} - \frac{5}{30} \quad \text{Answer 7p}$$

$$= \frac{13}{30}$$

Algebra has $P(A \cup B) = P(A) + P(B) - P(A \cap B)$
 $P(A \cap B) = 0$ if then A & B are mutually exclusive i.e. they are happening independent, there is no common.
 logical reason, $P(\text{red in bag} \rightarrow \text{white})$ doing it here, if we are drawing 1 card, it can happen that it's imp, same as while choosing 1st card, it can happen that it's green or red or white i.e. we are adding, same event is added for 2 times, so we are counting 1 time to eliminate that double counting.

⑦ A box has 6 black lined, 2 white & 3 blue shirts find the probability of drawing 2 black shirts if they are picked randomly?

$$\begin{bmatrix} 6B \\ 2W \\ 3BL \end{bmatrix}$$

15 shirts

Counting is told about replacement $\rightarrow$ No replacement

$$P(A) = \frac{6C_2}{15C_2} \rightarrow \frac{6C_2}{15C_2} \quad \text{[they haven't told that they are picked together so normally they say!]}$$

$$= \frac{6 \times 5}{15 \times 14}$$

$$= \frac{1}{7}$$

⑧ A box has 6 black lined, 2 white & 3 blue shirts. What is the probability that 2 red shirts & 1 blue shirt get chosen during a random selection of 3 shirts from the box?

$$\begin{bmatrix} 6B \\ 2W \\ 3BL \end{bmatrix}$$

15 shirts

$$P(A) = \frac{6C_2 \times 3C_1}{15C_3}$$

$$= \frac{15 \times 3}{350}$$

$$= \frac{3}{70}$$

$$\frac{6C_2 \times 3C_1}{15C_3} = \frac{15 \times 3}{350} = \frac{3}{70}$$

$$= \frac{4C_1}{15C_2} = \frac{3 \cdot 1 \cdot 2 \cdot 1}{15 \cdot 14 \cdot 13} = \frac{6}{2730} = \frac{1}{455}$$

$$\frac{3C_1 + 3C_2 + 3C_3}{15 \cdot 14 \cdot 13}$$

$$= \frac{12 \cdot 3C_1}{15 \cdot 14 \cdot 13}$$

$$= \frac{12 \cdot 1}{15 \cdot 14 \cdot 13}$$

$$= \frac{12}{2730}$$

Q. 10

Given 3 random selection, then we can find the probability of getting 3 shirts of 2 random colors.

$$P(A) = \frac{4C_2 \cdot 3C_1}{15C_3}$$

$$= \frac{6 \cdot 3}{455}$$

$$= \frac{18}{455}$$

$$= \frac{18}{455}$$

Given, finding the no. of ways to select 3 shirts of 2 colors.
 $\rightarrow$ 1st shirt $\rightarrow$ 4 ways
 $\rightarrow$ 2nd shirt $\rightarrow$ 3 ways
 $\rightarrow$ 3rd shirt $\rightarrow$ 2 ways

Q. 11 A box has 6 blue, 4 red, 2 white & 3 green shirts. What is the probability of picking at least 1 red shirt in 4 shirts that are randomly picked?

$$\frac{4C_1}{15C_4}$$

15 shirts

$$= \frac{4C_1 + 3C_1 + 2C_1 + 1C_1}{15C_4} = \frac{4 + 3 + 2 + 1}{1365} = \frac{10}{1365}$$

$$\frac{11 \cdot 10 \cdot 9 \cdot 8}{4!}$$

$$= \frac{7920}{24} = 330$$

$$= \frac{330}{1365}$$

$$= \frac{22}{91}$$

$$= \frac{22}{91}$$

$$= \frac{22}{91}$$

$$= \frac{22}{91}$$

$$= \frac{22}{91}$$

$$= \frac{22}{91}$$

$$= \frac{22}{91}$$

$$= \frac{22}{91}$$

$$= \frac{22}{91}$$

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$$= \frac{22}{91}$$

$$= \frac{22}{91}$$

$$= \frac{22}{91}$$

$$= \frac{22}{91}$$

$$= \frac{22}{91}$$

$$= \frac{22}{91}$$

Q. 12 Picking 4 shirts from 15 shirts.

$$P(A) = \frac{15C_4}{15C_4} = \frac{1365}{1365} = 1$$

$$P(B) = 1 - \frac{22}{91} = \frac{69}{91}$$

$$P(A \cap B) = \frac{69}{91}$$

⑥ On rolling a dice 2 times, the sum of 2 numbers that appear on the uppermost face is 8. What is the probability that the first throw of dice yields 4?

$$P(A) = \frac{1}{6} \text{ first 4 and 4}$$

$$= \frac{1}{6} \times \frac{1}{6}$$

$$= \frac{1}{36}$$

⑦ A box has 5 black & 3 green shirts. one shirt is picked randomly & put in another box. The second box has 3 black & 5 green shirts. Now a shirt is picked from 2nd box. What is the probability of it being a black shirt?

$$\left[\begin{array}{c} 5B \\ 3G \end{array} \right] \rightarrow \left[\begin{array}{c} 3B \\ 5G \end{array} \right]$$

Shirts Shirts

$$P(A) = \text{Picking 1B shirt from 1st box} \times \text{Picking 3B shirt from 2nd box}$$

$$= \frac{5}{8} \times \frac{3}{8} = \frac{15}{64}$$

$$P(B) = \text{Picking 1G shirt from 1st box} \times \text{Picking 5G shirt from 2nd box}$$

$$= \frac{3}{8} \times \frac{5}{8}$$

$$= \frac{15}{64}$$

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

$$= \frac{5}{8} + \frac{3}{8} - 0 \quad [\because A \text{ \& } B \text{ are mutually exclusive, i.e they are independent events}]$$

$$= \frac{8}{8} = 1$$

$$\frac{3}{2} \left[\frac{8}{2} \right]$$

⑧ What is the possibility of having 53 Thursdays in a non-leap year?

No leap year $\rightarrow$ Feb - 28 days

Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
1	1	1	1	1	1	1	1	1	1	1	1
2	2	2	2	2	2	2	2	2	2	2	2
3	3	3	3	3	3	3	3	3	3	3	3
4	4	4	4	4	4	4	4	4	4	4	4
5	5	5	5	5	5	5	5	5	5	5	5
6	6	6	6	6	6	6	6	6	6	6	6
7	7	7	7	7	7	7	7	7	7	7	7
8	8	8	8	8	8	8	8	8	8	8	8
9	9	9	9	9	9	9	9	9	9	9	9
10	10	10	10	10	10	10	10	10	10	10	10
11	11	11	11	11	11	11	11	11	11	11	11
12	12	12	12	12	12	12	12	12	12	12	12
13	13	13	13	13	13	13	13	13	13	13	13
14	14	14	14	14	14	14	14	14	14	14	14
15	15	15	15	15	15	15	15	15	15	15	15
16	16	16	16	16	16	16	16	16	16	16	16
17	17	17	17	17	17	17	17	17	17	17	17
18	18	18	18	18	18	18	18	18	18	18	18
19	19	19	19	19	19	19	19	19	19	19	19
20	20	20	20	20	20	20	20	20	20	20	20
21	21	21	21	21	21	21	21	21	21	21	21
22	22	22	22	22	22	22	22	22	22	22	22
23	23	23	23	23	23	23	23	23	23	23	23
24	24	24	24	24	24	24	24	24	24	24	24
25	25	25	25	25	25	25	25	25	25	25	25
26	26	26	26	26	26	26	26	26	26	26	26
27	27	27	27	27	27	27	27	27	27	27	27
28	28	28	28	28	28	28	28	28	28	28	28
29											
30											
31											

non leap year

365 days

52 weeks + 1 day

1 day

$$P(A) = \frac{1}{7}$$

[∵ from 52 weeks we are given 52 Thursdays, then over a day left, we can have 51 Thursdays. If that 1 day is Thurs, the p. of Thurs day is there is $\frac{1}{7}$]

Prob. of

① In a leap year, what is the prob. of getting the left 2 days as Tue-Wed?
366 days → 52 weeks & 2 days

∴ Prob. of the given event = $\frac{1}{7}$

7 Poss.

(Sat-Sun)
(Sun-Mon)
(Mon-Tue)
(Tue-Wed)
(Wed-Thu)
(Thu-Fri)
(Fri-Sat)

② In a drawer there are 4 white socks, 3 blue socks & 5 grey socks. Two socks are picked randomly. What is the probability for both the socks are same color?

4 white socks
3 blue socks
5 grey socks

$$P(A) = \frac{4}{12} \times \frac{3}{11} + \frac{3}{12} \times \frac{2}{11} + \frac{5}{12} \times \frac{4}{11}$$

$$= \frac{12 + 6 + 20}{12 \times 11}$$

[∵ no 2 socks selected from each color]
[No white socks are selected]

$$= \frac{38}{66}$$

③ Both socks are not same color

$$= \frac{4}{12} \times \frac{8}{11} + \frac{3}{12} \times \frac{4}{11} + \frac{5}{12} \times \frac{7}{11}$$

$$= \frac{32 + 12 + 35}{12 \times 11}$$

$$= \frac{84}{132} = \frac{7}{11}$$

$$P(A) = 1 - \frac{38}{66} = \frac{19}{33}$$

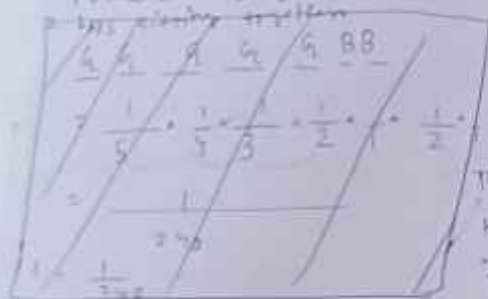
④ What is the probability of drawing two cards from a well shuffled pack of 52 cards?

$$P(A) = \frac{13}{52} \times \frac{12}{51}$$

$$= \frac{1}{17}$$

[nothing is mentioned so, no replacement & step selecting]

(6) What are the chances that no two boys are sitting together for a photograph if there are 5 girls & 2 boys?



5G + 2B
= 7 children

The no. of ways we can arrange 7 children = $7!$

The no. of ways 2 boys are sitting together

~~5G + 2B~~ 5G + 1B = 6

total ways = $6! \times 2!$

$\bar{A}$ = 2 boys are not sitting together

$\bar{A} = \text{total} - \text{total [i.e. 1]}$ [from Prob. of Prob. sitting together & not sitting together = total]

$A = \text{total} - \text{sitting together}$
 $= 7! - 6! \times 2!$
 $= 6! \times 5$

$$P(A) = \frac{6! \times 5}{7!} = \frac{5}{7}$$

(7) (i) (ii) (iii) (iv) (v) (vi) (vii) (viii) (ix) (x) (xi) (xii) (xiii) (xiv) (xv) (xvi) (xvii) (xviii) (xix) (xx) (xxi) (xxii) (xxiii) (xxiv) (xxv) (xxvi) (xxvii) (xxviii) (xxix) (xxx)

$A = 2$ boys sitting together

$$P(A) = 1 - P(\bar{A})$$

= 1 - (2 boys sitting together)

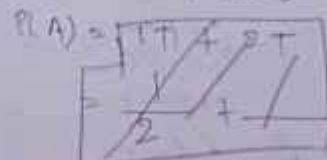
$$= 1 - \frac{6! \times 2!}{7!} \quad \left(\frac{\text{total we want}}{\text{total}} \right)$$

$$= \frac{7! - 6! \times 2!}{7!}$$

$$= \frac{6! \times 5}{7!}$$

$$= \frac{5}{7}$$

(8) When 2 coins are tossed simultaneously, what are the chances of getting at least one tail?



$$P(A) = \frac{1T + 2T}{4}$$

$$= \frac{1 \times \frac{1}{2} + 2 \times \frac{1}{4}}{1} = \frac{1}{2} + \frac{1}{2} = 1$$

no need

(9) (i) (ii) (iii) (iv) (v) (vi) (vii) (viii) (ix) (x) (xi) (xii) (xiii) (xiv) (xv) (xvi) (xvii) (xviii) (xix) (xx) (xxi) (xxii) (xxiii) (xxiv) (xxv) (xxvi) (xxvii) (xxviii) (xxix) (xxx)

$$P(A) = 1 - P(\bar{A}) = 1 - \frac{1}{4} = \frac{3}{4}$$

② Pipes & tanks = (H, N), (1, 2), (11, 2), (7, 2)

$$r(A) = \frac{3}{4}$$

⑩ Pipes & cisterns (1-2 hrs)

Tips & tricks

- Just invert (the time & work)

$$N \rightarrow \frac{1}{N}$$

eg Pipe fills tank in 8 hrs,
 $\therefore$ then in 1 hrs $\rightarrow \frac{1}{8}$ part of tank is filled (just like time & work)

- When it invert

$$T \xrightarrow{\text{invert}} n$$

$$n \rightarrow T$$

- whenever we are removing water we subtract
 removing $\rightarrow$ subtract

① Two pipes m & n can fill a tank in 22 hrs & 33 hrs respectively. In how much time will the tank be full, if both the pipes are opened simultaneously?
 $m \rightarrow \frac{1}{22}$ hrs $\frac{1}{22}$ part of tank is filled

$n \rightarrow 33$ hrs $\frac{1}{33}$ part of tank is filled

Both pipes are opened simultaneously.

$\therefore$ In 1 hrs

$$\begin{aligned} & \frac{1}{22} + \frac{1}{33} \quad \left(m \text{ can fill the tank in } 22 \text{ hrs, } n \text{ can fill the tank in } 33 \text{ hrs} \right) \\ & = \frac{33+22}{22 \times 33} \\ & = \frac{55}{22 \times 33} \\ & = \frac{5}{22 \times 3} \\ & = \frac{5}{66} \text{ part of tank will be full} \end{aligned}$$

The tank will be full in $\frac{66}{5}$ hrs, if both are opened simultaneously

② Pipe P can fill a tank in 38 hours. Pipe R can empty the full tank in 13 hours. If all the pipes are opened together, how much time will be needed to make the tank full?

P $\rightarrow$ Fill $\frac{1}{38}$ hrs $\frac{1}{38}$ part of tank will be filled

R $\rightarrow$ 13 $\rightarrow \frac{1}{13}$

R $\rightarrow$ empty $\frac{1}{13}$ hrs $\frac{1}{13}$ part of tank will be empty

If all of the pipes are opened,

$$\begin{aligned} & P + R \\ & = \frac{1}{38} - \frac{1}{13} = \frac{13-38}{494} = \frac{-25}{494} \end{aligned}$$

$$\frac{38 \times 13}{2 \times 13 \times 38} = \frac{1}{2}$$

$$= \frac{7+10+2}{18 \times 2}$$

$$= \frac{19}{36}$$

$\frac{19}{36}$ amt of tank will be filled in 1 hr.

the tank will be full in 14 hrs.

③ A tap can fill a tub in 24 hrs. Due to leak at the bottom of the tub, the tap fills the tub in 36 hours. If the tub is full, how much time will the leak take to empty it?

Tap $\frac{\text{fill}}{24 \text{ hrs}}$ $\frac{1 \text{ hr}}{24}$ amt of the tub is filled

at the leak $36 \times \frac{1}{36}$

$\frac{1}{24} - \frac{1}{36} = \frac{1}{36}$ (let the leak take empty it in x hrs)

$$\frac{1}{x} = \frac{1}{24} - \frac{1}{36}$$

$$= \frac{36 - 24}{36 \times 24}$$

$$= \frac{12}{576}$$

$$= \frac{1}{48}$$

$x = 48 \text{ hrs}$

④ P & Q filling the total amt of tank in 1 hr



④ Two pipes M & N can fill a tank in 22.5 & 15 minutes, respectively. If both the pipes are opened simultaneously, after how much time should N be closed so that the tank is full in 18 minutes?

M $\frac{\text{fill}}{22.5 \text{ mins}}$ $\frac{1 \text{ min}}{22.5}$ amt of tank is filled
N $\rightarrow 15$ $\frac{1}{15}$



When M & N were opened together, they together can fill the tank,

M	N
22.5	15
150	150
22.5	15
300	75
300	22.5
3	22.5
6.75	

$$= \frac{10}{22.5} + \frac{1}{15}$$

$$= \frac{30 + 45}{45}$$

$$= \frac{75}{45}$$

$$= \frac{5}{3}$$

$$\frac{22.5}{36} = \frac{1}{3}$$

$$\frac{1}{3} = \frac{1}{3}$$

$$\frac{1}{3} = \frac{1}{3}$$

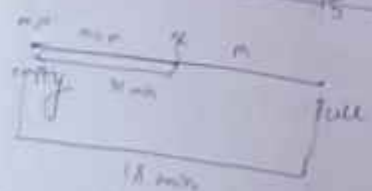
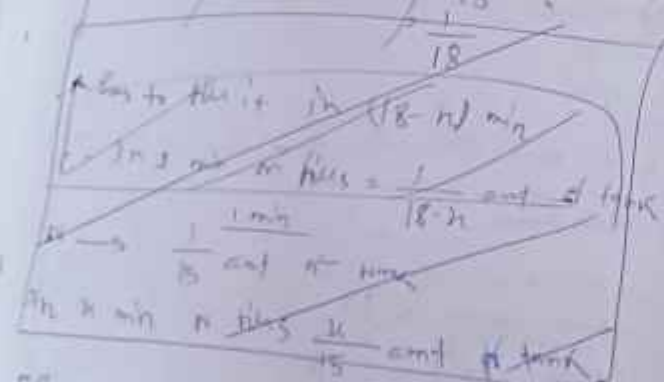
Let, after x min N is closed.

man is doing the tank alone.

32. Pipe m can fill the tank in 22.5 minutes. When the tank needs to be filled in 18 mins.

In 1 min $\frac{1}{18}$ amt of tank should be filled.
In the 18 min already filled $\frac{1}{5}$ amt of tank is left amt. i.e. $\frac{1}{5} - \frac{1}{18}$

$$= \frac{2-1}{18} = \frac{1}{18}$$



Now, m is closed after 3 mins.
So, m is m at the end he fills 18 mins
In 1 min m can fill $\frac{1}{18}$ amt of tank

$\therefore 3n = 18 - 3 = 15$
 $\frac{30^2 \times 18^2}{225} = 18$
 $\frac{30}{225} = 18$
 $\frac{4}{5}$ amt of the tank
 $\frac{4}{5}$ amt of tank is directly filled by m, so
the left amt = total - already filled
 $= 1 - \frac{4}{5}$
 $= \frac{1}{5}$ amt of tank is left to be filled

This left amt is filled by only n as we have considered m

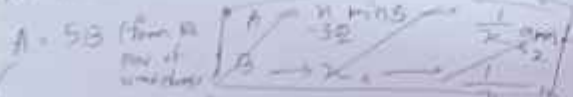
Now, n is on for 21 mins
In 1 min n fills $\frac{1}{15}$ amt of tank
If n $\frac{21}{15} = 1.4$

$\therefore \frac{21}{15} = 1.4$ i.e. n is on for 21 mins in 1.4
i.e. he can do $\frac{14}{15}$ amt of work
 $\therefore x = 3$ mins & n's work done is $\frac{4}{5}$

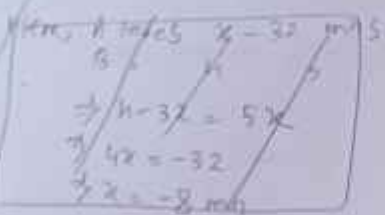
$\therefore$ n has to be closed after 3 mins, so that the entire tank gets filled in 18 mins.

⑤ Pipe A can fill a tank 3 times faster than pipe B & takes 32 minutes less than pipe B to fill the tank. If both the pipes are opened together then in how much time

the tank would be full?



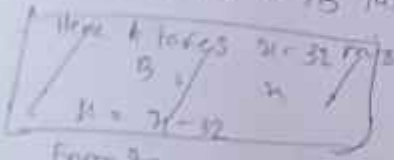
A is 5 times faster than B - i.e. if B = 1 unit
A = 5 times slower than A - i.e. A = 5 units



the time taken by B

B = 5A (B is 5 times slower than A)

(A takes 1 min, B takes 5 min)



From 7m,

$51 - 32 = \text{time taken by A}$

$19 - 32 = T$

$4T = 32$

$T = 8 \text{ min}$

A = 8 min

B = 5 * 8

= 40 min

$\frac{1 \text{ min}}{8} \text{ amt}$
 $\frac{1}{40} \text{ amt}$

[∴ In fact
time taken
is said so
we did it from
time taken per]

the total time taken by them,

$A + B$

$\frac{1}{8} + \frac{1}{40}$

$= \frac{5+1}{40}$

$= \frac{6}{40}$

$= \frac{3}{20}$

$= \frac{3}{20} \text{ min}$

$= \frac{3}{20} \text{ min}$

$= \frac{3}{20} \text{ min}$

$= \frac{3}{20} \text{ min}$

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$= \frac{3}{20} \text{ min}$

$= \frac{3}{20} \text{ min}$

② A tank has 3 taps P, Q & R. Taps P & Q can fill the tank in 1.5 & 2 hrs, respectively. Tap R can empty the completely filled tank in just half hour. Tap P is opened at 8 am, tap Q is opened at 9 am & tap R is opened at 10 am. At what exact time will the tank be empty?



let the tank will be empty after T hrs.

∴ P is on for T hrs

Q =

R =

P → $\frac{1 \text{ unit}}{1.5 \text{ hr}}$

Q → $\frac{1}{2}$

R → $\frac{1}{2}$

$\frac{1}{2}$

$\frac{1}{2}$

$\frac{1}{2}$

$\frac{1}{2}$



$$\frac{W}{P} + \frac{W}{S} + \frac{W}{R} = 0 \quad [W = \text{work done}]$$

$$\text{time done (Simply)} \quad [\text{time done} = \frac{\text{work}}{\text{rate}} \times \text{time}]$$

$\frac{W}{P}$

$\frac{W}{S}$

In 12 hrs P fills $\frac{2}{3}$ amt of tank

$$\therefore \frac{2T}{3}$$

like wise,

$$\frac{W}{S} = \frac{T-1}{2} \quad \therefore \frac{W}{R} = 2(T-2)$$

$$\therefore \frac{2T}{3} + \frac{T-1}{2} + 2(T-2) = 0$$

$$3 \quad \frac{4T + 3T - 3 + 12T - 12}{6} = 0$$

$$\therefore 19T = 15$$

$$\therefore T = \frac{15}{19}$$

$$= 4.2 \text{ hrs}$$

$$= 4 \text{ hrs } 12 \text{ min}$$

$$\begin{aligned} 1 \text{ hr} &\rightarrow 6 \text{ min} \\ 0.2 &\rightarrow 12 \text{ min} \end{aligned}$$

The tank will be empty after 4 hrs 12 min (after P was on)

$$\therefore 8 + 4 \text{ hrs } 12 \text{ min (P was on at 8 am)}$$

$$= 12 \text{ pm } 12 \text{ min} \quad \boxed{\text{but this isn't an ans}}$$

As we know that these 3 taps are filling the tank with those given flow. But we don't know about tank's capacity. i.e. it may happen that the tank is already filled & overflowing on the tank isn't full by P & S or maybe the tank is underflowing. I mean, we don't know what is tank's state if the water is overflowing/underflowing/full filled. In this cases then we can't do it by formula as in formula those cases i.e. when over/under flow happen, that we need to do is not considered. So we just need to understand & do it from our logic.

Given
P is opened at 8 am, (2 hrs before all three are opened together) & needs 15 hrs to fill the tank (i.e. $\frac{1}{15}$ tank can be filled by P, P alone can fill the tank in 15 hrs & full tank water is wasted).

At 9 am tap A is opened (1 hr before all three are opened together) & needs 2 hrs to fill the tank; but since P runs alone for the tank, let's consider all the water of A is wasted.

now at 10 am tap B is opened & needs $\frac{1}{2}$ hrs to empty the tank.

At 10 am the tank is full & all of 3 taps are opened together.

Total work done by all 3 taps (in 1 hr)

$$= \frac{20}{18} + \frac{1}{2} = 2$$

$$\frac{2}{3} = \frac{5}{6} \text{ (in 1 hr)}$$

So, when all 3 taps are open together they will empty a filled tank in $= 6 / \frac{5}{6}$ hr i.e. 1.2 hrs (1 hr 12 mins)

Therefore, the tank will be empty at 11:12 am

⑦ A cistern can be filled in 6 hrs by taps P & Q. If tap R also joins them, then cistern is filled in 5 hrs. Tap P can fill the cistern at twice the rate of tap Q. In what time tap Q & R fill the cistern?

P & Q $\rightarrow \frac{1 \text{ hr}}{6 \text{ hrs}} = \frac{1}{6}$ part of cistern
P & R $\rightarrow \frac{1 \text{ hr}}{5} = \frac{1}{5}$

$$\frac{1}{6} + \frac{1}{5} = \frac{1}{3}$$

$$\frac{1}{6} + \frac{1}{5} = \frac{1}{3}$$

$$\frac{1}{6} + \frac{1}{5} = \frac{1}{3}$$

$$\frac{1}{6} + \frac{1}{5} = \frac{1}{3}$$

Let's assume cistern fill in 1 hr
Q $\rightarrow \dots$
P $\rightarrow \dots$
R $\rightarrow \dots$

We know P & Q together fill the cistern in $\frac{1}{6}$ & hrs i.e. in 1 hr $\rightarrow \frac{1}{6}$ part of cistern
 $\therefore P + Q = \frac{1}{6}$ (the work done by P & Q together in 1 hr)
 $\frac{1}{6} + \frac{1}{5} = \frac{1}{3}$ ($\therefore$ Then 1 hr P does $\frac{1}{6}$ part of work)

Now similarly,
P & R $\rightarrow \frac{1}{5}$ — (ii)
 $\frac{1}{6} + \frac{1}{5} + \frac{1}{7} = \frac{1}{3}$
 $\frac{1}{6} + \frac{1}{5} = \frac{1}{3}$
 $\frac{1}{7} = \frac{1}{30}$

R can fill the cistern in 30 hrs

Step 1: In the current situation, the...

The person is eating 18 kg of rice in 2 days. So, he was done at p 12 times of...

Step 2: If p is eating more than 18 kg, then he can do 2 times more work than...

$$\frac{1}{x} = \frac{1}{y}$$

$$\frac{1}{x} = \frac{2}{y} \quad \text{--- (1)}$$

$$\frac{1}{x} = \frac{2}{y}$$

$$\frac{1}{x} + \frac{1}{y} = \frac{1}{6}$$

$$\frac{2}{y} + \frac{1}{y} = \frac{1}{6}$$

$$\frac{3}{y} = \frac{1}{6}$$

$$y = 18$$

So he can do 18 kg of rice in 2 days.

$$\frac{1}{18} = \frac{1}{30}$$

$$\frac{1}{18} = \frac{1}{30}$$

$$\frac{1}{18} = \frac{1}{30}$$

Step 3: If p is eating more than 18 kg, then he can do 2 times more work than...

Step 4: If p is eating more than 18 kg, then he can do 2 times more work than...

$$\frac{1}{x} = \frac{1}{y}$$

$$\frac{1}{x} = \frac{2}{y}$$

$$\frac{1}{x} = \frac{2}{y}$$

$$\frac{1}{x} + \frac{1}{y} = \frac{1}{6}$$

$$\frac{2}{y} + \frac{1}{y} = \frac{1}{6}$$

$$\frac{3}{y} = \frac{1}{6}$$

$$y = 18$$

So he can do 18 kg of rice in 2 days.

$$\frac{1}{18} = \frac{1}{30}$$

So he can do 18 kg of rice in 2 days.

So he can do 18 kg of rice in 2 days.

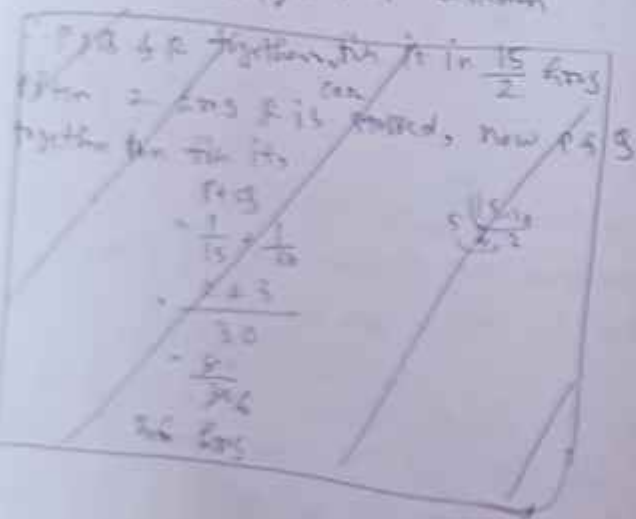
pipe R can empty a full tank in 30 hrs
 But two pipes P & S can fill a tank in
 15 hrs & 10 hrs respectively. From unknown
 opened all three taps after 2 hours stop
 worked is & closed pipe R. Due to this
 mistake how much time more would it take
 to fill the tank?

$$\frac{P \& S}{15 \text{ hrs}} \quad \frac{R}{30 \text{ hrs}}$$

$$\text{Total R} = \frac{1}{15} + \frac{1}{10} + \frac{1}{30}$$

$$= \frac{2+3-1}{30} = 1$$

$$= \frac{2}{30} \text{ out of cistern}$$



$\therefore P \& R$ can fill $\frac{2}{15}$ out of cistern in 1 hr

$\therefore$ In 2 hrs they can fill, $\frac{2 \times 2}{15}$

$$= \frac{4}{15} \text{ out of cistern}$$

$\therefore$ The left out of cistern now is $1 - \frac{4}{15}$

Now we know R is stopped & P & S left
 out will be filled by P & S

P & S together can fill ~~the tank~~

$$= \frac{1}{15} + \frac{1}{10}$$

$$= \frac{1}{6} \text{ out of cistern}$$

Let P & S together did the left work in x hrs

$\therefore$ In 1 hr they can do $\frac{1}{6}$ out of work

$$= \frac{1}{6} \times x = \frac{x}{6}$$

$$\therefore \frac{x}{6} = \frac{11}{15}$$

$$\therefore x = \frac{22}{5} \text{ hrs}$$

$$= 4.4 \text{ hrs}$$

$\therefore$ total time taken = 21 hrs

$$= 6.4 \text{ hrs (due to the mistake)}$$

If the mistake wouldn't happen then the tank would be filled in 6 hrs

$$P \& S = \frac{1}{6} \text{ out of cistern}$$

i.e. in 6 hrs the cistern can be filled

$$8-4 = 4$$

$$= 2 \times 2 \text{ fins}$$

2-4 fins more would be taken

20 mins

⑩ Arithmetic Progression (3-4 hrs)

• Basics

$$a_1, a_2, a_3, \dots, a_n \rightarrow \text{A.P. ?}$$

The only condition on the most imp condition for a series to be A.P. is the consecutive difference / the diff. b/w the consecutive terms should be equal throughout the series i.e.

$$\frac{a_2 - a_1}{a_3 - a_2} \dots a_n$$

$$\text{i.e. } a_2 - a_1 = a_3 - a_2 = \dots = \text{common} = a_n - a_{n-1}$$

Common difference = d

Let's say, the 1st no. of a series = a

In fact case if the common diff. is d , the 2nd term will be = $a + d$

$$\begin{aligned} \text{3rd} &= a + 2d \\ \text{4th} &= a + 3d \end{aligned}$$

as the diff. is increasing & that is the

Common diff.

$$\begin{aligned} \text{1st} & a, \text{ 2nd} a+d, \text{ 3rd} a+2d, \text{ 4th} a+3d \\ \text{1st term} & \quad \quad \quad \text{2nd term} \quad \quad \quad \text{3rd term} \quad \quad \quad \text{4th term} \end{aligned}$$

$$\text{the } n\text{th term } (a_n) = a + (n-1)d$$

Also denotes 2nd to 3rd term

where, n = total no. of terms in A.P.
 a = first term, a_n = n th term (last term)
 d = C.D.

$$n = \frac{\text{last term} - \text{first term}}{d} + 1$$

derived from this

Eg 1, 3, 5, 7, ... 15. Total no. of terms (n) in the series?

$$a = 1, \text{ C.D. } = 2, a_n = 15$$

$$n = \frac{15 - 1}{2} + 1$$

$$= 8 \quad (\text{i.e. 8th term is 15})$$

crosscheck (8th term should be 15)

$$\therefore a = 1, \text{ C.D. } = 2, n = 8$$

$$a_n = 1 + (8-1) \times 2$$

$$= 15$$

• Sum of an A.P. series

$$a, a+d, a+2d, a+3d, \dots, a+(n-1)d \rightarrow n \text{ terms}$$

$$= n \times d + (a + 3d) + (a + 3d) + \dots + (a + 3d)$$

$$= (a + a + \dots + n \text{ times}) + (d + d + 3d + \dots + (n-1)d)$$

$$= na + d(1+2+3+\dots+(n-1))$$

$$= na + d \cdot \frac{n(n-1)}{2}$$

$$= \frac{n[2a + (n-1)d]}{2}$$

Sum of the series = S_n

$$S_n = na + d \cdot \frac{n(n-1)}{2}$$

$$S_n = \frac{n}{2} [2a + (n-1)d]$$

$$= \frac{n}{2} [a + \{a + (n-1)d\}]$$

$$S_n = \frac{n}{2} [a + a_n]$$

$$S_n = \frac{n}{2} (\text{first term} + \text{last term})$$

[i.e. the sum of natural nos. $1+2+3+\dots+n$ is $\frac{n(n+1)}{2}$]
[Hence $\frac{n(n-1)}{2}$]
[Hence $\frac{n(n-1)}{2}$ and then $n \times d$]

• Arithmetic mean (A.M.)

If arithmetic series of odd no. of terms (say)

$$a, b, c \rightarrow b = \frac{a+c}{2} = \frac{a+n}{2}$$

$$a_1, a_2, a_3, \dots, a_n$$

$$a_1 + a_2 + a_3 + \dots + a_n$$

Let's say the n will be like,

a_1 a_2 a_3 a_4 a_5 a_6 a_7 a_8 a_9 a_{10}

Insert n arithmetic means into these 2 given nos.

$$a, m_1, m_2, m_3, \dots, m_{n-1}, b$$

considering

$$a, d = d = \frac{b-a}{n+1}$$

$$\therefore \text{2nd term} = m_1 = a + d$$

($\because$ If we know a & b we can get $m_1, m_2, \dots, m_n$)

$$m_1 = \frac{a+b-a}{n+1}$$

$$\therefore m_2 = \frac{a+2(b-a)}{n+1}$$

$$\therefore m_n = a + (n-1)d$$

$$\therefore m_n = \frac{a+(n-1)(b-a)}{n+1}$$

$$\therefore m_3 = \frac{a+3(b-a)}{n+1}$$

• Interesting thing about AP

If/Add/Subtract a particular number from all the terms of an AP the resulting series is again

$$1, 3, 5, 7, \dots$$

$$+2 \quad +2 \quad +2 \quad +2 \quad \dots$$

$$3, 5, 7, 9, \dots \text{ (AP)}$$

2) Multiply / Divide all terms of an AP by a constant $\rightarrow$ resulting series $\rightarrow$ AP

ex: 12, 3, 5, 7, ...
+ 2, +2, +2, +2, ...

2, 5, 10, 14, ... (AP)
9 4 4

3) Add 2 AP series $\rightarrow$ resulting series $\rightarrow$ AP

ex: 2, 4, 6, 8, ...
+ 3, 6, 9, 12, ...

5, 10, 15, 20, ... (AP)
5 5 5 5

1) 3rd and 7th terms of an AP are 19 & 29 respectively. Find its first term

$$\text{3rd term} = a + (3-1)d = 19$$

$$\rightarrow a + 2d = 19 \quad \text{--- (1)}$$

$$\text{7th term} = a + 6d = 29 \quad \text{--- (2)}$$

$$\text{Subtracting (1) from (2)} \\ 4d = 10 \\ d = 2.5$$

$$\therefore 4a = 56$$

$$\therefore a = 14$$

$$\therefore \text{First term} = 14$$

2) 5th term of an AP is 7 while 14th term is 17 - which of the following is its 10th term?

$$\text{8th term} = 17 \quad \text{or} \quad (10n-36) \quad \text{or} \quad n^2-25 \quad \text{or} \quad \text{none of the above}$$

$$\text{5th term} = 7 \quad \text{--- (1)}$$

$$\rightarrow a + 4d = 7$$

$$\text{--- (1)}$$

$$a + 4d = 7$$

$$a + 6d = 17$$

$$\rightarrow 2d = 10$$

$$\rightarrow d = 5$$

$$\therefore \text{10th term}$$

$$a + (n-1)d$$

$$= -13 + (10-1)5$$

$$= -13 + 5n - 5$$

$$= 5n - 18$$

$$= \frac{1}{2}(2 \times 5n - 2 \times 18)$$

$$= \frac{1}{2} \times (10n - 36)$$

$$\text{--- (2)}$$

$$\therefore a + 4d = 7$$

$$\rightarrow a + 20 = 7$$

$$\rightarrow a = -13$$

$$\text{--- (3)}$$

ie, the AP should satisfy the 5th term (7) & 7th term (17).

$$1 = 5n - 18$$

$$\rightarrow 5n - 18$$

$$= 7 \quad \text{(5th term) } \checkmark$$

$$5n - 18$$

$$= 17 \quad \text{(7th term) } \checkmark$$

1st way (through finding terms)

$$\begin{aligned} \text{1st way} & \quad \begin{aligned} & \text{1st way} \\ & \text{2nd way} \end{aligned} \\ & \quad \begin{aligned} & \text{1st way} \\ & \text{2nd way} \end{aligned} \end{aligned}$$

Sum of the above $X = 17$ $\therefore$ Ans $\rightarrow 17$

Q) The ratio of 2nd to 7th term of an AP is 1:3. 4th term is 9. Find 15th term. Also find the sum of this AP with 15 terms.

$$\begin{aligned} \text{Given, } \frac{\text{2nd term}}{\text{7th term}} &= \frac{1}{3} \\ \therefore \frac{a+d}{a+6d} &= \frac{1}{3} \\ 3(a+d) &= a+6d \\ 3a+3d &= a+6d \\ 2a &= 3d \\ a &= \frac{3}{2}d \quad \text{--- (1)} \\ \therefore \frac{3}{2}d &= \frac{1}{2}d \quad \text{(from (1))} \end{aligned}$$

$$\begin{aligned} \therefore a &= 3, d = 2 \\ \therefore 15^{\text{th}} \text{ term} &= 3 + (15-1) \cdot 2 \\ &= 3 + 14 \cdot 2 \\ &= 31 \end{aligned}$$

The AP series = 3, 5, 7, 9, 11, 13, 15, 17, 19, 21, 23, 25, 27, 29, 31

$$\begin{aligned} \text{1st way} & \quad \begin{aligned} & \text{1st way} \\ & \text{2nd way} \end{aligned} \\ & \quad \begin{aligned} & \text{1st way} \\ & \text{2nd way} \end{aligned} \end{aligned}$$

2nd way (the way we get the mean of a series)

$$\begin{aligned} 3, 5, 7, 9, 11, 13, 15, 17, 19, 21, 23, 25, 27, 29, 31 \\ \text{Mean} &= 17 \\ \text{A.M.} &= 17 \end{aligned}$$

3rd way (most shortest)

Whenever there is a series of odd no. of terms the A.M. is middle term.

Now, The middle of 15 terms = 8th term

$$\begin{aligned} \text{C.S.P.} & \quad \begin{aligned} & \text{C.S.P.} \\ & \text{C.S.P.} \end{aligned} \\ & \quad \begin{aligned} & \text{C.S.P.} \\ & \text{C.S.P.} \end{aligned} \end{aligned}$$

Here, the d is small, as well as the series is of small no. so we can do 1st & 2nd method but for long d & long no. of digits series, it will be most time consuming. So we'll always do this 3rd one.

Q. 12. The value of 10th term of the AP with 5th term is equal to 10 times its 10th term.

Ans. Let 10th term = 10th term
 $\Rightarrow \frac{a + 9d}{3} = 10 \left[\frac{a + 9d}{3} \right]$

$$3a + 15d = 5a + 45d$$

$$\Rightarrow 2a = -30d$$

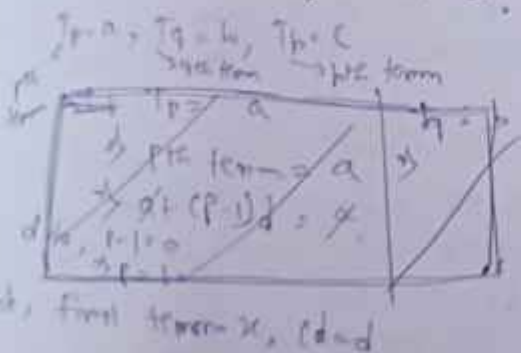
$$3a = -45d$$

$$\text{10th term} = a + 9d$$

$$= -15d + 9d$$

$$= -6d$$

Q. 13. If $T_p = a$, $T_q = b$, $T_r = c$. What is the value of $a(q-r) + b(r-p) + c(p-q)$?



$$\begin{aligned} T_p &= a & T_q &= b & T_r &= c \\ \Rightarrow \frac{a + (p-1)d}{3} &= \frac{b + (q-1)d}{3} & \Rightarrow \frac{c + (r-1)d}{3} &= \frac{a + (p-1)d}{3} \\ \Rightarrow \frac{a - x}{d} + 1 &= p & \Rightarrow \frac{b - x}{d} + 1 &= q & \Rightarrow \frac{c - x}{d} + 1 &= r \\ &= a(q-r) + b(r-p) + c(p-q) \\ &\Rightarrow a \left[\frac{b-x}{d} + 1 - \frac{c-x}{d} - 1 \right] + b \left[\frac{c-x}{d} + 1 - \frac{a-x}{d} - 1 \right] \\ &\quad + c \left[\frac{a-x}{d} + 1 - \frac{b-x}{d} - 1 \right] \\ &= a \cdot \frac{b-c}{d} + b \cdot \frac{c-a}{d} + c \cdot \frac{a-b}{d} \\ &= \frac{1}{d} [ab - pc + qb - qa + qc - rb] \\ &= \frac{1}{d} \cdot 0 \\ &= 0 \end{aligned}$$

Q. 14. The 12th term of an AP is -10 & the sum of first 14 terms is 36. Find the first term & common difference.

$$\begin{aligned} \text{Given: } (i.e., \text{ nos}) & \quad \Rightarrow S_{14} = 36 \quad (i.e., 14) \\ \Rightarrow 12\text{th term} &= -10 & \Rightarrow \frac{14}{2} [2a + 13d] &= 36 \times 18 \\ \Rightarrow \text{out } 11d &= -10 & \Rightarrow 2a + 13d &= 18 \\ &= -10 \div 2 & & \Rightarrow 2a + 13d = 18 \quad (1) \end{aligned}$$

$$\begin{aligned} 2a + 22d &= -20 \\ 2a + 3d &= 18 \end{aligned}$$

$$\begin{aligned} 3d &= -38 \\ d &= -2 \end{aligned}$$

What is the greatest possible sum of the AP $23, 21, 19, \dots$?

The terms are $23, 21, 19, 17, 15, \dots$
 $\downarrow \quad \downarrow \quad \downarrow$
 $-2 \quad -2 \quad -2$
 $\therefore 3, 5, \dots$ (increasing)

If we want it's a falling AP series. If we want to find the greatest possible sum, we need to find out the smallest positive no. after that the nos will become $\leq$ that will give "sum i.e. sum w/h get reduced". The smallest positive no. here is 1. As the series is of odd nos, the smallest positive is also an odd no.

If we want to get the greatest possible sum of that series (basically we don't know how many terms are there, so we don't know what is the n th term, so we can consider it as an series) we have to

from (1)

$$\begin{aligned} a &= -22 + 2d \\ a &= 12 \end{aligned}$$

consider the series till 1 (as if we consider more the sum will be negative, as 3rd condition)

now the series become,

$$23, 21, 19, \dots$$

the sum (or the greatest possible sum) = $\frac{n}{2} (23 + 1)$
 At the series -1

$$\begin{aligned} \text{Now } n &= \frac{1 - 23}{-2} + 1 \\ &= \frac{-22}{-2} + 1 \\ &= 12 \end{aligned}$$

$$\therefore \text{from (1), } \frac{12}{2} (23)$$

$$= 144$$

⑧ $1+3+4+5+7+7+10+11 \dots$ 10 terms?

(The 7th is, a series is given, up to 10 terms what is the sum of the series (S_n)?)

$$1+3+4+5+7+7+10+11 \dots 10$$

1st we need to check it is an AP or not (while this type of sum is given, we need to check it 1st).

we can write it like, It's a mixup of 2 series
 $1+3+5+7+9+11 \dots$ (increasing) and $4+7+10 \dots$ (increasing)

Proof (method)

$$(1+5) + (5+9) + \dots + (29+33) = 60 \text{ terms}$$

[Add consecutive 2 terms]

$$= 4+9 + 14+19 + \dots + 28+33$$

$$2 \times S_n = \frac{n}{2} [2a + (2n-1)d]$$

$$2 \times S_n = \frac{n}{2} [2 + (2n-1)5]$$

$$2 \times S_n = \frac{n}{2} [2 + 10n - 5]$$

$$2 \times S_n = \frac{n}{2} [10n - 3]$$

$$2 \times S_n = \frac{n}{2} [10n - 3]$$

$$2 \times S_n = \frac{n}{2} [10n - 3]$$

$$2 \times S_n = \frac{n}{2} [10n - 3]$$

Q1 First term of an AP is 3. The sum of first 5 terms is equal to 1/4 of the sum of next five terms. What is the 25th term?

Given, $a = 3$, $S_5 = \frac{1}{4} S_{10}$

Sum of 1st 5 terms = $\frac{1}{4}$ sum of next 5 terms

$\frac{5}{2} [2 \times 3 + 4d] = \frac{1}{4} \times \frac{5}{2} [2a + 4d]$

$6 + 4d = \frac{1}{4} \times 2(a + 2d)$

$6 + 4d = \frac{1}{2} (a + 2d)$

$12 + 8d = a + 2d$

$12 + 8d = 3 + 2d$

$9 = -6d$

$d = -\frac{3}{2}$

$\therefore a_{25} = a + 24d = 3 + 24 \times -\frac{3}{2} = 3 - 36 = -33$

$$12 + 8d = a + 2d$$

$$a - 6d = 12$$

$$\text{the sum of 25 terms}$$

$$S_{25} =$$

Proof:

$a_1, a_2, a_3, a_4, a_5, a_6, a_7, a_8, a_9, a_{10}$ (first 10 terms)

$a_{11}, a_{12}, a_{13}, a_{14}, a_{15}, a_{16}, a_{17}, a_{18}, a_{19}, a_{20}$ (next 10 terms)

$S_{10} - S_5$

$\therefore S_5 = \frac{1}{4} (S_{10} - S_5)$

$\therefore 4S_5 = S_{10} - S_5$

$\therefore 5S_5 = S_{10}$

$\therefore \frac{5}{2} [2 \times 3 + 4d] = \frac{10}{2} [2 \times 3 + 4d]$ [$a = 3$]

$$5 [6 + 4d] = 2 [6 + 8d]$$

$$30 + 20d = 12 + 16d$$

$$2d = -18$$

$$d = -9$$

$$\therefore a_{25} = a + 24d = 3 + 24 \times -9 = 3 - 216 = -213$$

Q. Sum of 3 nos. in an AP is 18 & sum of their squares is 158. What is the smallest no. Here we'll take the 3 terms as -

$a-d, a, a+d$ (In general we take them like $a, a+d, a+2d$ -) (If we take them like this, we will not be able to cancel them like this.)

$$\text{now, } a-d+a+a+d=18 \text{ (given)}$$

$$\Rightarrow 3a=18$$

$$\Rightarrow a=6 \text{ --- (1)}$$

$$\text{now, } (a-d)^2 + a^2 + (a+d)^2 = 158 \text{ (given)}$$

$$\Rightarrow a^2 - 2ad + d^2 + a^2 + a^2 + 2ad + d^2 = 158$$

$$\Rightarrow 3a^2 + 2d^2 = 158$$

$$\Rightarrow 3 \times 36 + 2d^2 = 158 \text{ (from (1))}$$

$$\Rightarrow 108 + 2d^2 = 158$$

$$\Rightarrow 2d^2 = 50$$

$$\Rightarrow d = \pm 5$$

now, the terms are,

$$\begin{cases} 1, 6, 11 & \text{(considering } +5) \end{cases}$$

$$\begin{cases} 11, 6, 1 & \text{(considering } -5) \end{cases}$$

In both cases the smallest no. is 1, so the smallest no. is 1.

Q. What is the sum of nos. in nth row of the following series? (The three terms are given use it for options, don't go after solving it as that's time consuming)

$$\begin{matrix} 1 & 2 & 4 \\ 1 & 3 & 4 \\ 5 & 6 & 7 & 8 & 9 \end{matrix}$$

(increasing)

$$i) n^2 + 1, ii) n^2 - 1, iii) n^3 + (n-1)^3$$

$$1 \rightarrow \text{sum} = 1$$

$$1+2+4 \rightarrow \text{sum} = 7$$

$$1+3+4+5+6+7+8+9 \rightarrow \text{sum} = 35$$

2. We try that any option get filled into these values

[no. no. of row]

$\Rightarrow n^2 + 1$ (if sum of the nth row is that, then the no. should satisfy for each of 1st $\rightarrow 1+1^2+1 \times$ now the each value, no. $(3, 2, -1)$)

$$2nd \rightarrow (2^2+1) \times \frac{5}{19} \times \text{(not satisfying)}$$

$$i) n^2 - 1$$

$$1st \rightarrow (2 \cdot 1)^2 - 1 = 2 - 1 = 1 \checkmark$$

$$2nd \rightarrow (2+2)^2 - 1 = 16 - 1 = 15 \neq 7 \times$$

(not satisfying)

$$\therefore n^3 + (n-1)^3 \rightarrow \text{answer}$$

$$iii) n^3 + (n-1)^3$$

$$1st \rightarrow (1^3 + (1-1)^3) = 1 = 1 \checkmark$$

$$2nd \rightarrow (2^3 + (2-1)^3) = 8 + 1 = 9 \neq 7 \checkmark$$

$$3rd \rightarrow (3^3 + (3-1)^3) = 27 + 8 = 35 \checkmark$$

(Satisfying)

24 (20) Word Problems on numbers or numbers system or Problems on numbers

Important note

Understand the Qn → Find the ans

① The sum of 2 numbers is 36 & their product is 248. What will be the sum of their reciprocals.

$$\begin{aligned} a+b &= 36 \\ ab &= 248 \\ \frac{a}{ab} + \frac{b}{ab} &= \frac{36}{248} \\ \frac{a+b}{ab} &= \frac{36}{248} \\ \frac{36}{248} &= \frac{9}{62} \end{aligned}$$

$$\frac{1}{a} + \frac{1}{b} = \frac{a+b}{ab} = \frac{36}{248} = \frac{9}{62}$$

② In a poultry farm having hens & peacocks can see 84 heads & 282 legs. How

many hens are there?

H → head
L → legs

$$H + P = 84 \quad \text{--- (I)}$$

$$2H + 4P = 282 \quad \text{--- (II)}$$

$$\Rightarrow H + 2P = 141 \quad \text{--- (III)}$$

$$H + P = 84 \quad \text{--- (I)}$$

$$\Rightarrow H + 2P = 141 \quad \text{--- (III)}$$

$$\Rightarrow P = -57$$

$$P = 57$$

$$H = 84 - 57$$

$$= 27$$

③ Summation of 5 consecutive numbers is found out to be 335. If we add the largest & smallest no. what will we get?

$$(n) + (n+1) + (n+2) + (n+3) + (n+4) = 335$$

$$\Rightarrow 5n + 10 = 335$$

$$\Rightarrow 5n = 325$$

$$\Rightarrow n = 65$$

$$2(n) + (n+4) = 2(65) + 4 = 134$$

Consecutive no. 5, 6, 7, 8, 9

26. @ Shre Rajwade's rising attention instead of multiplying by $\frac{3}{4}$, he divided by $\frac{3}{4}$. This led to a difference of 14. In the 2 answers, what is the number?

$$(m + \frac{3}{4}) - \frac{m}{(\frac{3}{4})} = 14$$

$$\Rightarrow \frac{3m}{4} - \frac{4m}{3} = 14$$

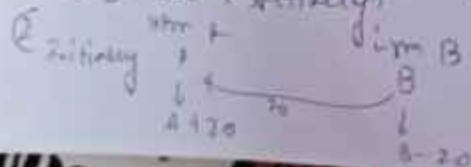
$$\Rightarrow \frac{3m - 16m}{12} = 14$$

$$\Rightarrow -13m = 16 \times 12$$

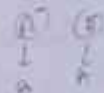
$$\text{neglect } \Rightarrow m = \underline{24}$$

1. didn't know which one is greater

27. Ramon has 2 urns. Both these urns have some pebbles. If 20 pebbles from urn B are shifted to urn A then, the no. of pebbles in each urn become interchanged. But if 10 pebbles from urn A are put into urn B, then number of pebbles in B are twice the no. in A. How many pebbles do A & B have respectively?



28. Shifting the no. of pebbles in each urn become interchanged.



$$\therefore A = B - 20 \quad \text{--- (i)}$$

$$\Rightarrow B - A = 20 \quad \text{--- (ii)}$$



$$B = 2A$$

$$\Rightarrow B - A = 2(A - 10)$$

$$\Rightarrow B - A = 2A - 20$$

$$\therefore 2A - B = 30 \quad \text{--- (iii)}$$

$$-A + B = 40 \quad \text{--- (iv)}$$

$$2A - B = 30 \quad \text{--- (v)}$$

$$\Rightarrow A = 70$$

$$\therefore 70 + B = 40$$

$$\Rightarrow B = 110$$

$$\therefore A = 70, B = 110$$

29. On exchanging the digits in unit's & ten's place, the difference b/w original & the new no.s become 27. The digit in the unit's place is 2 times the digit in the hundred's place. The digit in the ten's place is 3 times the digit in the

2. (Handwritten note: What is 75% of the original number?)



$$\begin{aligned} & 100 - 25 = 75 \\ & 75 \times 100 = 7500 \\ & 7500 \div 100 = 75 \end{aligned}$$

$$\begin{aligned} & 100H - 123H = 27 \\ & 23H = 27 \\ & H = 3 \\ & U = 6 \\ & 75 \div 9 \end{aligned}$$

$$\begin{aligned} & 114U (\text{original no.}) = \\ & \text{original no.} = 114U = 396 \Rightarrow 75\% \text{ of } 528 \\ & = \frac{75}{100} \times 528 \\ & = 396 \end{aligned}$$

$$\begin{aligned} & U = 24 \\ & T = 34 \end{aligned}$$

Application for both original & new

$$\begin{aligned} \text{from (I)} \quad H \times 100 + 34 \times 10 + 24 \times 1 \\ = 1324 \text{ (original no.)} \end{aligned}$$

$$\begin{aligned} \text{Now } H \times 100 + 7 \times 10 + U \times 1 \\ \text{exchange} \\ 2H \times 100 + U \times 10 + 7 \times 1 \\ \text{from (II)} \quad H \times 100 + 2H \times 10 + 34 \times 1 \\ = 1234 \text{ (new no.)} \end{aligned}$$

Q. In an area 2% families have 5 children each. But 8% have no children at all. Amongst the rest of the families 18% have 4 children & 2% have only one child. How many families live in the area, if 207 families have either 2 or 3 children each?

Let there are 100 families in the area.
 $2\% \times 100 = 2$ families have 5 children.
 $8\% \times 100 = 8$ families have no children.
 $100 - 10 = 90$

$$\begin{aligned} & 18\% \times 100 = 18 \text{ families have 4 children} \\ & 2\% \times 100 = 2 \text{ families have 1 child} \\ & \text{Total of 100 families} = 100 \text{ families} \end{aligned}$$

$$\begin{aligned} & 18\% \times 100 = 18 \text{ families have 4 children} \\ & 2\% \times 100 = 2 \text{ families have 1 child} \\ & \text{Total of 100 families} = 100 \text{ families} \end{aligned}$$

$$\begin{aligned} & 100 - 10 = 90 \\ & 90 - 18 = 72 \\ & 72 - 2 = 70 \\ & 70 \div 2 = 35 \end{aligned}$$

21. If the no. of families is x
 families with no. 8/9/1 child
 $18 + 2 + 18 + 24 = 62$

$= 55\%$

Total = 100
 Rem = 45%

- Rem 45% are 2/3 children
 Rem families are 2/3, i.e. 55%
 $= 45\%$

Then,

$45\% \text{ of } x = 297$
 $\Rightarrow 45x = 297$
 $\Rightarrow x = 660$

22. 1/4 part of tank is full of water.
 When 30 liters of water is taken out
 it becomes empty. The capacity of the
 tank is?

~~Initial part
 Full = $\frac{3}{4}$, empty = $\frac{1}{4}$
 Let capacity = x
 Given, $x - 30$~~



i.e. the tank has 30 lit
 full, i.e. $\frac{3}{4}$ of the tank
 is full of 30 lit water.



Let tank

$\frac{3}{4} \rightarrow 30 \text{ lit}$
 $1 \rightarrow ?$

$\frac{3}{4} \times ? = 30$

$\Rightarrow ? = 40 \text{ lit}$ (Capacity, i.e. max
 water the
 tank can contain)

mine

$\frac{3}{4} \rightarrow 30$

$1 \rightarrow ?$

$\frac{1}{4} = 30 \Rightarrow \frac{3}{4} = 30$

$\Rightarrow 30 \times 10$

$\Rightarrow 30 \times 10 = 40 \text{ lit}$ (true capacity)

23. A group wanted to renovate their club.
 Each member contributed an amount
 equal to twice the no. of members in the
 club. But the government contributed same

$$\begin{array}{r} 12 \\ 72 \\ \hline \end{array}$$

1000 oranges are divided among some boys & girls whose total number is 50 so that each boy gets 5 oranges & each girl gets 7 oranges the number of girls is?

$$B + G = 50 \quad \text{--- (1)}$$

$$5B + 7G = 250 \quad \text{--- (2)}$$

$$1B \rightarrow 5 \quad | \quad 1G \rightarrow 7$$

$$5B \rightarrow 5B \quad | \quad 5G \rightarrow 35$$

$$5B + 7G = 250 \quad \text{--- (2)}$$

$$B + G = 50 \quad \text{--- (1)}$$

$$5B + 7G = 250 \quad \text{--- (2)}$$

$$5B + 7G = 250$$

$$5B + 7G = 250$$

$$42G = 30$$

$$7G = 15 \quad \therefore B = 35$$

Problems on Trains (1-2 hrs)

Formula

$$\frac{D}{S \times T} \rightarrow \frac{D}{S/T} = \frac{D}{T/S}$$

$$D = S \times T$$

$$T = \frac{D}{S}$$

$$S = \frac{D}{T}$$

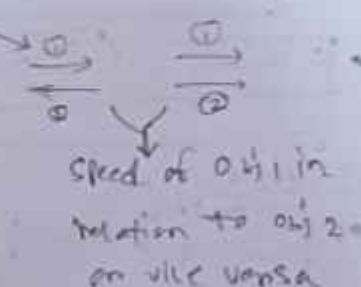
Types

1) Same
2) T
3) S
4) S

Trains $\rightarrow$ 1) length of the train in distance traveled will be considered
2) length of pole, car, man etc 0
(if mentioned some man only occurs units then only considers it.)

3) Relative Speed

However it's said the train is travelling towards this is respect to stationary obj



Speed of obj 1 in relation to obj 2 or vice versa

Relative Speed

when we are trying to find the speed or velocity of a moving object in relation to another moving obj.

Relative Speed

Normal speed of an obj

Car → (moving)

Tree (not moving)

Stationary obj

Speed is related to stationary obj = normal speed

Relative Speed of an obj

Car → (moving)

Car → (moving)

Relative Speed

Speed is related to moving obj = Relative Speed

A moving obj → Find out → Relative Speed

2 concepts

1. Same direction

2. Opposite direction

Relative Speed of T_1 w.r.t $T_2 = v_1 + v_2$

opposite direction → (+) add

Train (speed)

T_1

T_2

Train (speed)

relative speed of T_1 w.r.t $T_2 = |v_1 - v_2|$

Same direction

→ (-) subtract

2 trains with lengths 120 m & 110 m respectively are moving towards each other. Their speeds are 12 m/s & 23 m/s, respectively. What will be the time needed by the trains to cross each other?

T_1 120 m

T_2 110 m

12 m/s

23 m/s

The trains will completely cross each other

T_1 → T_2

Distance travelled by $T_1 = T_2$ distance = 110 m

$T_2 = 9.5$

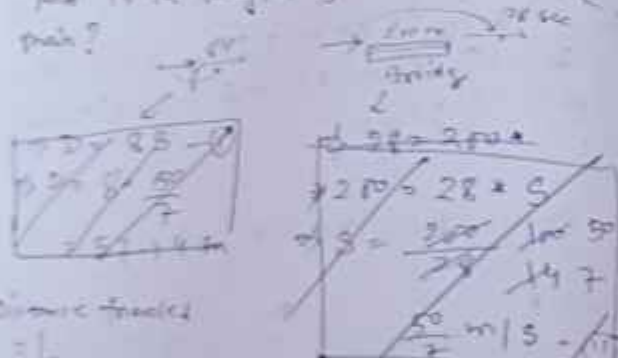
total distance travelled = 120 + 110 = 230 m
 $T_1 + T_2 = 12 + 23 = 35$
 $230 / 35 = 6.57$

while 2 trains are crossing each other asked how much total distance is travelled by them. Addition of their lengths

$$\begin{aligned} \text{Speed} &= 10 + 2.7 \\ &= 35 \text{ m/s} \\ \therefore 7 &= \frac{205 \times 10^3}{35.7} \end{aligned}$$

= 7.56

Q A train passes a stationary pole in 8 seconds the train also passes a 200 m long bridge in 28 seconds. Find the length & the speed of the train?



Distance traveled
= L

The train's

Speed

is same

both, so

Speed must

be same, also it's

the same train, so it'll have same speed.

Length of the train
has to be included in
distance traveled.

$$200 + L = \text{Distance traveled}$$

$$28.5 = \text{Time taken}$$

So?

$$S_A = S_B$$

$$\therefore \frac{L}{8} = \frac{200 + L}{28}$$

$$\therefore 28L = 400 + 22L$$

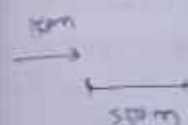
$$6L = 400$$

$$L = 80 \text{ m}$$

$$\therefore L = 80 \text{ m}$$

$$\therefore S_A = S_B = \frac{80}{8} = 10 \text{ m/s}$$

Q A train having length 150m passes a platform of 550m length. The time taken for it is 56 seconds. In how much time will this train take to pass the platform of 250m length?



$$D = 150 + 550$$

$$T = 56$$

$$S = \frac{700}{56} = 12.5$$

$$S = \frac{300}{24} = 12.5$$

$$\therefore \frac{2.5}{2} \text{ m/s}$$

$$L = 150, P = 250$$

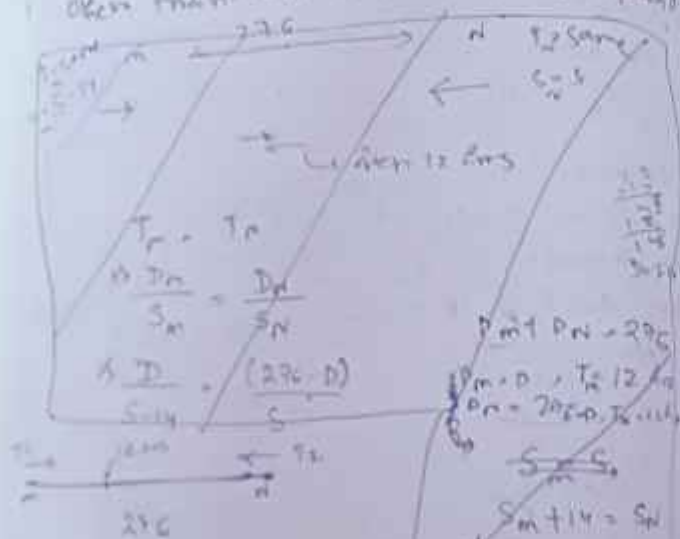
$$T = \frac{150 + 250}{12.5} \times 2$$

$$= \frac{400}{12.5} \times 2$$

$$= \frac{160}{5}$$

$$= 32.5$$

Q) Stations m & n are 220m apart. A train starts from m & B from n towards m & n respectively. They meet after 12 hrs. The train travelling from m to n is slower by 10 km/hr in comparison to the other train. What is speed of slower train?



$$T_r = T_n$$

$$\frac{D_m}{S_m} = \frac{D_n}{S_n}$$

$$\frac{D}{S_m} = \frac{(220 - D)}{S_n}$$

$$S_m = S_n - 10$$

$$\frac{D}{S_n - 10} = \frac{220 - D}{S_n}$$

$$D S_n = (220 - D)(S_n - 10)$$

$$D S_n = 220 S_n - 2200 - D S_n + 10 D$$

$$2 D S_n = 220 S_n - 2200 + 10 D$$

$$2 D S_n - 220 S_n = -2200 + 10 D$$

$$S_n (2 D - 220) = -2200 + 10 D$$

$$S_n = \frac{-2200 + 10 D}{2 D - 220}$$

Let P & Q are 100 m apart. A train starts from P & B from Q towards P & Q respectively. They meet after 12 hrs. The train travelling from P to Q is slower by 10 km/hr in comparison to the other train. What is speed of slower train?

$$S_m + 10 = S_n$$

$$\frac{D}{S_m} = \frac{100 - D}{S_n}$$

$$\frac{D}{S_n - 10} = \frac{100 - D}{S_n}$$

$$D S_n = (100 - D)(S_n - 10)$$

$$D S_n = 100 S_n - 1000 - D S_n + 10 D$$

$$2 D S_n = 100 S_n - 1000 + 10 D$$

$$2 D S_n - 100 S_n = -1000 + 10 D$$

$$S_n (2 D - 100) = -1000 + 10 D$$

$$S_n = \frac{-1000 + 10 D}{2 D - 100}$$

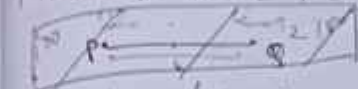
$$2 D + 14 = 23$$

$$2 D = 23 - 14$$

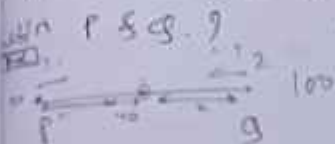
$$2 D = 9$$

$$D = \frac{9}{2} = 4.5 \text{ km/hr}$$

Q) From P & Q, two trains start moving towards each other at the same time. Their speeds are 20 km/hr & 100 km/hr respectively. When the 2 trains meet, 100 km has covered 40 m more than other train. Find the distance between P & Q.



Let P & Q are 100 m apart. A train starts from P & B from Q towards P & Q respectively. They meet after 12 hrs. The train travelling from P to Q is slower by 10 km/hr in comparison to the other train. What is speed of slower train?



$$t_1 = t_2$$

$$\frac{D_1}{S_1} = \frac{D_2}{S_2}$$

$$\frac{D_1}{20} = \frac{D_2}{100}$$

$$D_1 = \frac{20 D_2}{100}$$

$$D_1 = \frac{D_2}{5}$$

$$D_1 + D_2 = 100$$

$$\frac{D_2}{5} + D_2 = 100$$

$$\frac{D_2 + 5 D_2}{5} = 100$$

$$\frac{6 D_2}{5} = 100$$

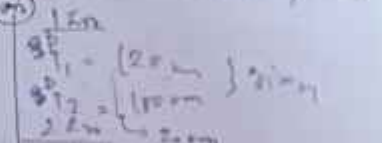
$$D_2 = \frac{100 \times 5}{6}$$

$$D_2 = \frac{500}{6}$$

$$D_2 = 83.33 \text{ km}$$

$$D_1 = \frac{D_2}{5} = \frac{83.33}{5} = 16.67 \text{ km}$$

Let P & Q are 100 m apart. A train starts from P & B from Q towards P & Q respectively. They meet after 12 hrs. The train travelling from P to Q is slower by 10 km/hr in comparison to the other train. What is speed of slower train?



$$S_1 = 20 \text{ km/hr}$$

$$S_2 = 100 \text{ km/hr}$$

$$D_1 = 20 \times t$$

$$D_2 = 100 \times t$$

Let P & Q are 100 m apart. A train starts from P & B from Q towards P & Q respectively. They meet after 12 hrs. The train travelling from P to Q is slower by 10 km/hr in comparison to the other train. What is speed of slower train?

$$D_1 + D_2 = 100$$

$$\frac{D_2}{5} + D_2 = 100$$

$$\frac{D_2 + 5 D_2}{5} = 100$$

$$\frac{6 D_2}{5} = 100$$

$$D_2 = \frac{100 \times 5}{6}$$

$$D_2 = \frac{500}{6}$$

$$D_2 = 83.33 \text{ km}$$

$$D_1 = \frac{D_2}{5} = \frac{83.33}{5} = 16.67 \text{ km}$$

- ② 2 trains A & B leave valvata for station at 8:00 pm & 8:30 pm respectively. A runs at 20 km/hr & B 120 km/hr. At what distance from valvata will the 2 trains meet?



distance same for both

$$30 \times T = 120 \times (T - \frac{1}{2})$$

$$\Rightarrow 3T = 4T - 2$$

$$\Rightarrow T = 2 \text{ hr}$$

$$T_A = 2 \text{ hr}, T_B = 2 - \frac{1}{2} = \frac{3}{2} \text{ hr}$$

$$D_A = D_B = 120 \times 2 = 180 \text{ km}$$

- ③ A train overtakes 2 boys who are running at the rate of 8 km/hr & 16 km/hr in the same direction as the train. The train completely passes them in 30 seconds & 10 seconds respectively. What is the length of the train?



When the train passes boy A then its speed becomes 8 km/hr & it completely passes A in 30 seconds.

We need to consider 2 cases for 2 boys. Case 1: From both cases the train is moving to the right.

$$\Rightarrow 30 \times T_1 = S_1 \times T_2 \quad \left[\text{Let speed of train} = S \right]$$

$$\frac{1}{30} \times (S - 8) \times 3600 = (S - 16) \times \frac{10}{60}$$

$$\Rightarrow 5S - 72 = 160 - 160$$

$$\Rightarrow S = 88 \text{ km/hr}$$

$$\text{distance traveled by train} = \text{length of train} (L)$$

$$= (88 - 8) \times \frac{30}{60}$$

$$= 80 \times \frac{30}{60}$$

$$= \frac{4}{3} \times 15$$

$$= \frac{4}{3} \times 15 = 0.8 \text{ m} \text{ or } 800 \text{ m}$$

② With stoppage, the speed of a train is 36 km/hr. However, without stoppage, it is 40 km/hr. Find out for how many times does the train stop per hour?
 Let's the train travels for 1 hr.
 If ~~no~~ stoppage it covers = 40 km
 If no stoppage it covers = 36 km

Because of stopping the train is wasting time.

The time for train is wasting = the time for which it is stopping

$$T = \frac{4}{40} \quad \text{(Always we have to take the ideal / good speed i.e., the speed while not stopping)}$$

$$\Rightarrow T = 0.1 \text{ hr}$$

$$\therefore 0.1 \times 60 = 6 \text{ min}$$

③ There are 2 trains of same length & moving in same direction. They are moving at 5 sec & 6 sec respectively. For how much time would they cross each other?

Here, $T = \frac{D}{P-S}$

$\frac{L}{P-S}$ → when they cross i.e. the distance travelled = $L_1 + L_2$ (crossing)
 $= L + L = 2L$

$$S_1 = \frac{L}{5} \quad | \quad S_2 = \frac{L}{6}$$

$$P-S = \frac{L}{5} - \frac{L}{6}$$

$$= \frac{L}{30}$$

$$T = \frac{2L \times 30}{L} = 60 \text{ sec}$$

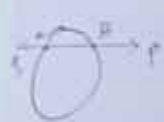
~~if we get a distance, that is involving a function, neglect.~~

neglect it, as we don't know which one is greater.

② Resurrection & Geometry

① Which of the following statement is true in regard to a circle?

- i) Every chord is secant X (independent sets)
- ii) Every secant is chord X
- iii) Secant is technically not a chord, but it contains a chord. ✓
- iv) Every secant line defines a unique chord. ✓



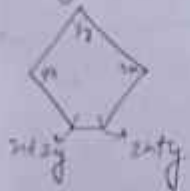
AA → chord
 BB → Secant
 AP → secant
 CP → secant

• Chord & Secant are 2 independent sets
 & mutually exclusive (no relation)

∴ only 2 & 3 - Independent

② If the expression shows one the degree
 & measures of the angle of the polygon find
 the value n.

Exterior angle
 extension =
 regular & irregular
 poly gon -
 (concept)



• Pentagon & hexagon

Interior angle (angles inside) $\text{Sum} = (n-2) \times 180^\circ$

n = no.
 of
 sides

True regular polygon [all sides & angles are equal]

Not regular → 2 dimensions

$$\text{Interior angle} = \frac{(n-2) \times 180}{n} \quad (\text{for regularity})$$

$$\begin{aligned}
 \text{Int. Ang. } 3n + \text{Ext. Ang. } 2n + \text{Int. Ang. } 2n &= (n-2) \times 180 \\
 \text{Int. Ang. } 10n &= (5-2) \times 180 \\
 \text{Int. Ang. } 10n &= 540 \\
 \text{Int. Ang. } &= 54
 \end{aligned}$$

③ One angle of a regular polygon measures
 177°. This polygon has total on 'n' sides,
 'n' is a multiple of which the following
 numbers?

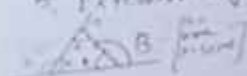
$$\begin{aligned}
 \text{Given, } \frac{(n-2) \times 180}{n} &= 177 \\
 \checkmark \quad 180n - 360 &= 177n \\
 \Rightarrow 3n &= 360 \\
 \Rightarrow n &= 120
 \end{aligned}$$

∴ It's the multiple of 3, 4, 12, etc.
All of the above

$$\begin{aligned}
 \text{④ 1 angle} &\rightarrow 177^\circ \\
 \text{there are } n \text{ angles} \\
 \therefore 177n &= (n-2) \times 180 \quad (\text{sum of all angles})
 \end{aligned}$$

④ Find the sum of the measures of one interior & one exterior angle of a regular n -gon.

- A: Interior angle
- B: Exterior angle



Any Interior-
any Exterior
 $= 180^\circ$

$\hookrightarrow$ valid only
for regular polygons



$$A + B = 180^\circ$$

Interior + Corresponding
Extensors
 $= 180^\circ$

$\hookrightarrow$ valid for all
polygons Regular
Polygons

$n = 960$
1/3 Polygon $\therefore$ any int. ext. $= 180^\circ$
 $\therefore 180^\circ$

⑤ What is the measure of the radius of the circle inscribed in a triangle whose sides measure 8, 15 & 17 units?



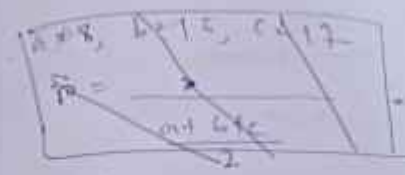
$\rightarrow$ Distance (touching 3 sides of tri)

Radius
 r : in radius

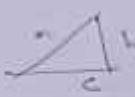
$$r = \frac{\text{area of } \Delta}{\text{Semi-perimeter of } \Delta}$$



$$\begin{aligned} \text{Semi-perimeter} &= \frac{\text{Perimeter}}{2} \\ \text{Perimeter} &= a + b + c \\ \therefore S &= \frac{a+b+c}{2} \end{aligned}$$



area of Δ
Heron's formula



$$\begin{aligned} A(\text{area}) &= \sqrt{S(S-a)(S-b)(S-c)} \\ S &= \frac{a+b+c}{2} \end{aligned}$$

$$\begin{aligned} S &= \frac{a+b+c}{2} = \frac{8+15+17}{2} \\ &= \frac{40}{2} \\ &= 20 \end{aligned}$$

$$\begin{aligned} A &= \sqrt{20(20-8)(20-15)(20-17)} \\ &= \sqrt{20 \times 12 \times 5 \times 3} \\ &= \sqrt{3600} \\ &= 60 \end{aligned}$$

$$r = \frac{60}{20} = 3$$

• For right angle Δ (sides, 1 perpendicular, 1 hypotenuse, 1 radius)



$$\text{radius} = \frac{\text{width}}{2}$$

$$r = \text{inradius} = \frac{\text{area}}{\text{semi-perimeter}}$$

$$= \frac{1}{2} \times \frac{b \times p + \text{area}}{\frac{b+p+h}{2}}$$

$$r = \frac{b \times p}{b+p+h}$$

Direct formula for r

$$r = \frac{b+p-h}{2}$$

Ex

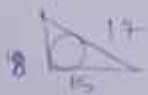
$$12^2 + 8^2 = 20^2$$

$$16^2 + 8^2 = 20^2$$

$\therefore$ is a right angle Δ

$$r = \frac{8+16-20}{2}$$

$$= \frac{4}{2} = 2$$



$$\text{Ex) } r = \frac{b+p}{b+p+h}$$

$$= \frac{8 \times 16}{8+16+20}$$

$$= \frac{128}{44} = 3$$

What is the measure of the radius of the circle inscribed in a triangle whose sides measure 8, 15, 17 units?

For Equilateral Δ (all sides are same)



ΔOQR

$$a^2 = h^2 + \left(\frac{a}{2}\right)^2$$

$$\Rightarrow h^2 = a^2 - \frac{a^2}{4}$$

$$\Rightarrow h^2 = \frac{3a^2}{4}$$

$$h = \frac{\sqrt{3}a}{2}$$

$$\text{height of equilateral } \Delta = \frac{\sqrt{3}a}{2}$$

now for Equilateral Δ ,

$$r(\text{inradius}) = \frac{\text{area}}{\text{semi-perimeter}} = \frac{\frac{\sqrt{3}}{4} a^2}{\frac{3a}{2}}$$

$$= \frac{\sqrt{3} a^2 a}{2 \times 3a \times 1/4}$$

$$r = \frac{a}{2\sqrt{3}}$$

$$\sin A = \frac{a}{c}$$

$$\Rightarrow a = \frac{16}{\frac{4}{5}}$$

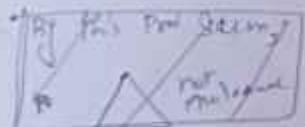
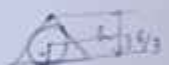
$$\sin A = \frac{a}{c}$$

$$\Rightarrow \frac{4}{5}$$

$$\Rightarrow \frac{8}{3}$$

$$\Rightarrow \frac{8}{3}$$

Radius of circle inscribed in $\triangle ABC$ is $\frac{1}{3}$



Find the radius of the circle inscribed in the triangle ABC, having sides 10 cm, 16 cm, 18 cm.

Meaning: half angle on the

$$\sin A = \frac{a}{c}$$

$$\sin A = \frac{a}{c}$$

$$\sin A = \frac{a}{c}$$

$$\sin A = \frac{a}{c}$$

$$\sin A = \frac{a}{c}$$

$$\sin A = \frac{a}{c}$$

$$\sin A = \frac{a}{c}$$

$$\sin A = \frac{a}{c}$$

$$\sin A = \frac{a}{c}$$

$$\sin A = \frac{a}{c}$$

$$\sin A = \frac{a}{c}$$

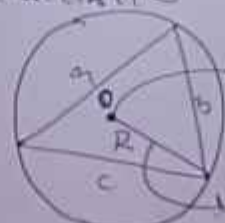
$$\sin A = \frac{a}{c}$$

$$\sin A = \frac{a}{c}$$

$$\sin A = \frac{a}{c}$$

$$\sin A = \frac{a}{c}$$

Circumference of circle



Circumference

Circumference

R (circumradius) = $\frac{\text{product of sides of } \Delta}{4 \times \text{area of } \Delta}$

$$R = \frac{abc}{4 \times A} \quad (A = \text{area of } \Delta)$$

• For right angle Δ

$$R = \frac{b \cdot c \cdot a}{2 \times \frac{1}{2} \times b \cdot c} = \frac{a}{2} = \frac{\text{hypotenuse}}{2}$$



• Equilateral Δ

$R = \frac{\text{product of sides}}{4 \times A}$

$$= \frac{a \cdot a \cdot a}{4 \times \frac{\sqrt{3}}{4} \times a^2} = \frac{a}{2\sqrt{3}}$$

$$R = \frac{a}{2\sqrt{3}}$$

$$R = \frac{\text{side}}{\sqrt{3}}$$



Q7 find the radius of the circle that circumscribes a triangle ABC is inscribed, having sides 13, 14, 15 cm.



ABC = not equilateral, not right angle
normal case
Circumradius $R = \frac{13 \times 14 \times 15}{4 \times A}$

$$= \frac{13 \times 14 \times 15}{4 \times 84} = \frac{25}{3} = 8.33$$

③ Elementary Statistics

• If items arranged in ascending / descending order -

1) $n \rightarrow \text{odd}$, median = $\left(\frac{n+1}{2}\right)^{\text{th}} \text{ term}$

2) $n \rightarrow \text{even}$, median = $\frac{\left(\frac{n}{2}\right)^{\text{th}} \text{ term} + \left(\frac{n}{2} + 1\right)^{\text{th}} \text{ term}}{2}$

• If median is odd, median $\rightarrow$ In half the
 n-odd, median $\rightarrow$ may not be in
 the given terms

• mode
 is term with most occurrence
 term having highest freq.

eg: 5, 5, 5, 5
 mode = 5

• mode will be definitely from the given terms

• mean ~~may~~ may not from the given terms

① The heights (in cm) of 9 students of a class are as follows:

150, 160, 145, 150, 147, 152, 149, 148, 145
 Find the median of this data.

145, 145, 147, 148, 149, 150, 152, 155, 160
 median = 149

Example: dog has 5 puppies the weights of
 5 puppies with whom they were born.
 their weights were 587g, 635g, 440g, 470g, 500g.
 Find the mean weight of the puppies?
 $\frac{587 + 635 + 440 + 470 + 500}{5} = 528.4$
 (not int)

$$\text{mean} = \frac{\text{observations sum}}{n} = \frac{\sum x}{n}$$

mean $\times n = \text{sum int}$ (not decimal)

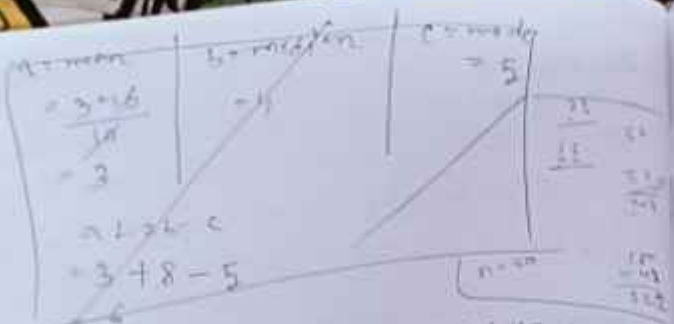
528.4 $\times$ 5 = int (from other options with

find the mode of the following set of numbers-

25, 27, 30, 32, 41, 22, 28, 33, 44, 51, 31, 32, 32
 - 3, 2

② In 50 numbers 10 are threes, 15 are
 fives, 18 are fives, & remaining are
 sixes. If a, b, c respectively represent the
 mean, mode & median of data, what is
 the value of $a + 2b - c$?

~~10 (three), b = 15 (five), c = 18 (five)~~
 mode = 5 (x)
 median = 5 (x)



10 (freq)
 15 (freq)
 78 (freq)
 7 (freq)

b (median)
 $n = 50$ (freq)

$$\frac{10}{2} + \frac{(10+1)15}{2} = \frac{10 + 165}{2}$$

$$= \frac{175}{2}$$

$$= \frac{175}{2}$$

$$((\text{mode}) = 5$$

$$a + b + c = \frac{175}{2} + 5 + 5 = 113.5$$

$$2a + b + c$$

$$= 4 + 6 + 5 = 15$$

$$= 3 + 3 + 3 = 9$$

$$3 \times \text{median} = \text{mode} + 2 \times \text{mean}$$

$$3 \times \text{median} = 6 + 2 \times 6$$

$$= \frac{6 + 12}{3}$$

$$= \frac{18}{3}$$

$$= 6$$

If the ratio of mode & median of a distribution is 6:5, then the ratio of its mean & median is?

$$\frac{\text{mode}}{\text{median}} = \frac{6}{5} = \frac{12}{10}$$

a mode is used/kids be distributed ultimately, 50 K=1

$$2 \text{ mode} = 12$$

$$\text{mode} = 6$$

$$\text{median} = \frac{3 \times 5 - 6}{2} = \frac{9}{2} = 4.5$$

$$\therefore \frac{\text{mode}}{\text{median}} = \frac{6}{4.5} = \frac{4}{3} = \frac{2}{1.5}$$

① Identify the statement which is not true?

a) It is necessary to have mean, mode & median for a given data set?

b) We can't have more than 1 mean, one median & one mode for a given data set.

c) Mean, mode, median has to be a value from given data set.

d) All the above are not true.

② The salary in rupees of 10 employees in a company are 50, 55, 60, 65, 70, 75, 80, 85, 90, 95.

What is the standard deviation in the above data?

$$s.d = G = \sqrt{\text{var}}$$

$$\text{var} = 62$$

$$G = \sqrt{\frac{\sum (x_i - \bar{x})^2}{n}}$$

$\downarrow$
 $\frac{\sum}{n}$
 var

$$\therefore \bar{x} = \frac{50 + 55 + 60 + 65 + 70 + 75 + 80 + 85 + 90 + 95}{10}$$

$$= \frac{700}{10}$$

$$= 70$$

$$G = \sqrt{\frac{(20)^2 + (15)^2 + (10)^2 + (5)^2 + 0 + (2)^2 + (5)^2 + (10)^2 + (15)^2 + (20)^2}{10}}$$

$$= \sqrt{\frac{20^2 + 15^2 + 10^2 + 5^2 + 0 + 2^2 + 5^2 + 10^2 + 15^2 + 20^2}{10}}$$

$$= \sqrt{\frac{1420}{10}}$$

$$= \sqrt{142}$$

$$= 11.916$$

(var = 142)

③ The mean of a set of data is 5. What will be the mean if the is subtracted from each?

Effect of mean / Effect of avg

$$\frac{x_1 + x_2 + x_3 + \dots + x_n}{n} = 5 \quad \text{--- (1)}$$

$$(x_1 - 5) + (x_2 - 5) + \dots + (x_n - 5) = ?$$

$$\therefore ? = (x_1 + x_2 + \dots + x_n) - 10n$$

$$\therefore ? = \frac{(x_1 + x_2 + \dots + x_n) - 10n}{n}$$

$$= \frac{5n - 10n}{n}$$

$$= -5$$

1.11) If initial avg is 31.5 then it is mixed with the data points

new avg = 32.5

Verify for $\frac{a}{b} \rightarrow \frac{a}{b} \times m$
 $\frac{a}{b} \rightarrow \frac{a}{b} \times m$
 $\frac{a}{b} \rightarrow \frac{a}{b} \times m$

in the sum 5 - 10 - 5

10) What value should be added to each of the data: 5, 10, 15, 20, 25, 30, 35 so that the mean & median become equal? 10, 15, 20, 25, 30, 35

Altered mean = $5 + 10 + 15 + 20 + 25 + 30 + 35 = 140$

Median = $\frac{20 + 25}{2} = 22.5$
 $\frac{140}{8} = 17.5$
 $17.5 + 2.5 = 20$
 $20 + 2.5 = 22.5$
 $22.5 + 2.5 = 25$
 $25 + 2.5 = 27.5$
 $27.5 + 2.5 = 30$
 $30 + 2.5 = 32.5$
 $32.5 + 2.5 = 35$
 $35 + 2.5 = 37.5$
 $37.5 + 2.5 = 40$
 $40 + 2.5 = 42.5$
 $42.5 + 2.5 = 45$
 $45 + 2.5 = 47.5$
 $47.5 + 2.5 = 50$
 $50 + 2.5 = 52.5$
 $52.5 + 2.5 = 55$
 $55 + 2.5 = 57.5$
 $57.5 + 2.5 = 60$
 $60 + 2.5 = 62.5$
 $62.5 + 2.5 = 65$
 $65 + 2.5 = 67.5$
 $67.5 + 2.5 = 70$
 $70 + 2.5 = 72.5$
 $72.5 + 2.5 = 75$
 $75 + 2.5 = 77.5$
 $77.5 + 2.5 = 80$
 $80 + 2.5 = 82.5$
 $82.5 + 2.5 = 85$
 $85 + 2.5 = 87.5$
 $87.5 + 2.5 = 90$
 $90 + 2.5 = 92.5$
 $92.5 + 2.5 = 95$
 $95 + 2.5 = 97.5$
 $97.5 + 2.5 = 100$
 $100 + 2.5 = 102.5$
 $102.5 + 2.5 = 105$
 $105 + 2.5 = 107.5$
 $107.5 + 2.5 = 110$
 $110 + 2.5 = 112.5$
 $112.5 + 2.5 = 115$
 $115 + 2.5 = 117.5$
 $117.5 + 2.5 = 120$
 $120 + 2.5 = 122.5$
 $122.5 + 2.5 = 125$
 $125 + 2.5 = 127.5$
 $127.5 + 2.5 = 130$
 $130 + 2.5 = 132.5$
 $132.5 + 2.5 = 135$
 $135 + 2.5 = 137.5$
 $137.5 + 2.5 = 140$
 $140 + 2.5 = 142.5$
 $142.5 + 2.5 = 145$
 $145 + 2.5 = 147.5$
 $147.5 + 2.5 = 150$
 $150 + 2.5 = 152.5$
 $152.5 + 2.5 = 155$
 $155 + 2.5 = 157.5$
 $157.5 + 2.5 = 160$
 $160 + 2.5 = 162.5$
 $162.5 + 2.5 = 165$
 $165 + 2.5 = 167.5$
 $167.5 + 2.5 = 170$
 $170 + 2.5 = 172.5$
 $172.5 + 2.5 = 175$
 $175 + 2.5 = 177.5$
 $177.5 + 2.5 = 180$
 $180 + 2.5 = 182.5$
 $182.5 + 2.5 = 185$
 $185 + 2.5 = 187.5$
 $187.5 + 2.5 = 190$
 $190 + 2.5 = 192.5$
 $192.5 + 2.5 = 195$
 $195 + 2.5 = 197.5$
 $197.5 + 2.5 = 200$
 $200 + 2.5 = 202.5$
 $202.5 + 2.5 = 205$
 $205 + 2.5 = 207.5$
 $207.5 + 2.5 = 210$
 $210 + 2.5 = 212.5$
 $212.5 + 2.5 = 215$
 $215 + 2.5 = 217.5$
 $217.5 + 2.5 = 220$
 $220 + 2.5 = 222.5$
 $222.5 + 2.5 = 225$
 $225 + 2.5 = 227.5$
 $227.5 + 2.5 = 230$
 $230 + 2.5 = 232.5$
 $232.5 + 2.5 = 235$
 $235 + 2.5 = 237.5$
 $237.5 + 2.5 = 240$
 $240 + 2.5 = 242.5$
 $242.5 + 2.5 = 245$
 $245 + 2.5 = 247.5$
 $247.5 + 2.5 = 250$
 $250 + 2.5 = 252.5$
 $252.5 + 2.5 = 255$
 $255 + 2.5 = 257.5$
 $257.5 + 2.5 = 260$
 $260 + 2.5 = 262.5$
 $262.5 + 2.5 = 265$
 $265 + 2.5 = 267.5$
 $267.5 + 2.5 = 270$
 $270 + 2.5 = 272.5$
 $272.5 + 2.5 = 275$
 $275 + 2.5 = 277.5$
 $277.5 + 2.5 = 280$
 $280 + 2.5 = 282.5$
 $282.5 + 2.5 = 285$
 $285 + 2.5 = 287.5$
 $287.5 + 2.5 = 290$
 $290 + 2.5 = 292.5$
 $292.5 + 2.5 = 295$
 $295 + 2.5 = 297.5$
 $297.5 + 2.5 = 300$
 $300 + 2.5 = 302.5$
 $302.5 + 2.5 = 305$
 $305 + 2.5 = 307.5$
 $307.5 + 2.5 = 310$
 $310 + 2.5 = 312.5$
 $312.5 + 2.5 = 315$
 $315 + 2.5 = 317.5$
 $317.5 + 2.5 = 320$
 $320 + 2.5 = 322.5$
 $322.5 + 2.5 = 325$
 $325 + 2.5 = 327.5$
 $327.5 + 2.5 = 330$
 $330 + 2.5 = 332.5$
 $332.5 + 2.5 = 335$
 $335 + 2.5 = 337.5$
 $337.5 + 2.5 = 340$
 $340 + 2.5 = 342.5$
 $342.5 + 2.5 = 345$
 $345 + 2.5 = 347.5$
 $347.5 + 2.5 = 350$
 $350 + 2.5 = 352.5$
 $352.5 + 2.5 = 355$
 $355 + 2.5 = 357.5$
 $357.5 + 2.5 = 360$
 $360 + 2.5 = 362.5$
 $362.5 + 2.5 = 365$
 $365 + 2.5 = 367.5$
 $367.5 + 2.5 = 370$
 $370 + 2.5 = 372.5$
 $372.5 + 2.5 = 375$
 $375 + 2.5 = 377.5$
 $377.5 + 2.5 = 380$
 $380 + 2.5 = 382.5$
 $382.5 + 2.5 = 385$
 $385 + 2.5 = 387.5$
 $387.5 + 2.5 = 390$
 $390 + 2.5 = 392.5$
 $392.5 + 2.5 = 395$
 $395 + 2.5 = 397.5$
 $397.5 + 2.5 = 400$
 $400 + 2.5 = 402.5$
 $402.5 + 2.5 = 405$
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 $572.5 + 2.5 = 575$
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 $597.5 + 2.5 = 600$
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 $617.5 + 2.5 = 620$
 $620 + 2.5 = 622.5$
 $622.5 + 2.5 = 625$
 $625 + 2.5 = 627.5$
 $627.5 + 2.5 = 630$
 $630 + 2.5 = 632.5$
 $632.5 + 2.5 = 635$
 $635 + 2.5 = 637.5$
 $637.5 + 2.5 = 640$
 $640 + 2.5 = 642.5$
 $642.5 + 2.5 = 645$
 $645 + 2.5 = 647.5$
 $647.5 + 2.5 = 650$
 $650 + 2.5 = 652.5$
 $652.5 + 2.5 = 655$
 $655 + 2.5 = 657.5$
 $657.5 + 2.5 = 660$
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 $662.5 + 2.5 = 665$
 $665 + 2.5 = 667.5$
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 $670 + 2.5 = 672.5$
 $672.5 + 2.5 = 675$
 $675 + 2.5 = 677.5$
 $677.5 + 2.5 = 680$
 $680 + 2.5 = 682.5$
 $682.5 + 2.5 = 685$
 $685 + 2.5 = 687.5$
 $687.5 + 2.5 = 690$
 $690 + 2.5 = 692.5$
 $692.5 + 2.5 = 695$
 $695 + 2.5 = 697.5$
 $697.5 + 2.5 = 700$
 $700 + 2.5 = 702.5$
 $702.5 + 2.5 = 705$
 $705 + 2.5 = 707.5$
 $707.5 + 2.5 = 710$
 $710 + 2.5 = 712.5$
 $712.5 + 2.5 = 715$
 $715 + 2.5 = 717.5$
 $717.5 + 2.5 = 720$
 $720 + 2.5 = 722.5$
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 $730 + 2.5 = 732.5$
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 $737.5 + 2.5 = 740$
 $740 + 2.5 = 742.5$
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 $745 + 2.5 = 747.5$
 $747.5 + 2.5 = 750$
 $750 + 2.5 = 752.5$
 $752.5 + 2.5 = 755$
 $755 + 2.5 = 757.5$
 $757.5 + 2.5 = 760$
 $760 + 2.5 = 762.5$
 $762.5 + 2.5 = 765$
 $765 + 2.5 = 767.5$
 $767.5 + 2.5 = 770$
 $770 + 2.5 = 772.5$
 $772.5 + 2.5 = 775$
 $775 + 2.5 = 777.5$
 $777.5 + 2.5 = 780$
 $780 + 2.5 = 782.5$
 $782.5 + 2.5 = 785$
 $785 + 2.5 = 787.5$
 $787.5 + 2.5 = 790$
 $790 + 2.5 = 792.5$
 $792.5 + 2.5 = 795$
 $795 + 2.5 = 797.5$
 $797.5 + 2.5 = 800$
 $800 + 2.5 = 802.5$
 $802.5 + 2.5 = 805$
 $805 + 2.5 = 807.5$
 $807.5 + 2.5 = 810$
 $810 + 2.5 = 812.5$
 $812.5 + 2.5 = 815$
 $815 + 2.5 = 817.5$
 $817.5 + 2.5 = 820$
 $820 + 2.5 = 822.5$
 $822.5 + 2.5 = 825$
 $825 + 2.5 = 827.5$
 $827.5 + 2.5 = 830$
 $830 + 2.5 = 832.5$
 $832.5 + 2.5 = 835$
 $835 + 2.5 = 837.5$
 $837.5 + 2.5 = 840$
 $840 + 2.5 = 842.5$
 $842.5 + 2.5 = 845$
 $845 + 2.5 = 847.5$
 $847.5 + 2.5 = 850$
 $850 + 2.5 = 852.5$
 $852.5 + 2.5 = 855$
 $855 + 2.5 = 857.5$
 $857.5 + 2.5 = 860$
 $860 + 2.5 = 862.5$
 $862.5 + 2.5 = 865$
 $865 + 2.5 = 867.5$
 $867.5 + 2.5 = 870$
 $870 + 2.5 = 872.5$
 $872.5 + 2.5 = 875$
 $875 + 2.5 = 877.5$
 $877.5 + 2.5 = 880$
 $880 + 2.5 = 882.5$
 $882.5 + 2.5 = 885$
 $885 + 2.5 = 887.5$
 $887.5 + 2.5 = 890$
 $890 + 2.5 = 892.5$
 $892.5 + 2.5 = 895$
 $895 + 2.5 = 897.5$
 $897.5 + 2.5 = 900$
 $900 + 2.5 = 902.5$
 $902.5 + 2.5 = 905$
 $905 + 2.5 = 907.5$
 $907.5 + 2.5 = 910$
 $910 + 2.5 = 912.5$
 $912.5 + 2.5 = 915$
 $915 + 2.5 = 917.5$
 $917.5 + 2.5 = 920$
 $920 + 2.5 = 922.5$
 $922.5 + 2.5 = 925$
 $925 + 2.5 = 927.5$
 $927.5 + 2.5 = 930$
 $930 + 2.5 = 932.5$
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 $967.5 + 2.5 = 970$
 $970 + 2.5 = 972.5$
 $972.5 + 2.5 = 975$
 $975 + 2.5 = 977.5$
 $977.5 + 2.5 = 980$
 $980 + 2.5 = 982.5$
 $982.5 + 2.5 = 985$
 $985 + 2.5 = 987.5$
 $987.5 + 2.5 = 990$
 $990 + 2.5 = 992.5$
 $992.5 + 2.5 = 995$
 $995 + 2.5 = 997.5$
 $997.5 + 2.5 = 1000$

1.11) If initial avg is 31.5 then it is mixed with the data points
 mean = 33.5, med = 33.5
 or med same (always)
 5 data points = med
 mean = 33.5
 new mean need to reduced as it's 35 now.
 so how much we need to reduce mean in the

$$\frac{35 - 33.5}{33.5} = 1.5$$

1.5 need to reduce will be reduced to get 33.5
 now if mean is reduced by 1.5, that implies from each data points 1.5 is reduced
 (effect of mean)

8 data points
 $1 \rightarrow 8 \times 1.5$ reduced
 $8 \rightarrow 8 \times 1.5$
 $\rightarrow 12$ total reduction

Ans = The val. which we need to replace total reduction

$35 - 12 = 23$

- 11) If the mean of 26 obs is 29 & on adding four more obs, the new mean becomes 32. What is the mean of the last four obs?

$$\frac{S_{26}}{26} = 29$$

$$\Rightarrow S_{26} = 29 \times 26$$

Given,

$$\frac{S_{26} + S_4}{30} = 32$$

$$\Rightarrow 29 \times 26 + S_4 = 32 \times 30$$

$$\Rightarrow S_4 = 32 \times 30 - 29 \times 26$$

$$= 960 - 754$$

$$= 206$$

$$\therefore S_4 = 206$$

$$\therefore \bar{A}_4 = \frac{S_4}{4} = \frac{206}{4} = 51.5$$

12) Mrs. A's Distances

Table

- It is same as time & work, time & wages
- $\frac{P}{T} = \frac{W}{T}$

- If 100 files take in 6 hrs
- In 1 hr $\rightarrow \frac{1}{6}$ part of work is filled

$$\frac{P}{T} = \frac{W}{T}$$

- Filling $\rightarrow \frac{1}{6}$
- Remaining $\rightarrow \frac{5}{6}$

- $P + T = 1$ - filling the total part of work in 1 hr

- Tap - leak = Filling the total amount of (blowing) (removing) tank in 1 hr

- If in the 10, 'time taken' on any other's POV is given, then solve the ans by that method i.e., that POV

- While doing sums, verify if looking too hard for this topics, but too easy this

Let's say you did not find the answer in this way, then you can try the other way.